UNIT-3

MACHINE LEARNING

2024

CONVOLUTIONAL NEURAL NETWORKS (CNNs)

A Convolutional Neural Network (CNN) is a special type of artificial neural network. used mainly for processing images, videos and visual data

It is designed to automatically detect important features such as edges, shapes, textures, and objects from input images.

It is one of the most important technique in Deep learning and computer vision.

Main Idea of CNN

A normal neural network treats every pixel separately, but CNN processes images using small filters called kernels.

- These filters move across the image and extract useful information.

FLATTENING in CNN

Flatening is the process of converting a multi-dimensional features map (2D or 3D date) into a single one-dimensional vector.

It is used in a convolutional Neural Network before sending data to the fully connected layer.

Why Flattering is Needed

After convolution and pooling operations, the output is still in matrix form.

- Features map size - 4x4 x3

But the fully connected layer accepts only a 1D vector.

So flattring converts.

- 4x4x3 → 48 values in one row.

Example → Suppose pooling layer output is

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

After Flattening

[1, 2, 3, 4]

Important Points

- Converts 2D / 3D data into 1D data
- Used between pooling layer and fully connected layer
- Does not lose data values
- Makes data ready for classification.

Advantage

- Simple conversion process
- Help connect CNN layer to dense layer
- Easy classification

Disadvantage

- Large feature maps create very large vectors
- Can increase computations.

Role in CNN → CNN layers extract features. while Flattering prepares those features for final classification

Subsampling in CNN

Subsampling is the process of reducing the size of feature map in Deep learning or ML while keeping the important information.

In CNN, subsampling is mainly done using the pooling layer.

It reduces

- Width
- Height
- Number of computations.

without losing important features.

Purpose of Subsampling

- Reduce image dimensions
- Reduce computation time
- Prevent overfiting
- keep important features only

- Subsampling is also known as pooling
- The main goal of subsampling is to make the network more efficiently by reducing the amount of data that must be processed in later layers.

Types of subsampling.

① Max Pooling.

Max pooling selects the largest value from each pooling region

For the example above.

max (1,3,5,6) = 6
max (2,4,1,2) = 4
max (7,8,1,2) = 8
max (3,4,5,6) = 6

Output → 6 4
8 6.

Advantages

- Preserves the strongest features.
- Works well for image recognition tasks

② Average Pooling.

Average Pooling computes the average value within each pooling region.

Example

→ (1+3+5+6)/4 = 3.75

Output → 3.75

Advantages :- • Retains more background information.

• Produces smoother feature maps.

Padding 2025, 2024

- Padding is a technique used in CNN, where extra pixels (usually zeros) are added around the border of an input image or feature map before applying a convolution operation.

- Padding helps control the size of the output feature map and ensures that important information near the image boundaries is not lost.

Why is Padding Needed.

Without padding, the size of the feature map decreases after every convolution.

For example:-

Input image size :- 5X5

kernel size :- 3X3

stride :- 1

Output size = $\frac{(5-3)}{1} + 1 = 3$

Output = 3X3

* The feature map becomes smaller. After several convolution layers, the image may shrink too much.

Padding solves this problem by adding extra pixels around the image.

Example of Padding (original image) 5X5

1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1

After Adding Padding = 1

0 0 0 0 0 0 0 0 0
0 1 1 1 1 1 1 0
0 1 1 1 1 1 1 0
0 1 1 1 1 1 1 0
0 1 1 1 1 1 1 0
0 1 1 1 1 1 1 0
0 1 1 1 1 1 1 0
0 0 0 0 0 0 0 0

The image size becomes → 7X7.

Stride

Stride is the number of pixels by which the convolution filter (kernel) moves across the input image at each step.

Example

- stride=1 → The filter moves 1 pixels at a time.
- stride = 2 :- The filter moves 2 pixels at a time. producing a smaller output feature map.

Effects of Stride.

- Smaller stride(1) :- Captures more details and produces a larger output.

- Larger stride (2 or more) :- Reduce computation and produces a smaller output.

In short
→ "stride is the number of pixels the convolution kernel moves across the input image during the convolution operation."

Convolutional Layer in CNN

A Convolutional Layer is the core building block of a Convolutional Neural Network CNN.

It is responsible for extracting important features such as edge, textures, shapes, and patterns from the input image.

A convolutional layer applies a set of small filters (kernels) to the input image.

Working of a Convolutional Layer.

1. Input Image. - The image is provided as input to the convolutional layer.

2 Kernel (Filter)

→ . A small matrix (e.g 3X3 or 5X5) is used to detect specific features.

3. Convolution Operation.

. The filter moves across the images

. At each position. element - wise multiplica -tion is performed between the filter and the image region.

. The result are summed to produced a single value.

4. Features Map

. The output generated after applying the filter is called a feature map.

Components of a convolutional layer.

- Input :- Image or feature map from the previous layer.
- Kernel (Filter). Detects patterns and features.
- Stride - Determines how many pixels the filters moves each step.
- Padding :- Adds extra pixels around the image to preserve dimensions.
- Activation function :- Introduces non-linearity.

Advantages.

- Reduces the no. of parameters compared to fully connected layers.
- Automatically extracts features
- Improves image recognition performance

Pooling Layer in CNN 2025

Pooling layer is a layer in a CNN that reduces the size of feature maps while preserving the most important information.

It helps decrease computation memory usage and overfitting.

The pooling layer performs a downsampling operation on the feature map generated by the convolutional layer.

Working of Pooling layer.

1. The feature map is divided into small regions (e.g. 2X2)
2. A pooling operation is applied to each region.
3. One value is selected from each ~~each~~ region to create a smaller feature map.

Advantages of Poolin

- Reduces feature map dimensions
- Decreases computational cost
- Reduces overfitting
- Makes the model more robust to small shifts and distortions.

Disadvantages.

- Some information may be lost during downsampling.
- Excessive pooling can reduce import -ant details

Loss Layer

Loss layer is the final layer in CNN. that calculate the error (loss) between the model's predicted output and the actual target output.

The loss layer measures how far the predicted result is from the correct result. The objective of training is to minimize this loss.

Working of a Loss Layer.

1. The CNN produces prediction
2. The prediction is compared with the actual target value.
3. A loss function computes the error
4. The error is propagated backward through the network.
5. Weights are updated to reduce the loss.

Common Loss functions.

1. Cross-Entropy Loss :- Used for classification problems.

For example, Actual class : Cat

Predicted probability :- Cat = 0.9
Dog = 0.1

A higher probability for the ~~higher~~ correct class results in lower loss.

2. Mean Squared Error (MSE)

Used mainly for regression tasks.

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

where.

y_i = Actual value

$\hat{y}_i$ = Predicted value

n = Number of samples.

Importance of the loss layer.

- Evaluates model performance
- Guides the learning process.
- Helps optimize network parameters

Inception Network. 2024

The Inception Network, also known as GoogleNet, is a deep CNN architecture developed by researchers at Google.

It won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) in 2014.

An Inception Network in CNN architecture that performs multiple convolution operations of different sizes (1x1, 3x3 and 5x5) in parallel and then combines their outputs.

This allows the network to capture features at different scales.

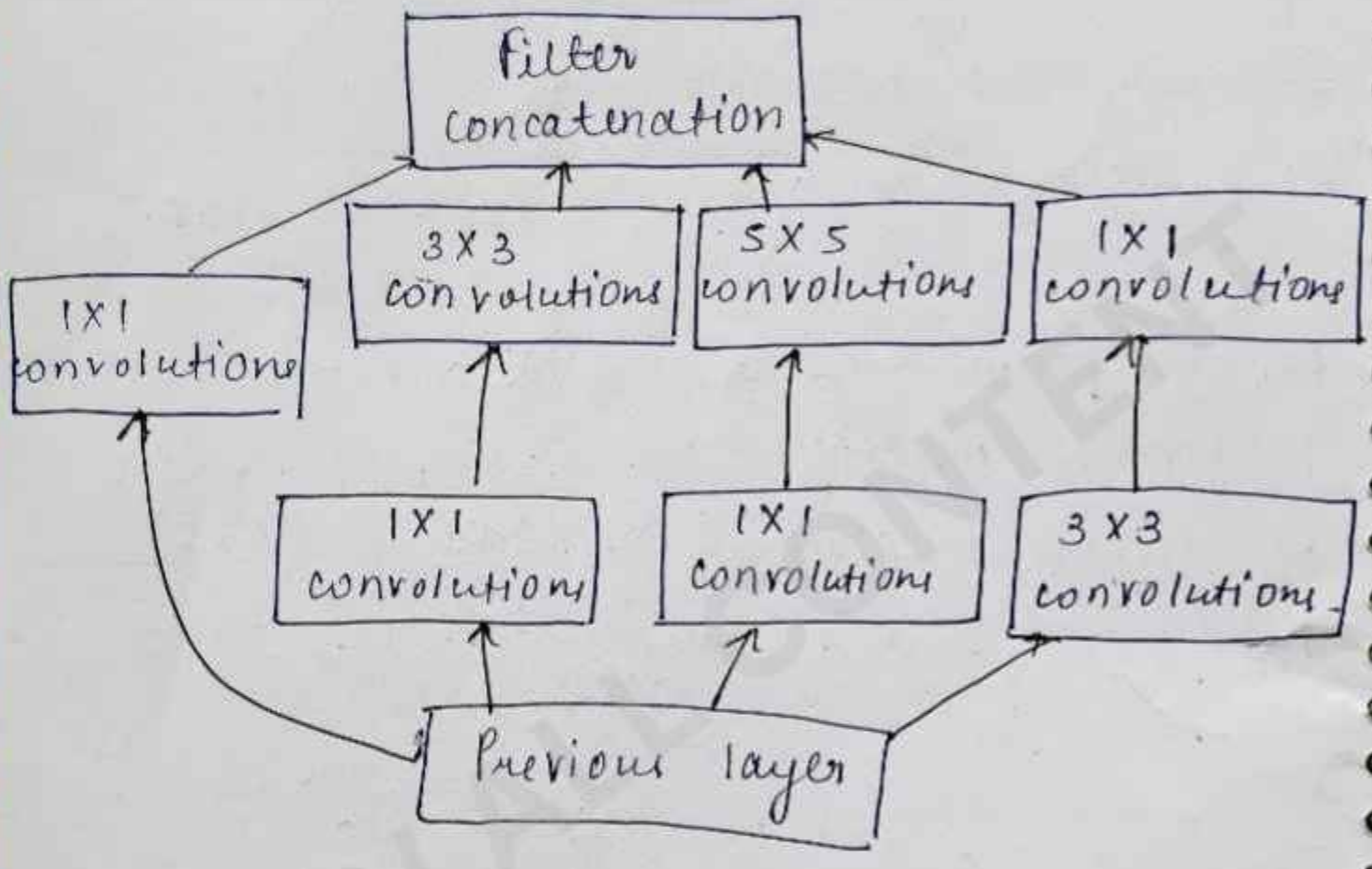
Inception Module.

The main building block of GoogleNet is the Inception Module.

It contains:

- 1x1 Convolution
- 3x3 Convolution
- 5x5 Convolution
- 3x3 Max pooling.

All these operations are performed in parallel, and their outputs are concatena-ted into a single feature map.



Inception Network.

Role of 1X1 convolution

The 1X1 convolution is used before 3X3 and 5X5 convolutions to :-

- Reduces the number of channels
- Lower computational cost
- Reduce memory usage

Transfer Learning.

Tranfer Learning is a ML technique in which a model trained on one task is reused for another related task. Instead of training a new model from scratch. the knowledge learned by a pre-trained model is transferred to the new problem.

Working.

1. Train a model on a large dataset (source task).
2. Save the learned features and weights
3. Use the pre-trained model for a new related task (target task).
4. Retrain only the final layers or fine-tune the model.

Types.

① Feature Extraction - Freeze pre-trained layers and trained only the new output layers.

(2). Fine-Tuning - Retrain some of the pre-trained layers along with the new layers.

Advantages.

1. Requires less training data
2. Reduce training time
3. Improves accuracy
4. Reduces computational cost.

Disadvantages.

1. Not effective if source and target tasks are very different
2. Choosing the wrong pre-trained model may reduce performance

Application

- Image Classification
- Object Detection
- face Recognition
- Medical Image Analysis.

One Shot Learning

One shot learning is a technique in which a model learns to recognize or classify an object from only one training example.

Unlike traditional ML which requires many examples. One-Shot Learning can learn from a very small amount of data.

Working

1. The model is trained to compare similarities between objects.
2. Give one example of a new class. it learns its features.
3. When a new sample arrives. the model compares it with the stored example.
4. If they are similar. it assigns the same class

Example → Suppose a face recognition system sees only one photo of a person. Later. when another photo of that person is shown. the system identifies the person by comparing facial features with the single stored images.

Application
→ Face Recognition
→ Signature Verification
→ Object Recognition.
→ security System.

Dimension Reductions. 2025

Dimension Reduction is the process of reducing the number of input features (variables) in a dataset while keeping as much important information as possible

for example :-

Student	Age	Height	Weight	Marks	Attendance
A	18	176	65	85	90
B	19	168	62	80	88

Here, Age, Height, Weight, Marks and Attendance are 5 dimensions (features) of some features are unnecessary or highly related, we can reduce them to fewer features. This process is called Dimension Reduction.

Need of Dimension Reduction.

Datasets in ML may contain many features

- Too many features can :-
  - Increase Training time
  - Require more memory
  - Cause Overfitting
  - Make Visualization difficult.

- Dimension Reduction helps simplify the data.

Types of Dimension Reduction.

(1) Feature selection.
Select only the most important features and remove the rest.

Example :- From . Age, Height. weight. marks, Attendance.

Keep - Attendance, marks. Age.

Remove less useful features.

Methods - Filter Methods
- Wrapper Methods
- Embedded Methods.

2. Feature Extraction.

Create new features by combining existing features.

Example: Instead of using
- Math Marks
- Physics Marks
- Chemistry Marks.

Create:
- Overall Performance Score.

Advantages:

1. Training is faster
2. Less storage required
3. Reduce overfitting
4. Better model performance
5. Easier data Visualization.

Disadvantages:

1. Some information may be lost
2. Reduce interpretability
3. Choosing the wrong technique can decrease accuracy.

Implementation of CNN like tensor flow

2024

A CNN is a deep learning model widely used for image classification, object detection, and computer vision tasks. TensorFlow provides the keras API to easily build, train and evaluate CNN models.

1. Import Required Libraries.

First import Tensorflow and keras libraries required for building the CNN model.

Python

```
import tensorFlow as tf
from tensorFlow.keras import
datasets, layers, models
```

Purpose - Provides tools for data handling model creation, training and evaluation

2 Load the Dataset

Load the training and testing data.

Purpose - The dataset contains input images and their corresponding labels.

3. Data Preprocessing

Normalize pixel values from 0-255 to 0-1

Purpose :- • Improves training speed
• Helps the model converge faster.
• Reduces numerical instability

4. Build the CNN Architecture.

Create the CNN model using convolution, pooling and dense layer

5. Compile the model

Specify optimizer. loss function, and evaluation metric.

Parameters.

• Optimizer - Updates weights efficiently
• Loss function - Measures prediction
• Accuracy - Evaluate classification performance

6. Train the Model

Train the CNN using the training dataset.

7. Evaluate the Model

~~Train~~ Test the trained model on unseen data.

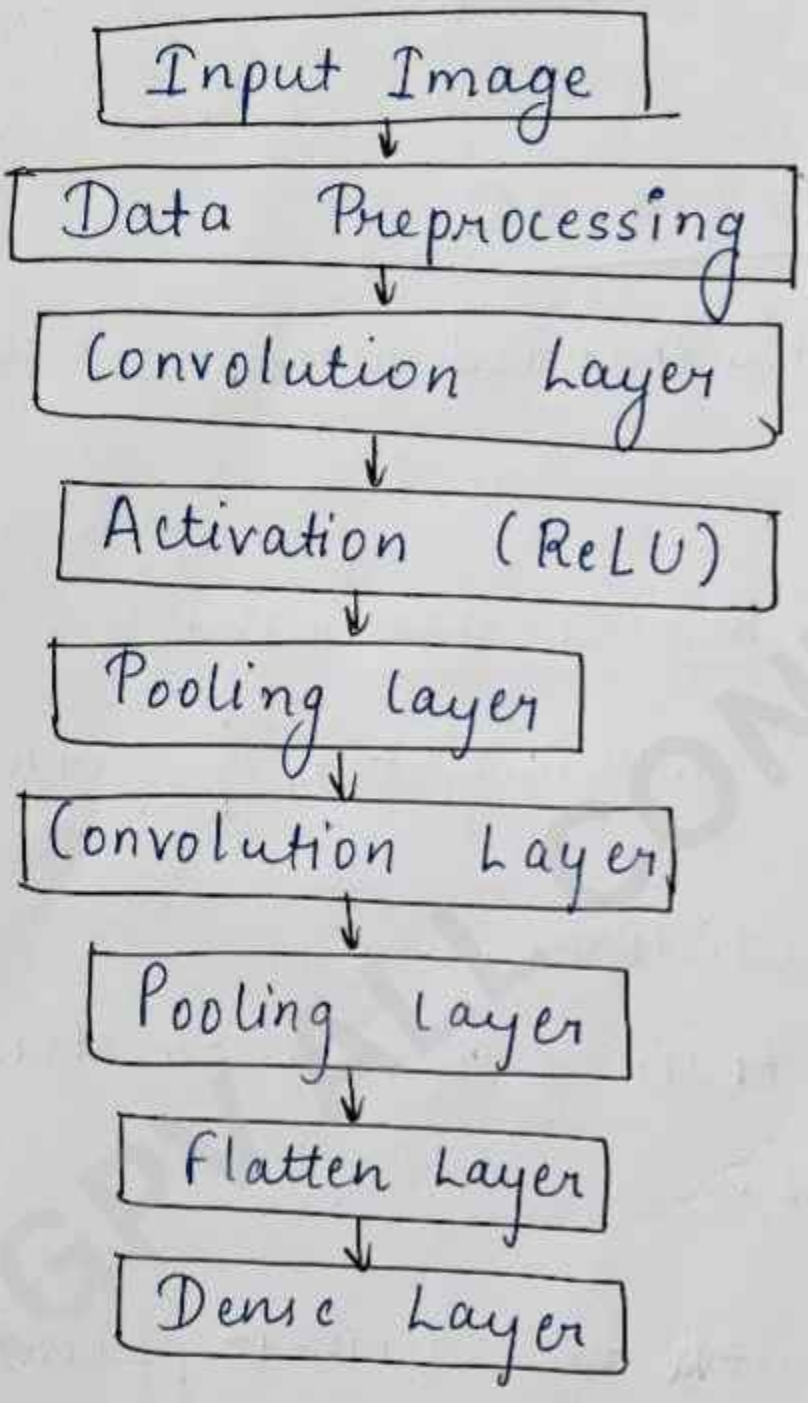
Purpose.

- Measures model accuracy
- Determines generalization performance.

8. Make Predictions.

- Used the trained model to classify new images.
- The class with the highest probability is selected as the final prediction.

Training Process Flow Diagram.



UNIT-4

Machine Learning -

Recurrent Neural Network 2024

A Recurrent Neural Network (RNN) is the type of artificial neural network specially designed to process sequential data.

Unlike traditional neural networks RNN can remember previous information and use it to make decisions about current inputs.

In simple words, RNN has memory which helps it understand the relationship between past and present data.

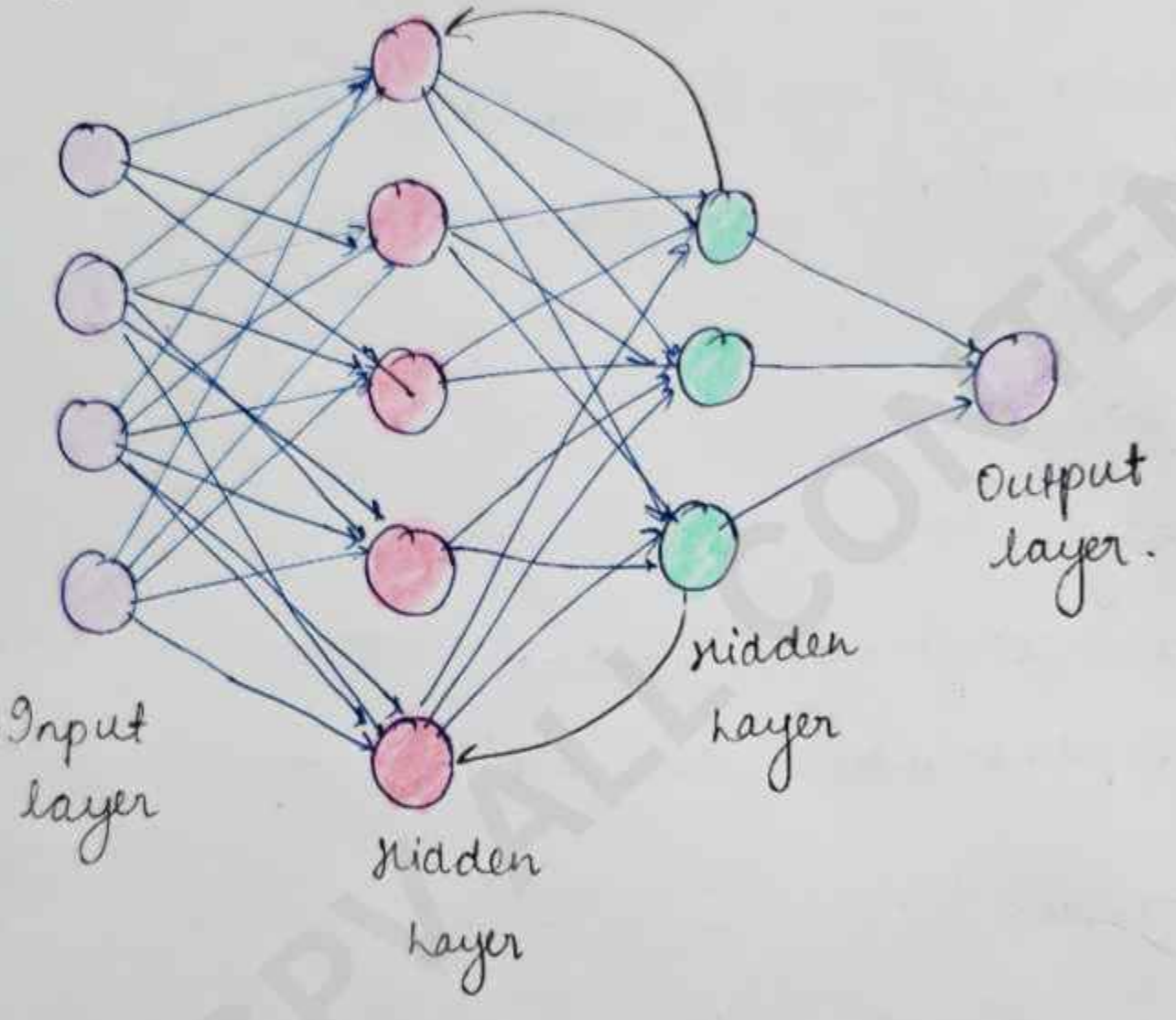
Need of RNN

Normal neural networks process each input independently and cannot remember previous information.

But many tasks require memory, such as.

- Language translation
- Speech Recognition.
- Text Predictions

RNN solves this problem by storing information from previous steps.



Recurrent Neural Network.

Example → Consider the sentence:-

"The movie was not good".

To understand the word "good". the model should remember the earlier word "not"

RNNs use previous content to make better predictions.

Advantages

- Handles sequential data
- Shares parameters across time steps

Limitations.

- Suffer from vanishing and exploding gradients during training.
- Difficult to learn long-term dependencies

To overcome RNN limitations, advanced architectures were developed:-

① LSTM (long-short Term Memory)
- Uses memory cells and gates
- Better at learning long-term dependencies

(2) GRU (Gated Recurrent Unit)

- Simpler than LSTM.
- Faster training with similar peryormance.

Long Short-Term Memory

2024, 2025

LSTM (Long Short-Term Memory) is a special type of Recurrent Neural Network (RNN) designed to remember information for a long period of time.

It was developed to solve the vanishing gradient problem of traditional RNNs.

Why LSTM is Needed?

In a simple RNN, information from earlier inputs may be forgotten after many time steps.

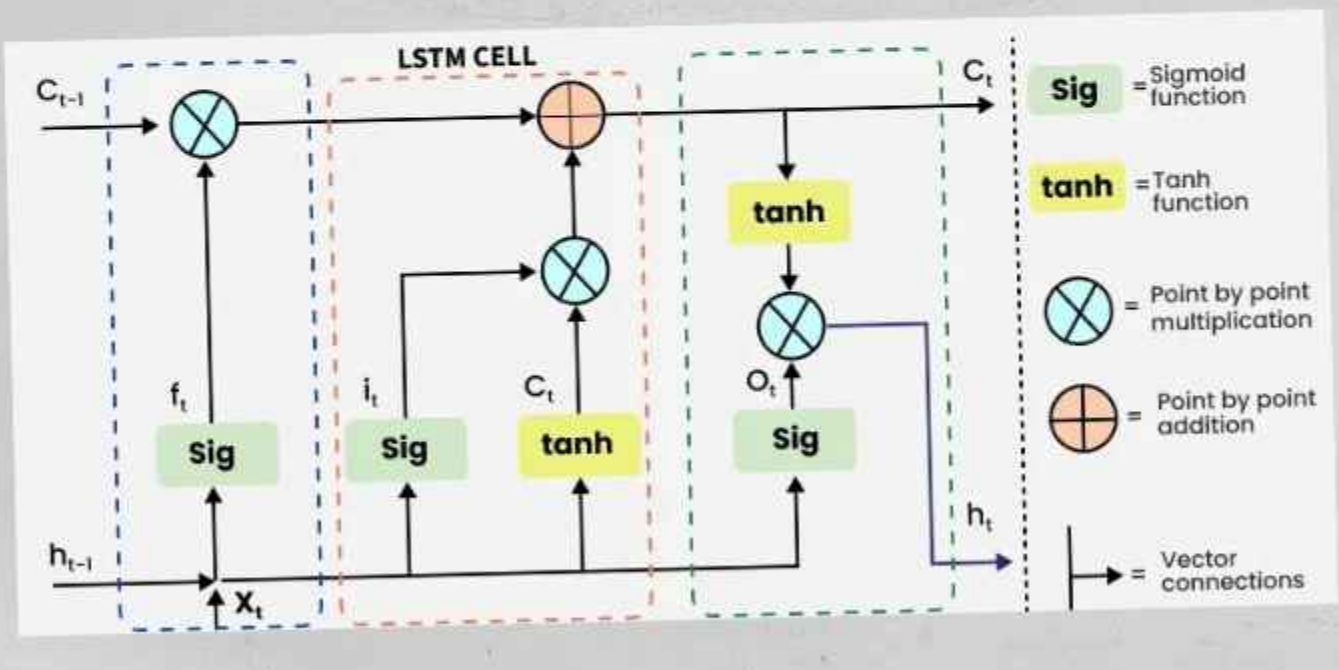
Example

→ "I grew up in France. I speak fluent French."

To understand why "French" is used, the model should remember "France" from earlier in the sentence.

RNNs often struggle with such long-term dependencies, while LSTMs can remember them.

Structure of LSTM.



An LSTM contains a cell state (memory) and three gates.

① Forget Gate - Decides what information should be removed from memory.

② Input Gate - Decides what information should be removed from memory.

③ Output Gate - Decides what information should be ~~stored~~ sent as output.

Working of LSTM.

Step 1 → Removes unnecessary information from the cell state

step 2 - Add useful new information to the cell state.

step 3 - Update the memory by combining old and new information.

step 4. Produce the final output and hidden state.

Advantages

- Remember information for a long time
- Solve vanishing gradient problem.
- Good for sequential data.

- Better performance than simple RNN.

Disadvantages.

- More complex than RNN
- Requires more computation and memory.
- Training can be slower.

Gated Recurrent Unit

A Gated Recurrent Unit (GRU) is a type of Recurrent neural network (RNN) architecture designed to handle sequential data and mitigate the vanishing gradient problem that affects traditional RNNs.

- GRUs use gates to control how information flows through the network, allowing them to remember long-term dependencies more effectively.

Components of a GRU

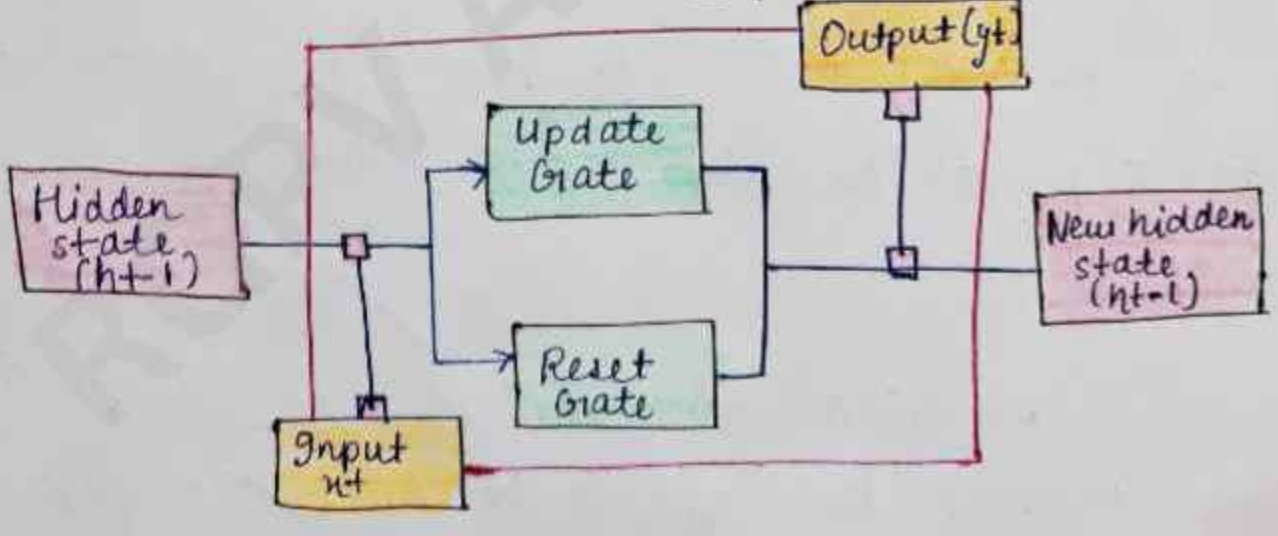
A GRU has two gates :-

1 Update Gate (z_t) • Determines how much of the previous hidden state should be retained.

• Acts similarly to the combination of the forget and input gates in an LSTM

2. Reset Gate (r_t) • Controls how much past information to forget when computing the new candidate state.

Structure



Advantages

- Simpler Architecture than LSTM.
- Fewer parameters, reducing computation and memory requirements.
- Effective for sequence modeling tasks such as:-
  - → Natural Language Processing (NLP)
  - → Speech Recognition
  - → Time-series forecasting
  - → Machine Translation

Attention Model

The Attention model is a mechanism that helps a neural network focus on the most important parts on the input while generating an output.

It was introduced to overcome the limitations of traditional Recurrent Neural Network (RNNs) and Encoder-Decoder models, which struggle with long sequence

Why Attention model is needed.

In a basic Encoder-Decoder model:-

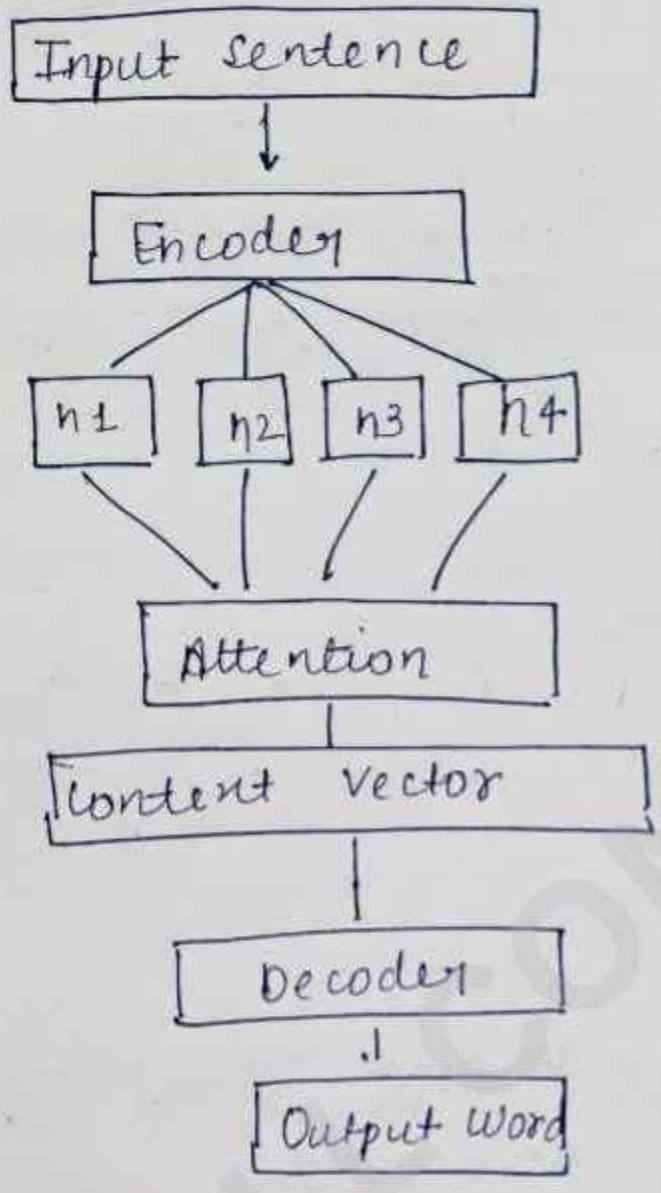
- The encoder converts the entire input sentence into a single fixed-size vector.
- The decoder uses this vector to generate the output.

→ For long sentences, important information may be lost because everything is compressed into one vector.

Example

Input → "The book that I bought yesterday is very interesting."

- When translating this sentence, the model should pay more attention to words like book, bought, and interesting at different decoding steps.
- Attention allows the model to look back at all encoder outputs instead of relying on only one vector.



Simple Diagram of Attention model.

where. h1, h2, h3, h4 are hidden states.

Advantages.

1 Handles long sentence better
2 Improves translation accuracy
3 Focuses on relevant words.
4 Reduces information loss.
5 Forms the foundation of the Transformer architecture.

Short Definition

→ Attention model is a mechanism that enables a neural network to dynamically focus on the most relevant parts of the input sequence while generating each output, improving performance on long-sequence tasks.

Markov Decision Process (MDP)

A Markov Decision Process (MDP) is a 2025 2024 mathematical framework used in Reinforce-ment learning (RL) to model decision-making problems where outcomes depends on the current state and action taken.

Markove Property:-

The future depends only on the current state, not on the sequence of past states.

$$P(S_{t+1} \mid S_t) = P(S_{t+1} \mid S_1, S_2 \ldots S_t)$$

$$P(S_{t+1} \mid S_t) = P(S_{t+1} \mid S_1, S_2 \ldots S_t)$$

This is called markov Property.

Components of MDP

An MDP is represented as :-

(S, A, P, R, γ)

1. State (S) :- Set of all possible state of the environment.

Example :- Position in a maze

2. Action (A) :- Set of action available to the agent.

Example :- Move Up, Down, left, Right

3. Transition Probability (P) :- Probability of moving from one state to another after taking an action.

$P(s', | s, a)$.

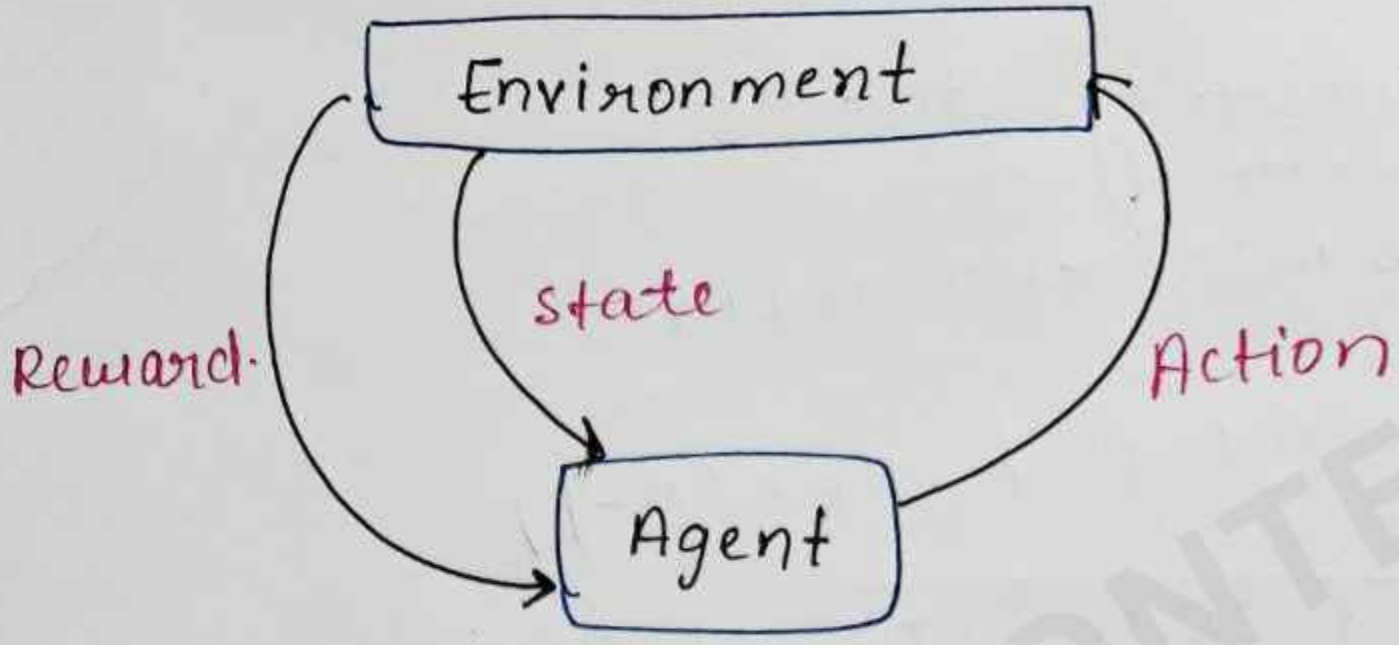
4. Reward Function (R) :- Reward received after performing an action

$R(s, a)$

5. Discount Factor (γ) :- Determines the Importance of future rewards

$0 \leq \gamma \leq 1$

- $\gamma = 0$ → only immediate reward matter
- γ close to 1 → future rewards are important.



Markov Decision Process

Importance of MDP

- Foundation of Reinforcement Learning.
- Models sequential decision-making
- Used in Robotics. game playing, recommendation systems. and self-driving cars!

Reinforcement Learning:-

Reinforcement learning is a type of machine learning in which an agent learns by interacting with an environment. The agent performs actions and receives rewards or punishment. and it learns to choose actions that maximize the total reward over time.

Basic component.

1] Agent → A learner or decision maker.

2] Environment → The place where the agent operates.

3] Action The choices made by the agent.

4] Reward Feedback given to the agent for its actions. (positive or negative)

5] State The current situation of the environment.

Working of Reinforcement Learning:

1. The agent observes the current state of the environment.
2. It choose an action
3. The environment responds with a reward or penalty.
4. The agents updates its strategy to maximize future rewards.

Example:

A robot learning to walk.

- If it moves correctly, it gets a positive reward
- If it falls, it get negative reward.

→ Over time, the robot learns the best way to walk.

Application

- Robotics
- Game playing (like chess or video games)
- Self - driving cars.
- Recommendation systems.

Advantages

- Learns from interaction with the environment.
- Can solve complex decision - making problems.
- Improves performance over time.

Disadvantages

- Requires a large amount of training time
- Designing the reward system can be difficult
- Sometimes it may learn inefficient behaviors

Bellman Equation.

The Bellman Equation is a fundamental equation in Reinforcement learning (RL) that expresses the value of a state as the immediate reward plus the discounted value of future ~~reward~~ states.

State Value function.

$$V(s) = R(s) + \gamma \sum_{s'} P(s'|s) V(s')$$

Where:

- $V(s)$:- Value of state s
- $R(s)$:- Immediate reward
- γ = Discount Factor (0-1)
- $P(s'|s)$ = Probability of moving to next state s'.

Meaning ⇒ Value of Current state = Immedi-ate Reward + Discount-ed future Reward.

The Bellman equation breaks a complex problem into smaller subproblems, making it easier to find the optimal policy.

Importance.

- Foundation of Reinforcement learning
- Used in Value Iteration and Policy Iteration.
- Helps an agent find the optimal strategy

Value Iteration & Policy Iteration

2024

Both are methods used to solve a Markov Decision Process (MDP) and find the optimal policy

1. Value Iteration

Value Iteration repeatedly updates the value of each state until the values converge

Steps

1. Initialize all state values.
2. Update values using the Bellman equation.
3. Repeat until values stop changing significantly
4. Derive the optimal policy from the final values

Advantages.

- Simple to implement
- Find the optimal policy

Disadvantages.

- May requires may iterations.

2. Policy Iteration

Policy Iteration directly improves a policy until it becomes optimal.

Two main Points

a) Policy Evaluation :- Calculate the value of states under the current policy.

b) Policy Improvement :- Choose better actions based on the computed values.

- Repeat these two steps until the policy no longer changes.

steps

1. Start with a random policy
2. Evaluate the policy
3. Improve the policy
4. Repeat until convergence.

Advantages

Usually converges in fewer iterations

Often faster than value iteration

Disadvantages.

- Each iteration can be computationally expensive

Difference

Value Iteration	Policy Iteration
1 Updates values directly	1 updates policy directly
2 One step (value update) repeated	2 Two steps :- evaluation + improvement
3 More Iteration	3 Fewer Iteration
4 Simpler	4 Slightly more complex

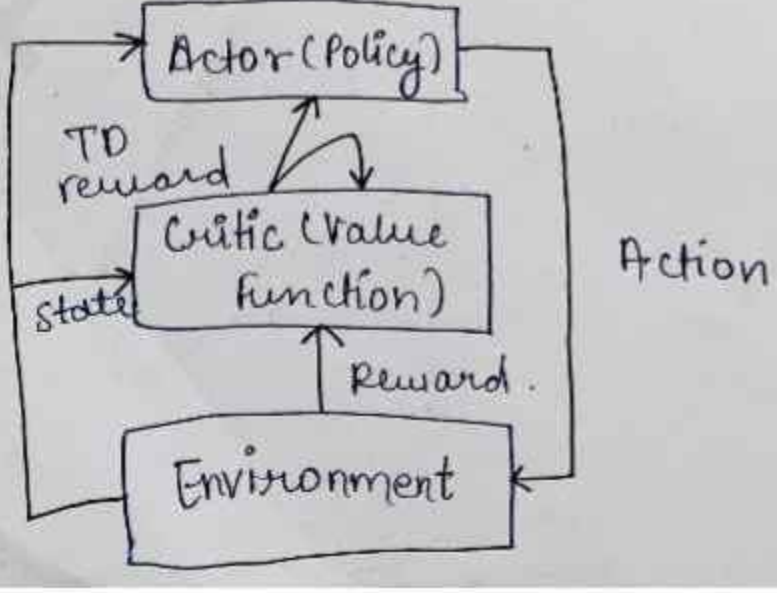
NOT FOR COMMERCIAL CONTENT

Actor - Critic Model 2024

A Actor - Critic Model is a Reinforcement Learning (RL) method that combines Policy - Based Learning and Value - Based learning

It has two component.

1. Actor :-
   - Decides which action to take
   - Learns the policy $\pi(a|s)$.
   - Interacts with the environment.
2. Critic :-
   - Evaluates the action chosen by the actor.
   - Estimates the value function $V(s)$.
   - Provides feedback to improve the actor.



Steps

1. Actor observes the current state
2. Actor selects an action
3. Environment returns reward and next state.
4. Critic evaluates how good the action was.
5. Critic sends feedback.
6. Actor updates its policy and improves future actions.

Example.

Imagine a robot learning to walk.

- Actor - Decides how to move its legs
- Critic - Checks whether the movement improved balance and speed.
- if good → positive feedback
- if bad → negative feedback

Over time, the robot learns to walk efficiently

Advantages

learns faster than many basic RL methods.

Handles large state spaces

combines strength of policy and value methods

suitable for continuous actions.

Application

- Robotics
- self driving cars
- Game Playing
- Resource allocation

Q - Learning

2025

Q- Learning is a model - free Reinforcement learning algorithm that learns the best action to take in each state by estimating Q- values (Quality Values).

A Q- value represents the expected future reward of taking an action in a parti - cular state.

Q - Value

Q (s, a).

where

- s = state
- a = Action
- Q(s,a) = Expected reward for taking action a in state s.

Q- Learning update Equation.

$$Q(s,a) = Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s,a)]$$

Where

- α (alpha) = Learning rate
- γ (gamma) = Discount factor
- r = Immediate reward
- s' = Next state
- $\max Q(s', a')$ = Maximum future Reward.

Working steps.

1. Observe current state s.
2. Choose an action a.
3. Receive reward(r) and next state (s').
4. Update the Q-value using the Q-learning equation.
5. Repeat until the optimal policy is learned.

Q-Table Example.

State	Left	Right
S_1	2	5
S_2	3	8

The agent chooses the action with the highest Q-Value.

Example.

In S1, choose Right because 5 > 2.

Advantages.

- Simple and easy to implement
- Does not require a model of the environment.
- Learns the optimal policy through experience.

Disadvantages.

- Large state spaces require large Q-tables.
- Learning can be slow for complex problems.

Application.

- Robotics
- Game Playing
- Path Finding
- Resource Management.

SARSA

(State - Action - Reward - State - Action)

SARSA is a model-free Reinforcement Learning algorithm that learns the action value function Q(s,a) based on the action actually taken by the agent

The name SARSA comes from:.

$$S \rightarrow A \rightarrow R \rightarrow S \rightarrow A$$

- State
- Action
- Reward
- Next state
- Next Action

working steps

1. Observe current state s.
2. Choose action a
3. Take action and receive reward r.
4. Observe next state s'
5. Choose next action a'
6. Update Q-value
7. Repeat

PERSONAL CONTENT

2024

Q Describe the significance of n-gram precision and brevity penalty in BLEU score calculation.

Ans → BLEU - (Bilingual Evaluation Understudy) is a metric used to evaluate the quality of machine-translated text by comparing it with one or more reference translations.

N-gram Precisions → N-gram precisions measures how many words or sequences of words (n-gram) in machine-generated translation match those in the reference translation.

- 1-gram (unigram) - Checks individual words
- 2-gram (Bigram) - Checks pair of consecutive words.

- 3-gram (Trigram) - Check sequences of three words
- 4-gram. Check sequences of four words.

Importance →

- Evaluates the accuracy of translated words and phrases.
- Higher-order n-grams capture fluency and correct word order.
- Prevents a system from getting a high score by simply using correct words in the wrong sequence.

Example: Reference :- The cat is on the mat

Candidate :- The cat is on mat.

- Many words match, so unigram precision is high, but some higher-order n-grams are missing.

Q **Brevity Penalty(BP)** :- A translation system could achieve high precision by producing very short translations containing only common matching words. To avoid this. BLEU uses a Brevity Penalty.

Formula

$$BP = \begin{cases} 1 & c > r \\ e^{(1-r/c)} & c \leq r \end{cases}$$

where c = length of candidate translation
r = length of reference translation

Importance

Penalize translations that are too short

Encourages complete translations

Balance precision with adequate translation length.

Significance in BLEU score

- N-gram precision measures translation accuracy and fluency.
- Brevity Penalty ensures the translation is not excessively short
- Together, they provide a balanced evalution of machine translation quality

NOT FOR COMMERCIAL CONTENT

MACHINE LEARNING

UNIT - 01

Introduction to Machine learning.

Machine learning is a branch of Artificial Intelligence (AI) that allows computers to learn from data and make decisions or predictions. without being explicitly programmed.

Instead of writing rules. we give data. and the computers learn pattern from it.

Machine learning is a subset of Artificial Intelligence.

It is useful when

- Data is large
- Rules are complex
- Patterns are difficult to write manually.

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Artificial Intelligence

Any technique that enables computers to mimic human intelligence, using logic, if-then rules, decisions trees and machine learning (including deep learning)

Machine learning

↑ Subset of AI that includes that include statistical techniques that enable machines to improve at tasks with experience. This include deep learning.

Deep learning

The subset of ML that allow software to train itself to perform tasks, like speech and image recognition, by exposing multilayered neural networks to large data.

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Important terms of Machine learning.

① Algorithm → A step-by-step method used by a computer to solve a problem or learn from data.

Example → Decision Tree, Linear Regression.

② Dataset → A collection of data used to train and test a machine learning model.

Example → Images, numbers

③ Training Data → The data used to ~~check~~ teach the model so it can learn patterns.

④ Test Data → Data used to check how well the model works after traning.

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Scope of Machine Learning.

The scope means the area where machine learning can be used.

① Healthcare → It helps doctors detect diseases, analyze medical images, and predict patient risks.

② Finance and Banking → Used for fraud detection, credit scoring, and stock market prediction.

③ E-commerce → Recommends products to users based on their interests (like online shopping recommendations).

④ Self-Driving Cars → It helps vechicles to recognize roads, objects and traffic signals.

⑤ Natural Language Processing → Used in chatbots, language translation, and voice assistants.

⑥ Education → Personalized learning systems and intelligent tutoring systems.

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Limitations of Machine Learning

Despite its advantages, Machine Learning also has some limitations.

① Needs Large Amount of Data → Machine learning models require a lot of data to learn properly.

② High Cost → Training machine learning models requires powerful computers and infrastructure.

③ → Time-consuming → Training and testing models can take a long time.

④. Data Quality Problem → If the training data is incorrect or biased, the model will give wrong results.

⑤ Security Risks → Machine learning systems can be attacked or manipulated with fake data.

⑥ Predictor variables → It is a feature(s) of the data that can be used to predict the output.

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Advantages / scope of ML.

① Automation of Tasks.

Machine learning can automatically perform tasks without human intervention, it reduces manual work and increases efficiency.

② Handles Large Data.

ML can analyze and process very large amounts of data quickly and accurately.

③ Improves with experience.

ML systems learn from data and improves their performance over time.

④ Accurate Predictions.

ML models can make accurate predictions and decisions based on patterns in data.

⑤ Wide Application.

Machine learning is used in many fields such as healthcare, finance, education, transportation, and e-commerce.

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Types of Machine Learning.

→ A machine can learn to solve a problem by any one of the following three approaches :-

These are the ways in which a machine can learn :-

- Supervised learning
- Unsupervised learning
- Reinforcement learning.

1 Supervised Learning :-

→ It is a type of machine learning in which the model is trained using labelled data.

→ This means the input data already has the correct output (target), and the algorithm learns the relationship between input and output.

→ The machine learns from examples with correct answers.

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Example → Suppose we want to predict wheather an email is spam or not spam.

Email features	Output
Contains many ads	spam
From known person	Not spam

The machine studies many such examples and learn to classify new emails.

How supervised Learning works.

1. Training data is given with input and correct output.
2. The algorithm learns the pattern between input and output.
3. The model is trained.
4. The trained model predicts the output for new data.

Types of Supervised Learning.

1 Classification → Output is a category or class.

- Example → Spam detection, disease detection.

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2→ Regression → Output is a continuous value.

Example → Predicting house price or temperature.

Application of supervised learning.

- Email spam detection
- Face Recognition
- Medical Diagnosis
- Stock price predictions.
- Image classification.

2. Unsupervised Learning.

Unsupervised learning is a type of machine learning in which the model is trained using data without labeled outputs.

The algorithm analyzes the data and finds hidden patterns or structures in the dataset automatically

Example → Suppose we have data of many customers but we do not know their categories. The algorithm studies the data and groups of customers with similar behavior together. This process is called clustering.

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Types of Unsupervised learning.

1 Clustering. - It groups similar data points together.

Example :- Customer segmentation

2. Association :. It defines and finds relationships between variables in the data.

Example → Market basket analysis (people who buy bread often buy butter).

Common Algorithm.

- K-means Clustering
- Hierarchical clustering
- DBSCAN
- Apriori Algorithm.

Application.

① Customer segmentation
② Recommendation systems.
③ Fraud detection.
④ Market analysis.

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Advantages.

1. No labeled data is required
2. It helps discover hidden patterns in data.
3. Useful for analyzing large datasets.

Disadvantages.

1. Results can be difficult to interrupt
2. Accuracy is harder to evaluate.
3. Sometimes it may finds irrelevant patterns.

③ Reinforcement Learning.

Reinforcement learning is a type of machine learning in which an agent learns by interacting with an environment. The agent performs actions and receives rewards or punishment. and it learns to choose actions that maximize the total reward over time.

Basic component.

1] Agent → A learner or decision maker.
2] Environment → The place where the agent operates.

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3] Action The choices made by the agent.

4] Reward Feedback given to the agent for its actions. (positive or negative)

5] State The current situation of the environment..

Working of Reinforcement Learning.

1. The agent observes the current state of the environment.
2. It choose an action
3. The environment responds with a reward or penalty.
4. The agents updates its strategy to maximize future rewards.

Example.

A robot learning to walk.

- If it moves correctly, it gets a positive reward
- If it falls, it get negative reward.

→ Over time, the robot learns the best way to walk.

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Application

- Robotics
- Game playing (like chess or video games)
- Self - driving cars.
- Recommendation systems.

Advantages

- Learns from interaction with the environment.
- Can solve complex decision - making problems.
- Improves performance over time.

Disadvantages

- Requires a large amount of training time
- Designing the reward system can be difficult.
- Sometimes it may learn inefficient behaviors

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Regression

REGRESSION

Regression is a method in machine learning that finds the relationship between independent variables (inputs) and a dependent variable (output) to predict a continuous value.

Regression is used when the output is a number not a category.

Example.

Some common examples :-

- Predicting house price based on size and location.
- Predicting temperature for tomorrow.
- Predicting salary based on experience.
- Predicting sales of a product.

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How Regression works.

1. The model takes training data (input + output)
2. It finds the mathematical relationship
3. bitween them
4. It creates a best-fit line or curve
5. This line is used to predict new values

Example formula (Linear Regression).

$$y = mx + c$$

where

y = predict output
m = slope
x = input variables
c = intercept

Types of Regression.

1. Linear Regression
2. Polynomial Regression
3. Support vector Regression
4. Decision tree regression
5. Random forest regression.

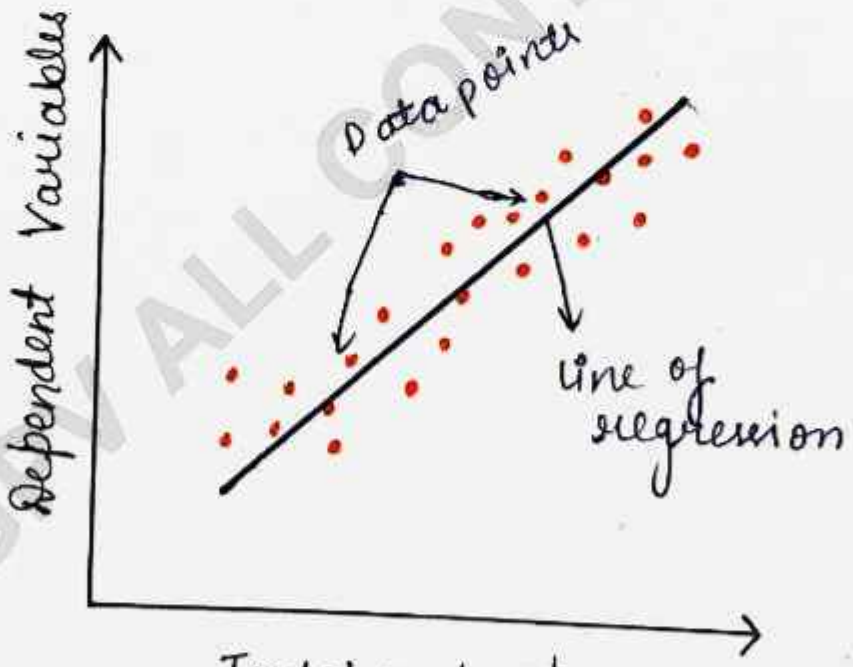
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1. LINEAR REGRESSION

Linear Regression is a regression algorithm used to model the linear relationship between independent variables (X) and a dependent variable Y

It predicts values using a straight line formula.

$$Y = mX + c$$



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Working

1. The algorithm takes training data.
2. It finds the best-fit straight line that minimize prediction error.
3. This line is used to predict new values.

Example.

Predicting house price based on area.

Area = 1000 sq ft

Price = ₹ 30 Lakh

Linear regression finds the relationship between area and price

POLYNOMIAL REGRESSION.

Polynomial Regression is a type of regression used when the relationship between (X) and output (Y) is not linear (not a straight line) but curved.

It is actually a extension of linear regression, but it models data using polynomial equation.

Formula → $Y = a + bx + cx^2 + dx^3 \ldots kx^n$

Y = predicted output

x = input variable

a, b, c, d = coefficients.

n = degree of polynomial.

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Example - (degree 2)

$$Y = a + bx + cx^2.$$

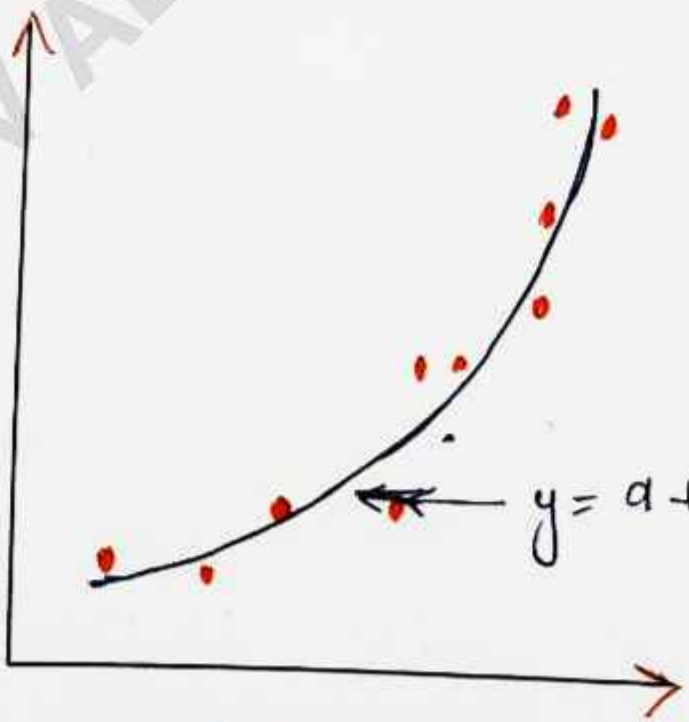
Use of Polynomial Regression.

Linear regression fails when data is curved.

Example :- Growth of plant over time
Population growth.
Temperature variation.

These follow a curve, not a straight line.

Graph →



Polynomial model.

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Working -

1. Take input data (X, Y)
2. Convert X into polynomial features
   $X \rightarrow X, X^2, X^3 \ldots$
3. Apply linear regression on these new features.
4. The model fits a curve instead of a straight line
5. Use the curve to predict new values.

Degree of Polynomial.

The degree (n) controls the shape of the curve

Degree 1 → Straight line (linear)

Degree 2 → U-shape curve (parabola)

Degree 3 → More complex curve.

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SUPPORT VECTOR REGRESSION.

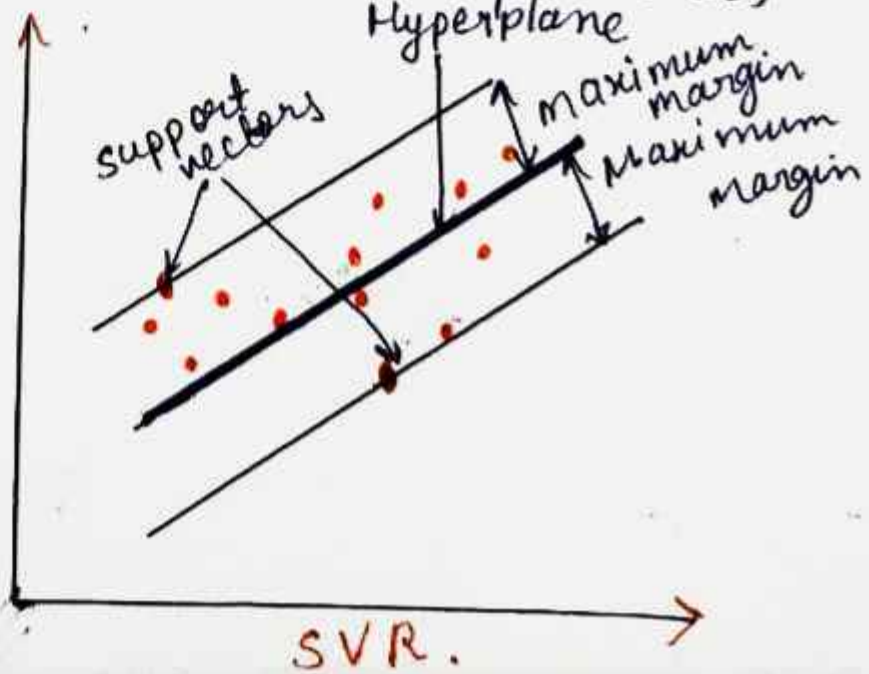
SVR is a regression algorithm based on support vector machines (SVM) that predicts continuous values by finding a function that fits the data within a fixed error margin (ε)

- Instead of minimizing total error, SVR tries to keep error within a limit.

SVR does not fit all data points exactly. It tries to:

- Fit a line (or curve)
- Keep most data points within a margin (ε)

→ This margin is called the epsilon (ε) tube.



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DECISION TREE REGRESSION.

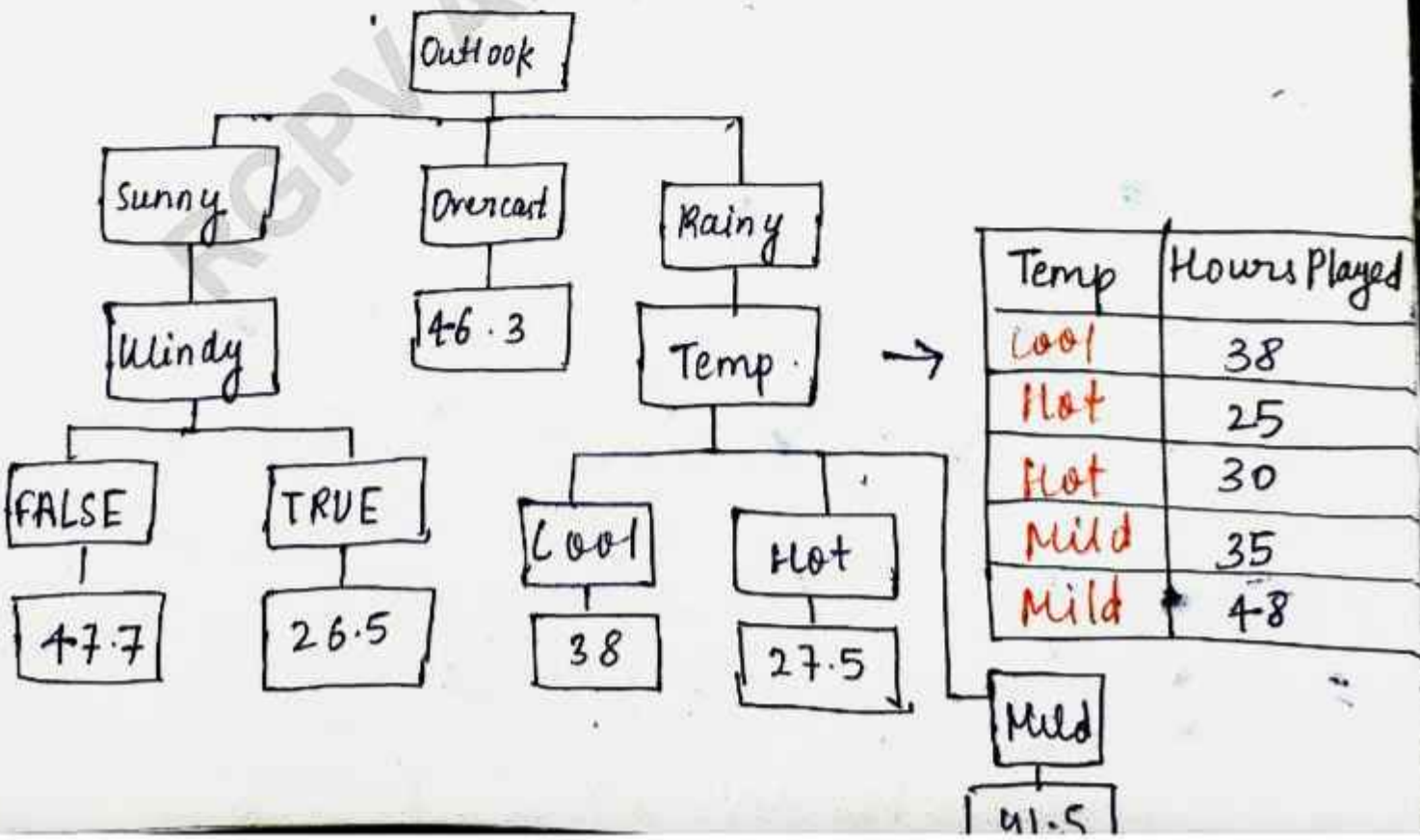
It is a machine learning algorithm that predicts continuous values by splitting the dataset into smaller parts using a tree like structure.

It creates decisions step-by-step and give a numerical output.

Instead of fitting a line (like linear regression)

- It Divides data into regions.
- Assign a constant value to each region.

So prediction is based on which region the input falls into.



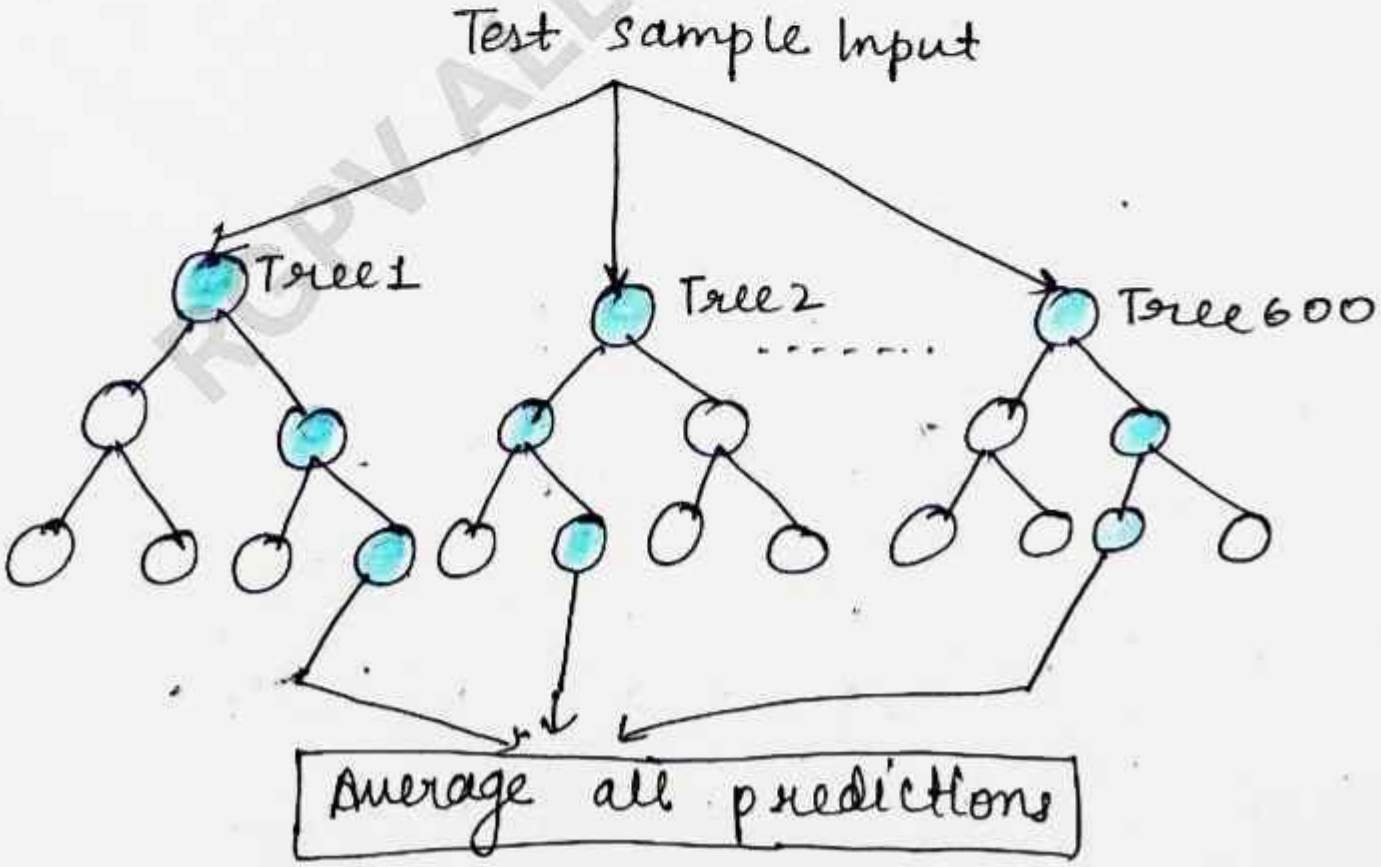
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RANDOM FOREST REGRESSION

Random forest Regression is an algorithm that predicts continuous values by combining the result of multiple decision trees.

- Instead of using one tree, it uses a forest (many trees).

- one decision tree ⇒ may overfit.
- Many trees together ⇒ better accuracy.
- final prediction ⇒ average of all tree predictions



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PROBABILITY

Probability is the mathematical foundation of Machine learning used to model uncertainty, randomness, and prediction confidence.

In real-world data, Data is noisy. Outcomes are uncertain and exact answer is not possible.

So, ML uses probability to say:
This prediction is 80% likely.

$$0 \leq P(E) \leq 1$$

0 → event will never happen
1 → event will definitely happen

Basic Probability Terminology.

① Random Experiment: An experiment whose outcome cannot be predicted exactly.

Example ① Tossing a coin
② Rolling a dice
③ Selecting a customer randomly.

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② Sample Space (S) :- The set of possible outcomes of an experiment

Example → Rolling a dice.

$S = \{1, 2, 3, 4, 5, 6\}$

③ Event (E) - A subset of a sample space.

Example → Even (A) → getting an even number.

$A = \{2, 4, 6\}$.

Common type of Probability.

① Classical Probability → It is used when all outcomes are equally likely.

$$P(E) = \frac{\text{favorable outcomes}}{\text{Total outcomes}}$$

Example → Probability of getting Head in a coin toss

$$P(H) = \frac{1}{2}$$

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② Empirical (experimental) Probability.
Based on past data and observation.

$$P(E) = \frac{\text{No. of time event occurs}}{\text{Total Trials}}$$

Example→ Outcome of 100 customers, 60 purchased.

$$P(\text{Purchase}) = \frac{60}{100} = 0.6$$

③ Subjective Probability.
Based on personal experience or judgement.

Example→ Probability of stock market rising tomorrow.

PROBABILITY DISTRIBUTION.

It describes how the values of a random variable are distributed along with their probabilities.

A distribution tells us which values occur, how often, and with what probability

Importance→
It help us to → ① Understand data behavior
② Identify patterns & trends
③ Make prediction and soon

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STATISTICS

Statistics is the foundation of Machine Learning because it helps us understand data, make predictions, and evaluate models.

Statistics is the science of collecting, analyzing, Interpreting, and presenting data.

In Machine Learning, statistics is used to

- Understand patterns in data
- Make Predictions
- Handle Uncertainty
- Evaluate model performance.

→ In simple words :-

ML = Data + Statistics + Algorithms.

Types of Statistics.

① Descriptive ~~data~~ Statistics.

② Inferential Statistics.

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(1) Descriptive Statistics.

Descriptive Statistics is the branch of statistics used to Organize data, Summarize data and describe the main features of data

- It does not make predictions.
- It only tells us what the data look likes.

Importance.

Before building any ML model. we must.

- Understand the dataset
- Detect patterns
- Find errors or outliners.

(2) Inferential Statistics:-

It is the branch of statistics that uses sample data to draw conclusions, make predictions, or take decisions about a population.

sample → subset of population, small & manageable, denoted by n.

main component → Estimation, Hypothesis Testing, Regression analysis, Analysis of Variance

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LINEAR ALGEBRA.

Linear Algebra is a branch of mathematics that deal with vectors, matrices and linear equations.

It is very important in fields like machine learning, data science, computer graphics and engineering.

☆ Linear algebra studies :-

Vectors → quantities with direction and magnitude.

Matrices → rectangular arrays of numbers

Linear equations → equation forming straight lines.

Example of linear equation:

$$2x + 3y = 5.$$

Example of linear Algebra in Machine Learning.

1. Dataset and Data Files.
2. Images and Photographs.
3. Linear Regression.
4. Principle component Analysis.
5. Deep learning.
6. Recommender system.

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CONVEX OPTIMIZATION.

Convex Optimization is a very important topic in machine learning and mathema-tics, especially for solving problem where we want the best (optimal) solution.

Convex optimization is a type of optimization problem where :-

① The objective function is convex

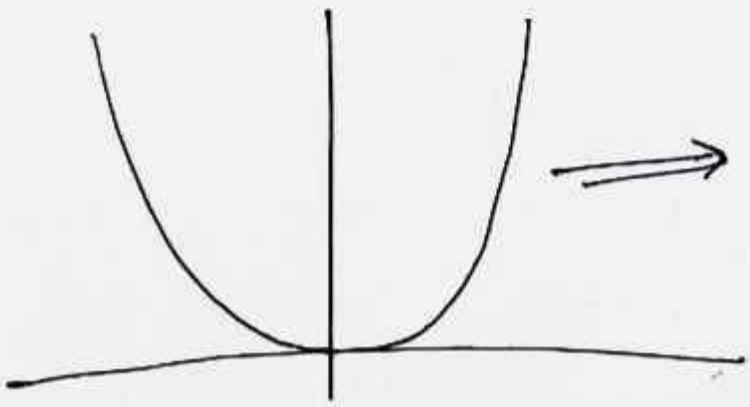
② The feasible region is also convex.

convex function → A function is convex if the line between any two points line above or on the curve.

Mathematical condition

$$f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta) f(x_2)$$

Example = $f(x) = x^2$



$\Longrightarrow y = x^2$

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DATA VISUALIZATION.

Data Visualization is the process of transfor-ming raw data into graphical representati-ons like charts, graphs, and dashboards to make information.

Easy to understand, Insightful and Actionable.

It is a key part of EDA (Exploratory Data Analysis) in Machine Learning.

Need of Data Visualization.

Data visualization is used when:

① Easy Understanding → Complex data becomes simple.

② Identify patterns → Trends and patterns are easy to see.

③ Better decision making → Helps in analyzing information quickly.

④ Saves times Visual data is faster to read than tables.

⑤ Detect errors - Outliners or unusual values become visible.

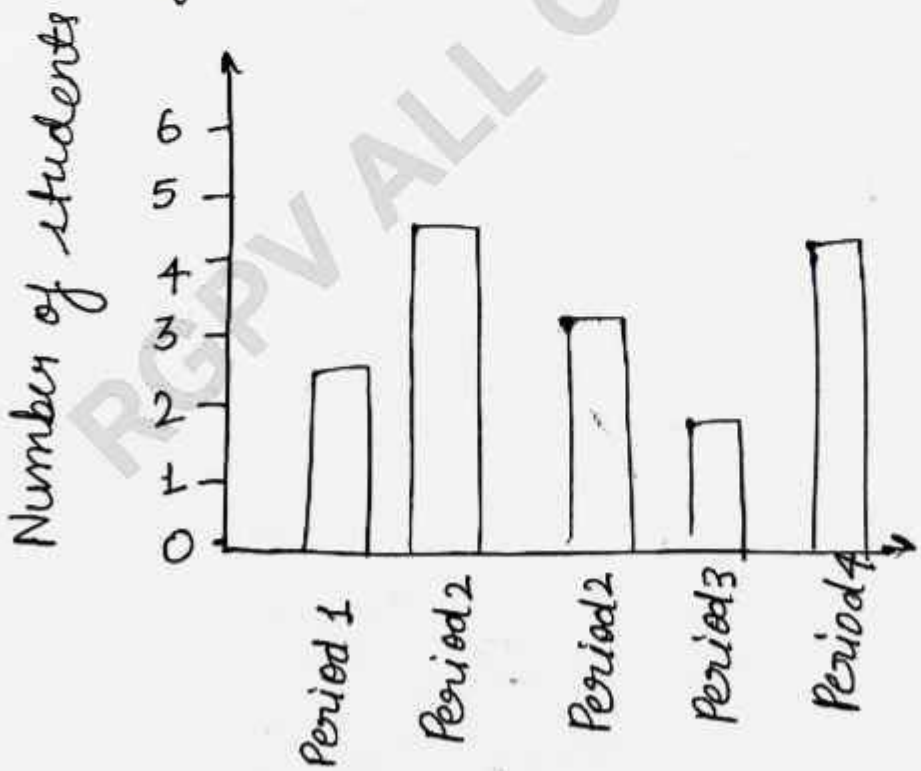
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Common types of Data Visualization

1. Bar Chart
2. Line Chart
3. Pie Chart
4. Histogram
5. Scatter plot.

BAR CHART

The Bar Chart is a graphical representation of data using rectangular bars where the length or height of each bar represents the value of the data.



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A bar chart has two axes :-

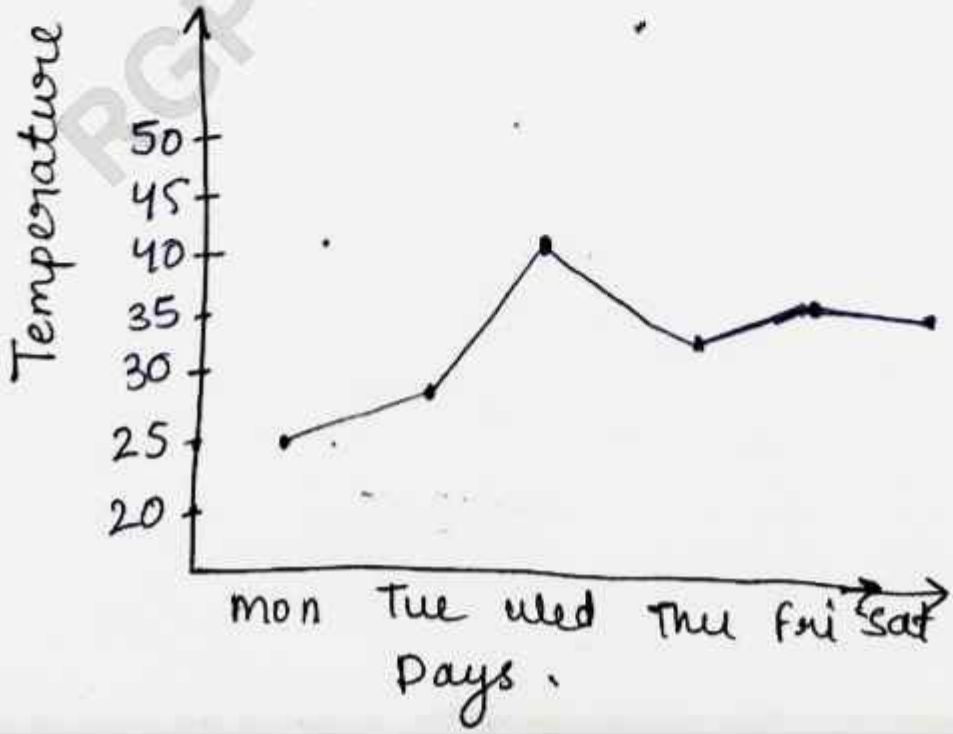
- X-axis (Horizontal axis) → shows categories (subject, products, years, etc).
- Y-axis (Vertical axis) → shows values or numbers

LINE CHART

A line chart is a type of graph that shows changes in data over time by connecting data points with a continuous line.

A line chart represents data using points connected by straight lines to show trends or changes.

Example → Temperature change during a week.



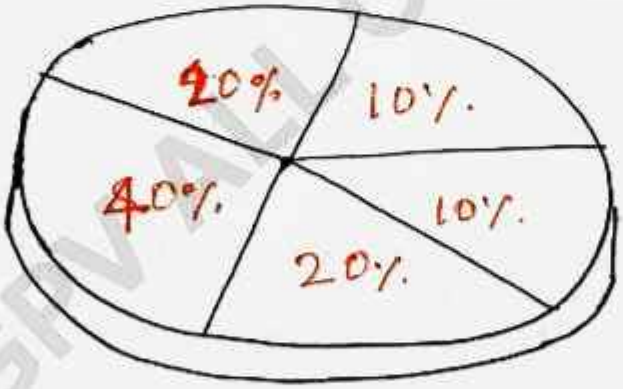
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PIE CHART

A Pie chart is a circular graph used to represent data as parts of a whole. The circle is divided into different slices. and each slice shows the percentage or proportion of a category.

A pie chart shows how a whole quantity is divided into different parts.

Example. Distribution of students ~~on~~ in different streams.



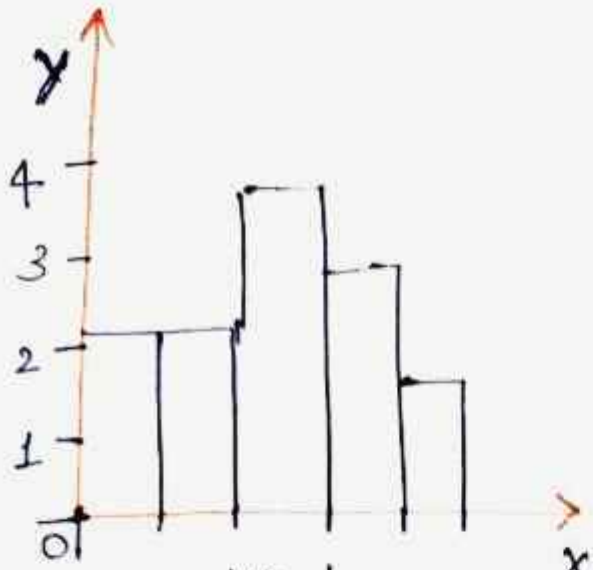
Percentage category.
40% CSE
10% IT
10% Civil
20% AI
20% ME.

HISTOGRAM

A histogram is a type of graph used to represent the frequency distribution of continuous numerical data using rectangular bars.

→ A histogram shows how often data values occur within certain ranges (intervals)

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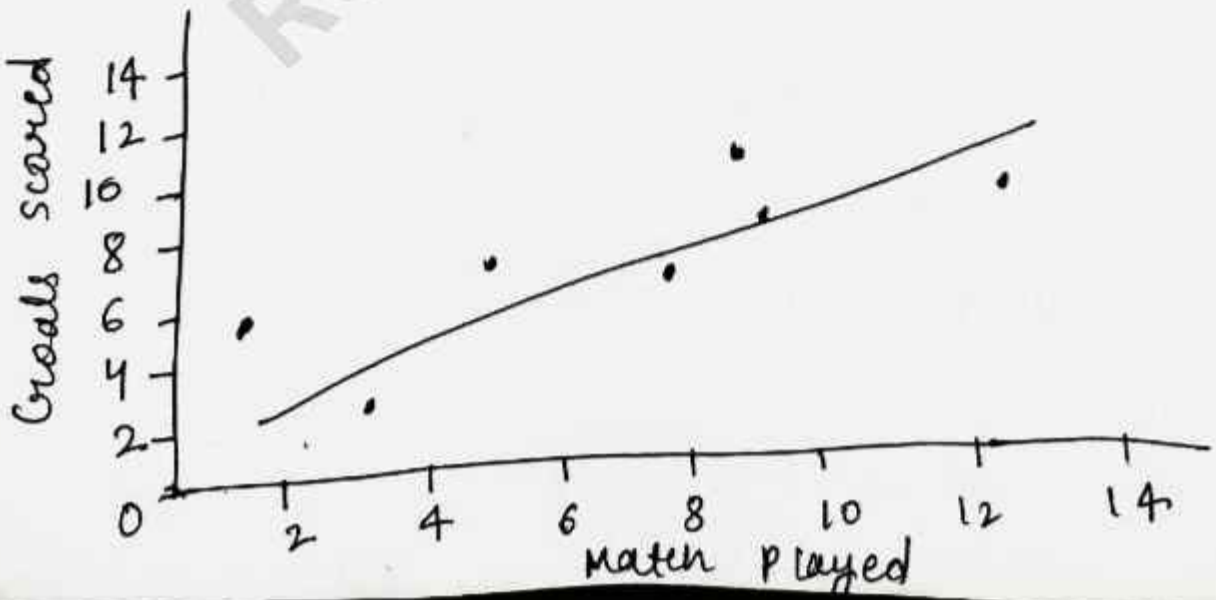


Histogram

SCATTER PLOT.

A scatter plot is a type of graph used to show the relationship between two variables by plotting data points on a coordinate plane.

A scatter plot displays data as points to show how two variables are related.



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HYPOTHESIS FUNCTION.

A Hypothesis function is a mathematical function used by a machine learning model to predict the output value from input data

Hypothesis function = a formula used by the model to make predictions.

It represents the relationship between input (features) and output (target values)

Example→ Predicting house price based on size:

Input → Size of house

Output → Price of house.

The model uses a hypothesis function to estimate the price.

Example of hypothesis function→

Linear Regression.

HYPOTHESIS TESTING

Hypothesis Testing is a statistical method used to decide whether a statement about data is true or false based on sample data

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Hypothesis testing checks if a claim about data is correct.

Types of Hypothesis

① **Null Hypothesis** → The default assumption that there is no effect or no difference

Example → H_0:- The new ML model does not improve accuracy.

② **Alternative Hypothesis** (H_1 or H_a)

The opposite statement that we want to prove.

Example: The new ML model does not improve accuracy.

Example in Machine Learning.

Suppose we build a new ML algorithm and want to test if it performs better.

- H_0 → New model accuracy = Old model accuracy.
- H_1 → New model accuracy > Old model accuracy.

After testing → If result is statistically significant → New model is better.

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DATA DISTRIBUTION

Data Distribution describes how data values are spread or arranged across different ranges or categories.

Data distribution shows how frequently different values appear in a dataset.

Example → Marks of students in a class may be distributed like :-

Marks Range	Number of students
0 - 20	2
21 - 40	5
41 - 60	10
61 - 80	8
81 - 100	3

This table shows how the marks are distributed among students.

Importance of Data Distribution.

Data distribution is important because it helps to :-

RGPV ALL CONTENT

- Understand the pattern ~~the~~ of data.
- Identify trends and variations.
- Detect outliners (unusual values)
- Choose the correct machine learning model.
- Perform statistical analysis

DATA PREPROCESSING

Data Preprocessing is the process of cleaning, transforming and organizing raw data before using it in a machine learning algorithms.

It means preparing raw data so that it can be used effectively for machine learning.

Example → If a dataset has missing values, duplicate records, or incorrect formats, preprocessing is used to fix these issues.

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Need for Data Preprocessing.

Data Preprocessing is important because real-world data may have.

1. Missing values
2. Noisy data (errors)
3. Duplicate data.
4. Duplicate Data formats.
5. Irrelevant information.

Without preprocessing, machine learning models may produce incorrect predictions.

DATA AUGMENTATION

Data augmentation is a technique used in machine learning and deep learning to increase the size of a dataset by creating new training data from existing data.

It is mainly used when available dataset is small.

It means generating additional data from existing data to improve model training.

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Example → If we have 100 images of cats, we can create more images by rotating, flipping and cropping.

NORMALIZATION OF DATA SETS.

Normalization is a data preprocessing technique used to rescale numerical features so they fall within a common range.

It converts different-scale data into a uniform scale without changing relationships between values.

The scale of normalization usually between 0 and 1.

Importance of Normalization –

In datasets, features can have very different ranges.

- Age → 18 to 60
- Salary → 10,000 to 1,00,000.

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- Machine learning models may give more importance to larger values (like salary).
- Normalization fixes this by bringing everything to the same scale.

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UNIT-2

MACHINE LEARNING

1. Linearity Vs non Linearity
2. Activation functions like sigmoid, ReLU.
3. Gradient Descent
4. Multilayer Network
5. Back Propagation.
6. Auto encoders
7. Batch Normalization.
8. L1 and L2 Regularization.
9. Tuning Hyper parameters.

Important Topics

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UNIT-2

Machine Learning.

Linearity Vs non linearity

Linearity A model is linear when the relationship between input and output is a straight line (or a linear combination of inputs)

Mathematical form -

$$y = w_1 x_1 + w_2 x_2 + \ldots + b$$

Main Points.

Output changes proportionally with input

It is easy to compute and interpret.

Decision boundary = straight line (2D) or plane (3D).

Example →

- Predicting salary based on experience.
- linear regression.

RGPV ALL CONTENT

Non-Linearity

A model is non-linear when the relationship is curved or complex, not a straight line.

Example → $y = x^2$
or
$y = \sin(x)$

Main Points.

- It captures complex patterns
- Decision boundary - curve or irregular shape.
- Needed for real-world problems like images, speech.

Importance of Non-Linearity.

If we only use linear models:-

Even multiples layers behaves like a single linear model

suppose → $y = w_2(w_1 x)$

This is still linear!

So, without non-linearity, deep learning is useless.

RGPV ALL CONTENT

Activation function.

Neuron → A neuron is the smallest unit of a neural network that.

Take input.
Processes it
Produces an output

It is inspired by human brain neuron

Activation function - An activation function is a mathematical function applied to neuron's output :-

$$y = f(Wx + b)$$

Y = Activation (Σ (weight * input) + bias).

Activation function decides, whether neuron should activated or not by calculating weighted sum and adding bias with it to introduce non-linearity in the output of neuron.

It used to determine the output of neural network like yes or no, It maps the resulting values in between 0 to 1 or -1 to 1 etc. (depending upon the function).

RGPV ALL CONTENT

Activation function are an extremely important feature of the artificial neural networks.

Without Activation Function

- Neural network becomes linear.
- Cannot solve complex problems.

With Activation function.

Introduces non-linearity.

And Enables:

- Image recognition
- Speech processing
- NLP tasks.

Important Types of Activation function

1. Sigmoid Activation function
2. ReLU (Rectified Linear Unit)

RGPV NOTES CONTENT

① Sigmoid.

The sigmoid activation function is a mathematical function that converts any real value into a value between 0 and 1.

It is also called the logistic function

It is easy to work with and has all the nice properties of activation functions, it's non-linear, continuously differentiable, monotonic, and has a fixed output range.

Mathematical formula.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

where

- x = input to the neuron
- e = Euler's constant

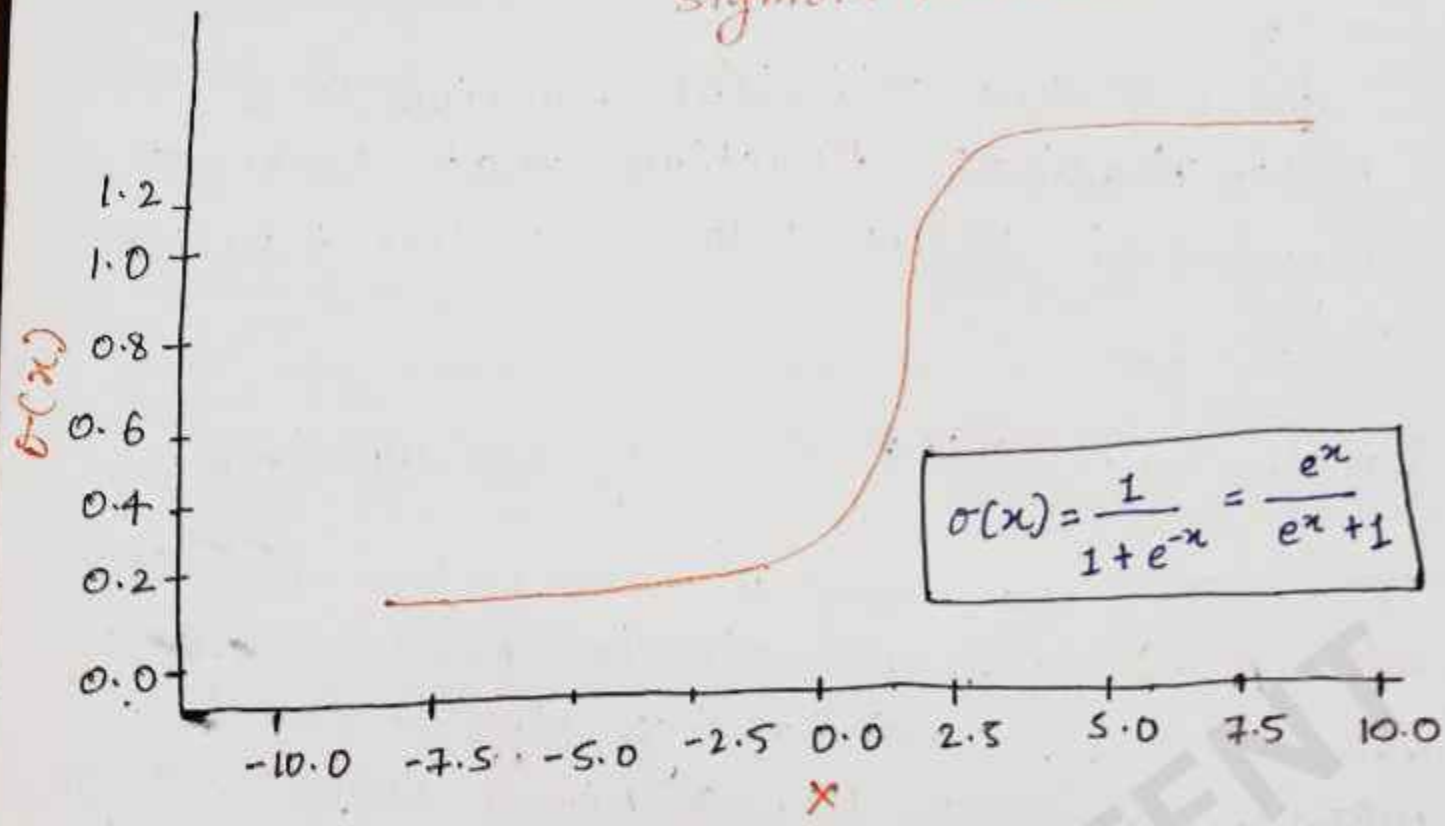
Output range →

$$0 < \sigma(x) < 1$$

- Output is always between 0 and 1
- Never exactly 0 or 1

RGPV ALL CONTENT

Sigmoid function



$$\sigma(x) = \frac{1}{1+e^{-x}} = \frac{e^x}{e^x+1}$$

Derivative

$$\sigma'(x) = \sigma(x)(1-\sigma(x))$$

Easy to compute
Used in backpropagation.

Advantages / Pros

1→ Produces values between 0 & 1, so it can be interpreted as probability. and useful in binary classification.

2- The function is continuous and smooth.

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3. Good in output layer especially in Yes / no problems and True / False predictions.

4. It is Monotonic Function means always increasing.
   - Higher Input → higher output.
   - no sudden jumps

Disadvantages / Cons.

(1) Vanishing Gradient Problem occurs means for large positive or negative values.
Gradient becomes very small (≃0).

(2) A extreme values:
   - Output becomes almost constant (0 or 1)
   - because of this model stops learning.

RGPV ALL CONTENT

(2) ReLU

- ReLU (Rectified Linear Unit) is a non-linear activation function
- It provides the same benefits as sigmoid but with better performance.
- It transforms input as:

$$f(x) = \max(0, x)$$

- If input is negative → output is 0.
- If input is positive output is same value

Importance.

→ Without activation

- Neural network becomes linear.

→ With ReLU.

- It introduces non-linearity.
- Helps model learn complex patterns.

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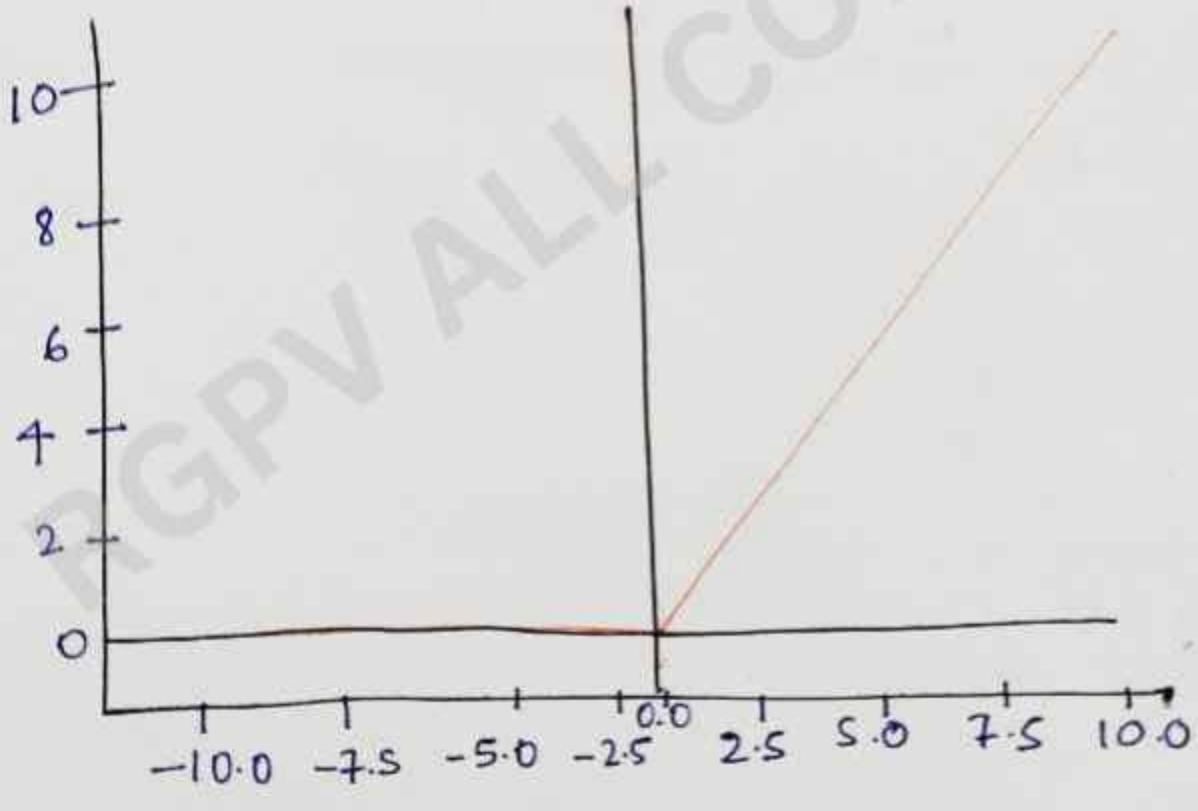
$$f(x) = \max(0, x).$$

where

- x is the input to the neuron.
- The function returns x if x is greater than 0.
- If x is less than or equal to 0, the function return 0.

The formula can also written as:

$$f(x) = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$



ReLU Activation function

RGPV ALL CONTENT

Parametric ReLU (PReLU)

→ Learns slope automatically for negative values.

ELU (Exponential Linear Unit.

→ Smooth version of ReLU.

where is ReLU used

→ Mostly in

- Hidden layers of neural networks
- Deep learning models.

RGPV ALL CONTENT

What are Weights and Bias in Neural Networks.

A neuron (like a small calculator) works like this :-

$$\text{Output} = (w \cdot x) + b$$

where.

x : input
w = weight
b = bias.

Weight

Weight is a numerical parameter in a neural network that determines the strength and importance of an input feature in producing the output.

weight tells - How much influence does the input have on the final result?

Representation

for single input neuron :-

$$y = w \cdot x$$

+ for multiple inputs.

$$y = w_1x_1 + w_2x_2 + w_3x_3 \ldots\ldots w_nx_n$$

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Where.

x_1, x_2, x_3 = inputs

w_1, w_2, w_3 = corresponding weights.

Each input has its own ~~input~~ weight.

Role of Weight

Weights play a very important role in neural network.

Feature Importance

High weight - input is very important

Low weight - input is less important.

Decision Making

- Output of neuron depends mainly on weights
- They help the model decide correct predictions.

Pattern learning

- Neural network learns pattern by adjusting weights.

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Real life example

→ Suppose we are predicting student performance.

Inputs:

Study hours = x_1

Sleep = x_2

Practice = x_3

Weights

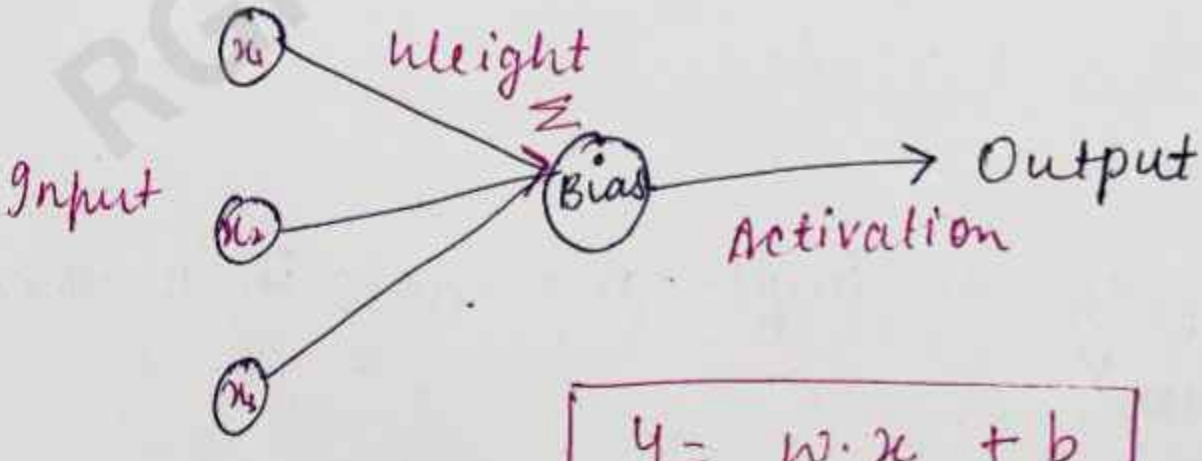
$w_1 = 3$

$w_2 = 1$

$w_3 = 2$

$y = (3 \times x_1) + (1 \times x_2) + (2 \times x_3)$

Study has heighest importance (weight = 3)



$$y = w \cdot x + b$$

RGPV ALL CONTENT

Bias

A bias is an additional parameter in a neural network that is added to the weighted sum of inputs to shift the output.

In simple words → Bias is a value that helps the model adjust the output independently of input.

Mathematical Representation -

For a neuron.

$$y = w \cdot x + b$$

for multiple inputs.

$$y = w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n + b$$

where b = bias.

Role of Bias

Bias play an important role in neural networks.

① shift the output

Bias move the output up and down.

REPRODUCTION ALL CONTENT

② Increases Flexibility. Helps model to fit data better.

③ Work without Input
Even if all inputs are zero, output can still be non-zero.

Real life example.

Suppose predicting salary:.

Inputs:.

Experience = x

weight = 5

Bias = 10 (base salary

$$y = 5x + 10$$

→ Even if experience is 0. salary = 10.

→ Bias = fixed starting value.

Bias in neural network.

- Every neuron has its own bias
- Bias is added after multiplying weights.

RGPV ALL CONTENT

Flow

① Multiply inputs with weights
② Add Bias
③ Apply activation function.

* Activation function → Applied to the output of a neuron ~~should be~~ ~~activated~~ to determined the final output.

It decides whether a neuron should be activated (ON) or not.

Loss FUNCTION

A function that measures the difference between actual output and predicted output.

Loss function tells how wrong the model's prediction is.

Why loss function is important.

Loss function is very important beause

① Measure Error - Tells how far prediction

RGPV ALL CONTENT

is from correct answer.

② Guides learning → Helps model improve by reducing error.

③ Used in optimization

- Works with Gradient Descent to update weights.

It tells the model.

- If prediction is close to actual value → Low Loss.

If prediction is far from actual value → High Loss

- The main goal of training is to minimize the loss.

Simple Example → Suppose actual house prince = ₹ 50 lakh

Model predicted = ₹ 45 lakh.

Then error → difference between actual and predicted.

Loss function calculate this error mathematically.

Common Loss functions.

1. Mean Squared Error (MSE).

This is used in Regression.

PGP VAIL CONTENT

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

- squares the error
- Large errors get more penality

2. Mean Absolute Error (MAE)

used in Regression

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \tilde{y}_i|$$

- Uses absolute difference
- Less affected by outliners.

3. Binary Cross Entropy.

Used in Binary classification (Yes/No, Spam/Not spam).

$$L = -(y\log(p) + (1-y)\log(1-p))$$

Difference Between Lose and cost

- Loss = error fore one training example
- Cost = average lose for all examples

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GRADIENT DESCENT

Gradient Descent is an optimization algorithm used in Machine learning (ML) to find the best values of parameters (weights and biases) that minimize the error of a model.

- It helps the model learn from data step by step.

Simple Idea.

→ Imagine you are standing on a mountain and want to reach the lowest point.

you cannot see the whole mountain. so you:

1. Check the slop around you.
2. Move a little downward.
3. Repeat until you reach the bottom.

This is how Gradient Descent works in ML.

Main Purpose / Role in ML

Gradient Descent is used to minimize the cost function (or Loss function).

- High cost → model predictions are bad.
- Low cost → model predictions are good.

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formula

$$\theta = \theta - \alpha \frac{\partial J(\theta)}{\partial \theta}$$

where:

- θ = parameter (weight)
- α = learning rate
- $J(\theta)$ = cost function.

★ we subtract because we want to move toward the minimum error.

Steps of Gradient Descent.

1. Initialize Random weights
2. Calculate prediction
3. Find error using cost function.
4. Compute Gradient (slop)
5. Update weights.
6. Repeat until error become minimum.

PGV ALL CONTENT

Types of Gradient Descent.

1. Batch Gradient Descent.
2. Stochastic Gradient Descent.
3. Mini-Batch Gradient Descent.

(1) Batch Gradient Descent

In this complete training dataset is used to calculate the gradient of the cost function before updating the model parameters.

In this method. the algorithm processes all training examples together as a single batch and then performs one update of weights in each integration.

Working process.

- Take the entire training dataset
- Calculate predicted output for all examples.
- Compute total error (cost)
- Calculate gradient using all data.
- Update weight once
- Repeat until convergence.

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Characteristics

- Use full data set in every integration
- Produces smooth and stable convergence.
- Computationally expensive for large datasets.
- Require high memory.

Advantages

1. Accurate, gradient calculation.
2. Stable movement toward minimum
3. Easy mathematical analysis.

Disadvantages.

1. Very slow for large datasets.
2. High computations time
3. Not-suitable for real-time learning

PGRV ALL CONTENT

Example :- If a dataset contains 50,000 records. Batch Gradient Descent uses all 50,000 records before updating weights once.

2. Stochastic Gradient Descent (SGD)

:- It is a variation of gradient Descent where only one randomly selected training example is used to calculate the gradient and update the model parameters at a time.

- The word "stochastic" means random.
- Instead of waiting to process the entire dataset. SGD updates weights immediately after processing each training example.
- This makes learning faster but less stable.

Working Process.

1. Select one training examples.
2. Calculate predictions and error
3. Compute gradient
4. Update weights immediately.
5. Repeat for all examples.

RDPV ALL CONTENT

Characteristics

- Uses one data point at a time
- Fast updates
- Random fluctuations during learning.
- Less memory usage

Advantages

1. Faster training
2. Work well for huge datasets.
3. Can escape local minima
4. Suitable for online learning

Disadvantages.

1. Noisy updates
2. Convergence is less stable
3. Error graph fluctuate

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Example → If dataset has 50,000 records :-

- Weights update 50,000 times in one epoch.
- Each update uses only one record.

3. Mini - Batch Gradient Descent.

- Mini Batch Gradient Descent is a Hybrid approach that combines Batch Gradient Descent and Stochastic Gradient Descent.

- In this method, the training dataset is divided into small groups called mini-batches. and each mini - batch is used to update the weights.

- It balance the speed of SGD and the stability of Batch GD.

- This is the most commonly used optimizat-ion method in modern deep learning.

working process.

1. Diride data set into small batches
2. Select one mini - batch
3. Calculate gradient for that batch
4. Update weights
5. Repeat for remaining batch

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Characteristics.

- Uses small subset of data
  - faster computation
  - more stable than SGD
  - Efficient memory usage

Advantages

- fast and efficient.
- Stable convergence
- Suitable for GPU Processing
- Most practical method.

Disadvantages.

- Batch size selection is important.
- Improper batch size may reduce performance.

Example

→ If dataset has 50,000 records and batch size is 500:

- Data Divided into 100 mini-batches.
- Weights update 100 times in one epoch.

RAHUL CONTENT

MULTILAYER NETWORK

A Multilayer Network means a neural network that has more than one layer between input and output

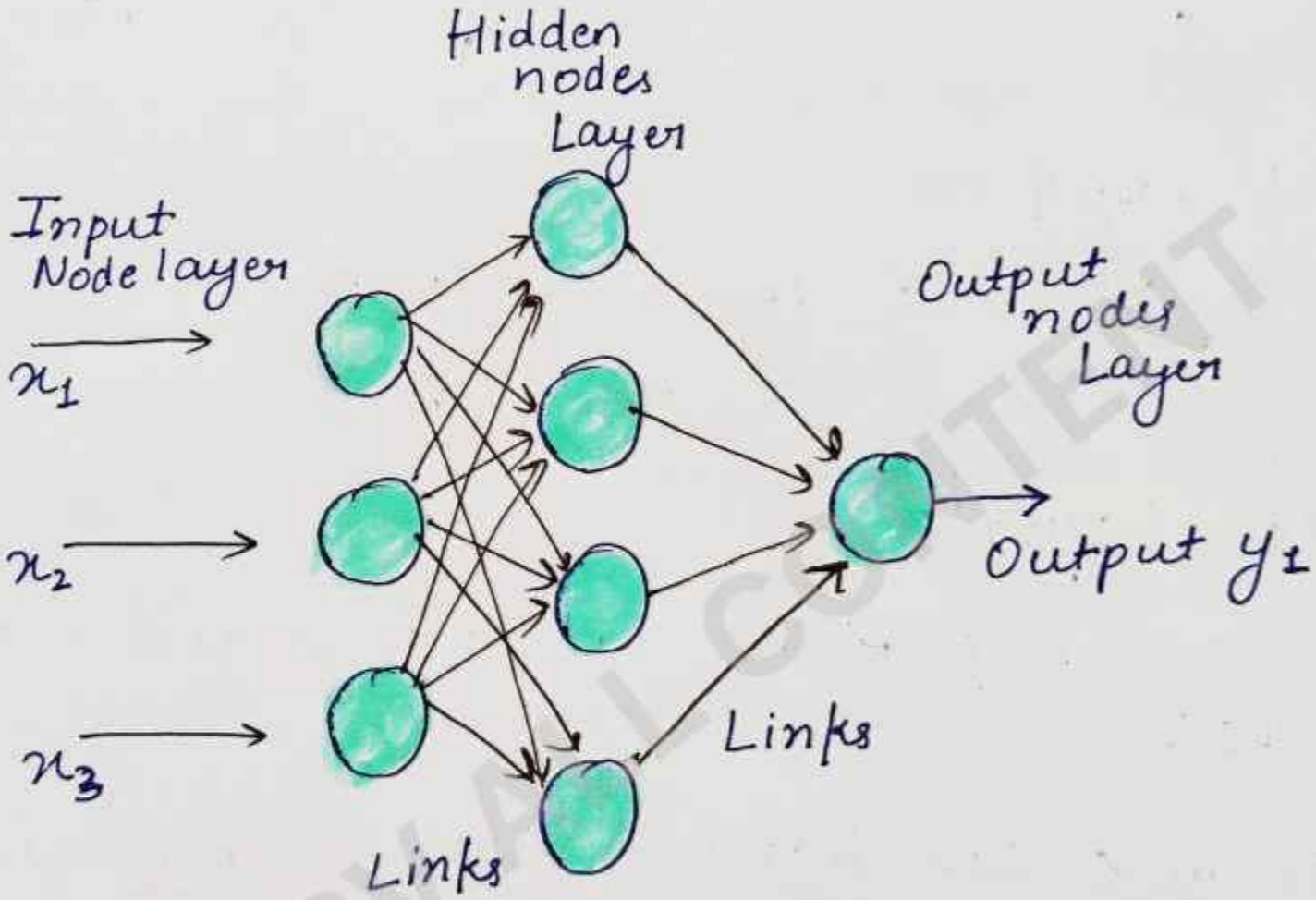
It is also called a Multilayer Neural Network or Multilayer Perceptron (MLP)

Structure of Multilayer Network

A multilayer network mainly has 3 types of layers:-

1. Input layer
   - Take input data from the user
   - Example:- image pixels, numbers, text etc.
2. Hidden Layer(s)
   - These layers perform calculations and learn patterns.
   - A network can have one or many hidden layers.
   - More hidden layers = deeper learning.
3. Output Layer:
   Gives the final result or prediction.
   Example:- . Cat or Dog classification..
   . Spam or Not Spam.

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Working of Multilayer Network.

Suppose we want to identify whenever an image is a cat or dog.

Step 1. Input Layer - The image data is given as input.

Step 2:- The Hidden layers:- detect edges, shapes, ears, eyes, patterns etc.

- Each neuron performs calculations using: weights, bias, activation function.

Step 3 Output Layer:- Finally, the network predicts: Cats or Dog.

Mathematical Idea

→ Each neuron calculates:-

$$y = f(\Sigma wx + b).$$

where:-

- x = input
- w = weight
- b = bias
- f = activation function
- y = output.

Why Multilayer Network are used.

Single-layer networks can solve only simple problems.

RGPV ALL CONTENT

- Multilayer networks can solve :-
- image recognition.
- speech recognition.
- language translation.
- medical diagnosis.
- Recommendation system etc.

Advantages

1. Learns complex patterns
2. High accuracy.
3. Works well for large datasets.
4. Used in Deep learning.

Disadvantages.

1. Requires more training time.
2. Need large data
3. Computationally expensive
4. Can overfit sometimes.

Real-Life Application.

- Face Recognition
- self-driving cars
- Voice assistants
- Handwriting recognition.
- Fraud detection.

FOR ALL CONTENT

BACK PROPAGATION

Back Propagation is a learning Algorithm used in neural networks to reduce error and improve accuracy.

It helps the network learn by calculating error, sending the error backward and updating weights.

It is mainly used in Multilayer Neural Networks.

- **Back** :- error moves backward.
- **Propagation** :- spreading information.

So Back Propagation means sending the output error backward through the network to improve learning.

Steps of Back Propagation.

① **Forward Propagation** → Input is passed through the network to get output

Example:-

Input → Hidden layer → Output.

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2. Calculate Error

~~to~~ compare : Actual Output and Predicted output.

Error formula :-

Error = Actual Output - Predicted Output.

3. Backward Propagation.

The error is sent backward from :-

- Output layer to
- Hidden layer.

The network checks :-

- which weights caused more error.

4. Update weight

Weight are adjusted to reduce error.

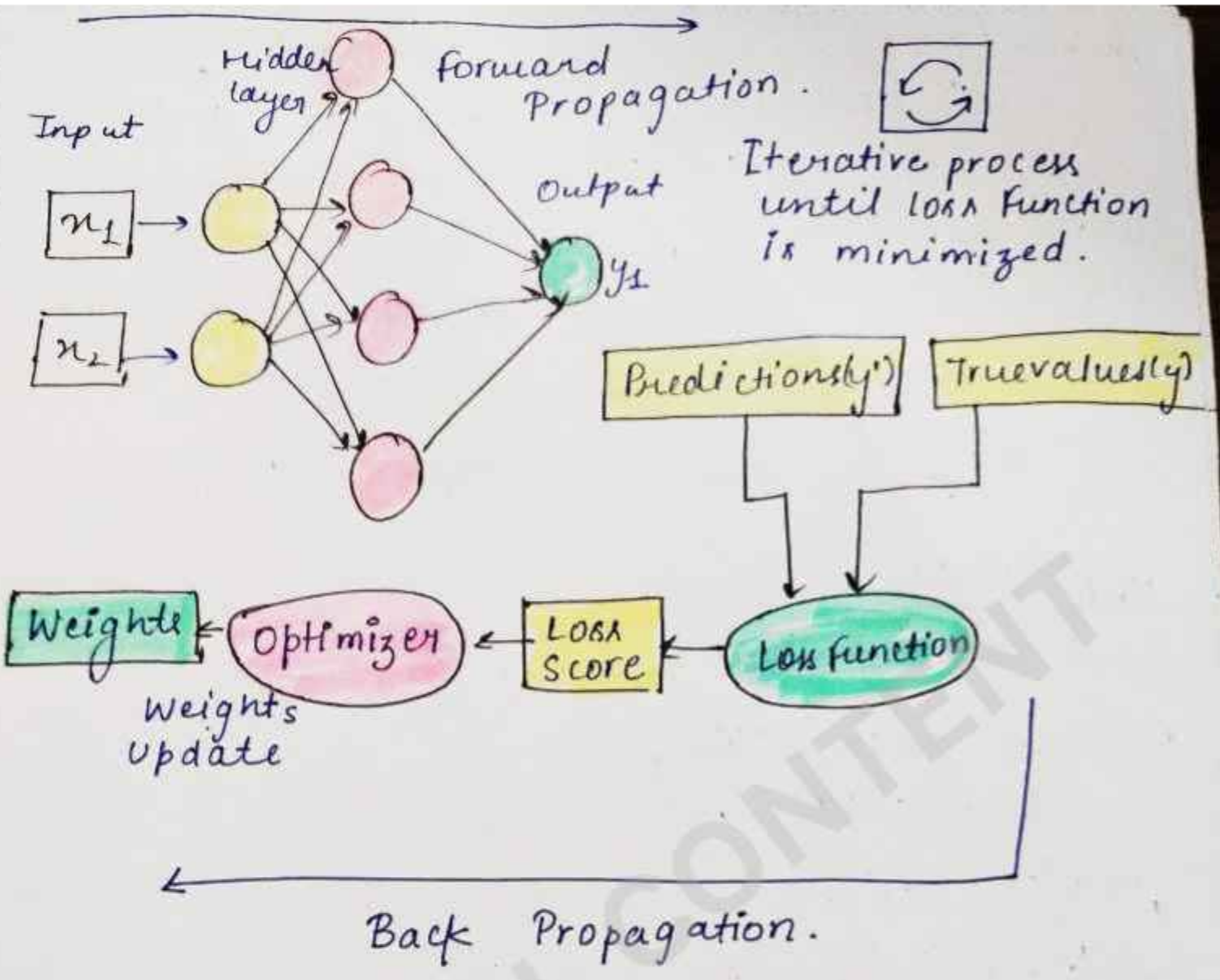
Weight update formula :-

$$W_{new} = W_{old} - \eta \frac{\partial E}{\partial W}$$

where :-

- W → weight.
- η → learning Rate.
- E → error.

SIGNAL CONTENT



Importance:-

- It helps the neural network:-
- learn from mistakes.
- Improve predictions.
- Increase accuracy.

★ without back propagation, deep learning would not work properly.

AGRWALL CONTENT

Advantages

- Improve Accuracy
- Learn complex patterns
- Work well in deep learning.
- Reduce error automa-tically.

Disadvantages.

- Training can be slow ~~Learn complex patterns~~
- Needs large data.
- Require High computation power.

Applications.

Back Propagation is used in:-

- Image Recognition.
- Speech Recognition
- Medical dignosis
- Self-driving cars
- Chatbots

RGPV ALL CONTENT

AUTO ENCODER

- An Autoencoder is a type of neural network used to learn important features of data, Compress Data and Reconstruct the original data.
- It works automatically, so it is called Auto Encoders.

Main Idea

An autoencoder tries to:
Convert input data into a smaller form and then rebuild the original data from it.

Example
- Compressing an image.
- Removing noise from Photos.
- Features extraction.

Structure of Autoencoder.

An autoencoder has 3 main parts:-

1. Encoder
   - Compresses input data into smaller representation.
   - Extract important features.

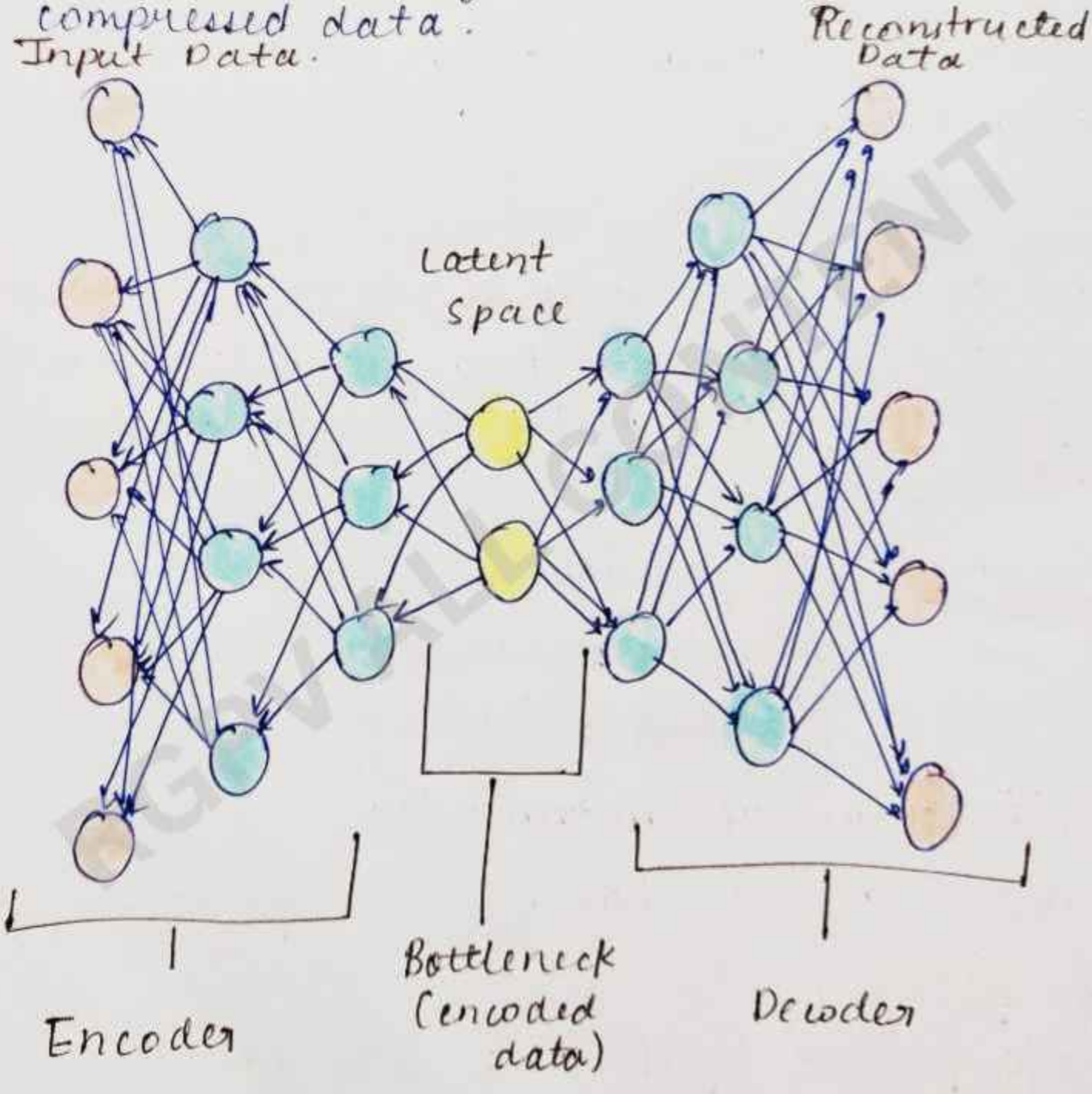
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2. Bottleneck (Latent space)

- Small compressed form of data.
- Stores only important information.

3. Decoder

- Reconstructs original data from compressed data.



Working of Autoencoder.

Suppose an image is given as input.

step 1 Encoder - The encoder compress the image.

- Large image → smaller feature representation.

Step 2 Bottleneck - Only important information is stored.

step 3 Decoder - The Decoder rebuilds the image from compressed data.

step 4 Compare Output - The reconstructed image is compared with original image.

The network learns by reducing reconstruction error.

RGPV ALL CONTENT

Types of Auto encoders

(1) Convolution Auto encoders
(2) Sparse Auto encoders
(3) Deep Auto encoders

(1) Convolution Auto Encoders

It uses Convolutional Neural Network (CNN) layers for learning image features.

A convolutional autoencoder:

★ Compresses the image using convolutional layers.
★ Learns important image features.
★ Reconstructs the original image.

→ It is mainly used for:-

- Image compression.
- Noise removal
- Feature extraction
- Image reconstruction

It has two main parts

(1) Encoder - The encoder:-

- uses convolution and pooling layers
- reduces image size.
- extracts important features.

Example → Image → feature Maps → Compressed Representation.

② Decoder :- The decoder :

- Reconstructs the image.
- uses deconvolution / upsampling layers.
- produces output similar to original image.

② Sparse Autoencoder :-

- In this only a few neurons are active at one time.
- It learns only important features of data
- Even if many neurons exist, the network forces most neurons to remain inactive. This creates efficient learning.

Features

- learn useful patterns.
- Reduces unnecessary learning.
- Improves feature extraction.

RGPV ALL CONTENT

(3). Deep Autoencoder

A Deep Autoencoder is an encoder with many hidden layers.

It is used for learning complex patterns in large datasets

features.

- It learns deep features
- Provide Better compression of data.
- It also Handles complex data.

Advantages of Autoencoder.

- Reduces data size
- Learns features automatically.
- Remove noise
- Useful for dimensionality reduction.

Disadvantages of Autoencoder.

- Training can be slow
- Needs large data.
- Sometimes reconstructed output is blurry.

REPLY ALL CONTENT

BATCH NORMALIZATION

Batch Normalization is a technique in ML used to make training neural networks faster, more stable, and more reliable.

Batch normalization (often called BatchNorm) normalizes the inputs of each layer so that they have.

- mean ≈ 0
- variance ≈ 1

This is done for each mini-batch during training

Need of Batch Normalization.

During data traing, the distribution of activations changes as weights updated - this is called internal covariate shift.

BatchNorm helps by :-

- stabilizing these distributions
- allowing higher learning rates
- reducing sensitivity to initialization
- acting as a mild regularizer.

RGPV ALL CONTENT

How it Works:

for each feature in a mini-batch:-

1. Compute mean

$$\mu_B = \frac{1}{m} \sum_{i=1}^{m} x_i$$

2. Compute Variance.

$$\sigma_B^2 = \frac{1}{m} \sum_{i=1}^{m} (x_i - \mu_B)^2$$

3. Normalize

$$\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$$

4. Scale and shift (learnable parameters):

$$y_i = \gamma \hat{x}_i + \beta$$

- γ (gamma) = scale
- β (beta) = shift
- ϵ (epsilon) = small constant for numerical stability.

Where it is used.

BatchNorm is commonly applied:

- after linear/dense layers.
- after convolution layers (in CNNs)
- before activation functions (most common practice)

PRIVATE CONTENT

Training Vs Inference.

- Training - uses batch statistics (mean & variance of current mini-batch)
- Inference - uses running averages collected during training

Benefits / Advantages.

- Faster convergence
- Most stable gradients
- Less overfitting (slight regularization effect)
- Enable deeper networks.

Limitations / Disadvantages.

- Depends on batch size (small batches - noisy estimates)
- Less effective in some architecture. (e.g. RNNs).
- Alternative exist like
  - Layer Normalization
  - Group Normalization

ROYAL CONTENT

DROPOUT

Dropout is a regularization technique. used to reduce overfitting in neural networks.

It works by randomly turning off some neurons during training.

During training, dropout randomly removes some neurons from the network temporarily.

These neurons

- do not participate in Forward propagation
- do not participate in Back propagation

→ This forces the network to learn more about features

Need of Dropout

Without dropout

- neural networks may memorize training data.
- model become overfitted
- poor performance on new idea.

Dropout helps the model generalize better.

PGP ALL CONTENT

L1 and L2 Regularization.

Regularization:- It is a technique in ML and Deep learning used to reduce overfitting

Overfitting happens when a model learns the training data too well and perform poorly on new data.

Regularization controls the complexity of the model by adding a penality to the loss function.

Need of Regularization.

→ Suppose a model has very large weights

→ Large weights make the model
- too sensitive to small changes.
- memorize training data.
- perform poorly on test data.

→ Regularization keeps weights smaller and simpler

BCA ALL CONTENT

General formula.

The new loss function becomes.

Total loss = Original Loss + Regularization Term

The regularization term penalizes large weights.

L1 Regularization.

L1 Regularization adds the sum of absolute value of weights to the loss function.

$$\text{Loss} = \text{Original Loss} + \lambda \sum_{i=1}^{n} |w_i|$$

where

w_i = weights.

λ = regularization strength

Large λ → Stronger penality

How L1 Works

L1 pushes some weights exactly to 0.

That means.

- unimportant features are removed
- only important features remain

so L1 performs automatic feature selection.

NOT FOR ALL CONTENT

Example of L1

Suppose weights are.

- [5, 0.2, 0.01, 7]

→ After L1 regularization:

- [4.5, 0, 0.6]

→ some weight become zero.

Properties of L1

1. Sparse model → many weight become zero.
2. Feature selection → Unnecessary features are removed automatically.
3. Simpler model → Model becomes easier to understand.

Advantages

- Reduces overfitting
- Removes useless features
- Good for high-dimensional data
- Creates compact models

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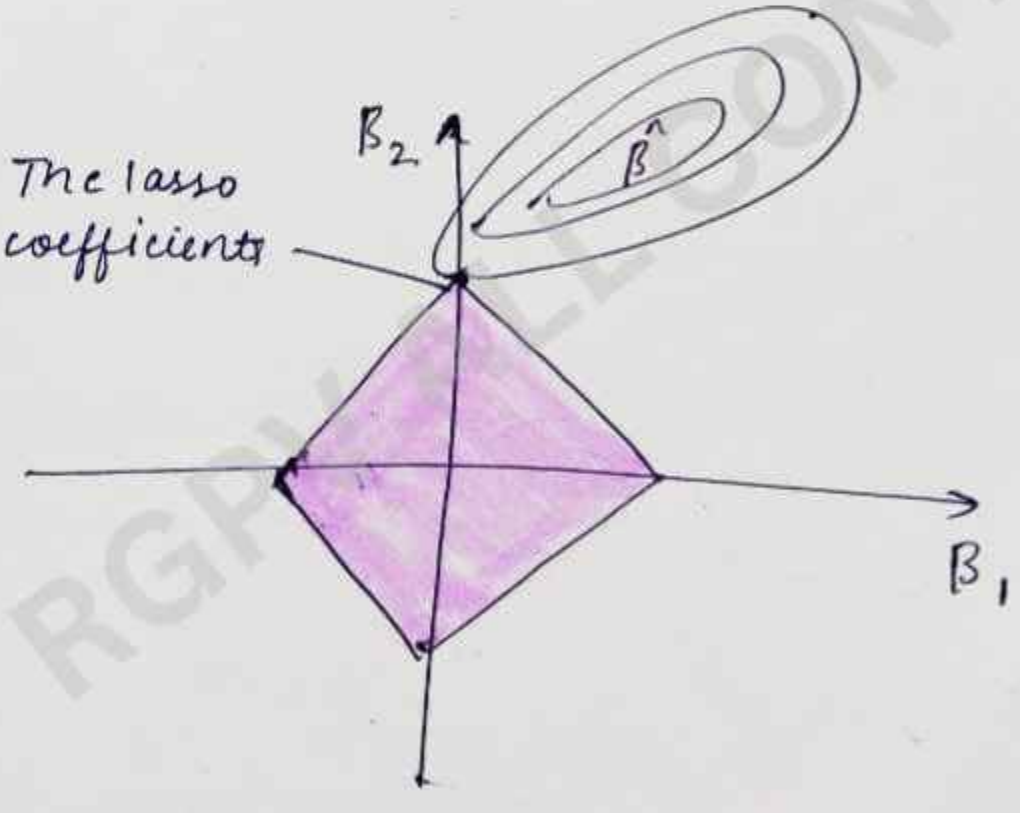
Disadvantages.

- Training may become unstable
- Problem with highly correlated features
- Can remove useful features accidentally.

Lasso Regression.

L1 regularization in linear regression is called lasso Regression.

Lasso means → Least Absolute Shrinkage and Selection operator.



LASSO

RGHT CONTENT

L2 Regularization.

L2 regularization adds the ~~square of~~ sum of squared weights to the loss function.

formula.

$$\text{Loss} = \text{Original loss} + \lambda \sum_{i=1}^{n} w_i^2$$

How L2 Works

L2 reduces weights gradually.

Weights become:-

- small
- close to zero.

But usually not exactly zero.

Example :- Original weights

[5, 0.2, 0.01, 7]

After L2:

[4.2, 0.15, 0.008, 5.8]

All weights shrink smoothly

Properties of L2

1. Weight Shrinkage → Large weights are reduced.

RGPV ALL CONTENT

2. Smooth learning - Training becomes more stable.

3. No feature Removal - Features stay in the model.

Advantages of L2:-

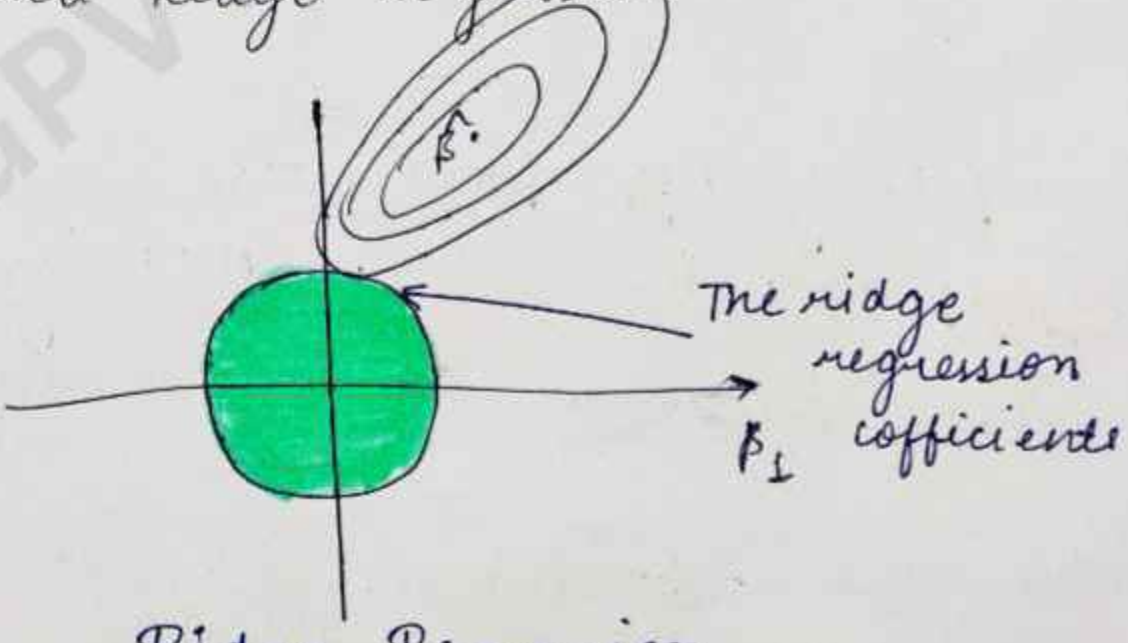
- Prevent Overfitting effectively
- Stable training
- Works well in deep learning
- Handles correlated features better

Disadvantages of L2:-

- Does not perform features selection.
- Keeps all features in model.

Ridge Regression

L2 regularization in linear regression is called Ridge Regression.



Ridge Regression

RGPV FULL CONTENT

Hyperparameter Tuning

Hyperparameter tuning is a process in ML used to find the best values of hyperparam -eters so that a model gives maximum accuracy and performance.

Hyperparameters are settings that are chosen before training the model.

They control:

- learning process
- Model complexity
- training behavior

They are not learned automatically from data.

Examples of Hyperparameters.

Hyperparameters		Meaning
1 Learning Rate	→	Controls step size during training
2 Batch Size	→	Number of samples processed together.

PGPV ALL CONTENT

3 Number of Epochs → How many time data passes through model.

4 Number of Hidden Layers → Depth of neural network

5 Number of Neurons → Units in each layer

6 Regularization Parameter (λ) → control overfitting

Importance

→ Correct Hyperparameters help to :-

- Improve accuracy
- Reduce overfitting
- Speed up training
- Increase model stability.

→ Bad Hyperparameters may cause :

- poor accuracy
- slow learning
- overfitting.

Example

Suppose learning rate is :-

- Too high :- model may overshoot minimum.
- Too low :- training becomes very slow.

So we find the best value.

PGDALECONTENT

Types of Parameters

① Model Parameters → Learned automatically during training.

Examples :- weights & biases.

② Hyperparameters

Set manually before training

Example-learning rate, epochs, batch size.

Techniques of Hyperparameter Tuning.

Some Best techniques of Hyperparameter tuning are given below :-

① Manual search

② Grid search

③ Random search.

④ Bayesian Optimization.

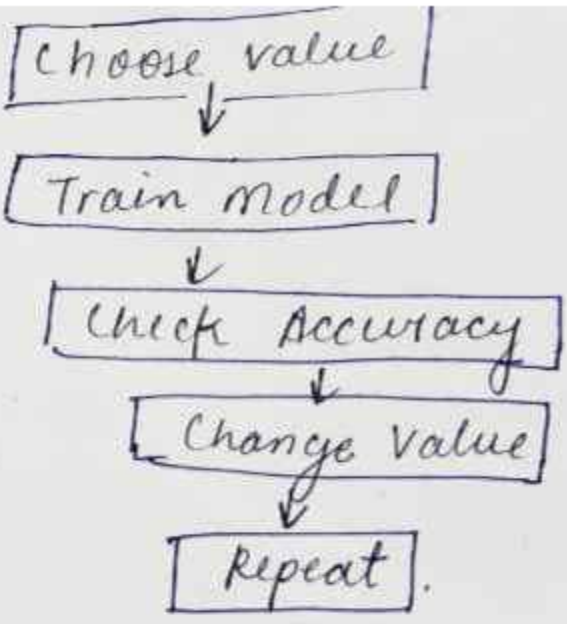
1 Manual search

In this technique, the user manually changes hyperparameters and checks performance.

Example :- learning rate 0.1
- then 0.01
- then 0.001

RCP ALL CONTENT

- It is simple and easy for beginners
- But time consuming and depend on user experience.



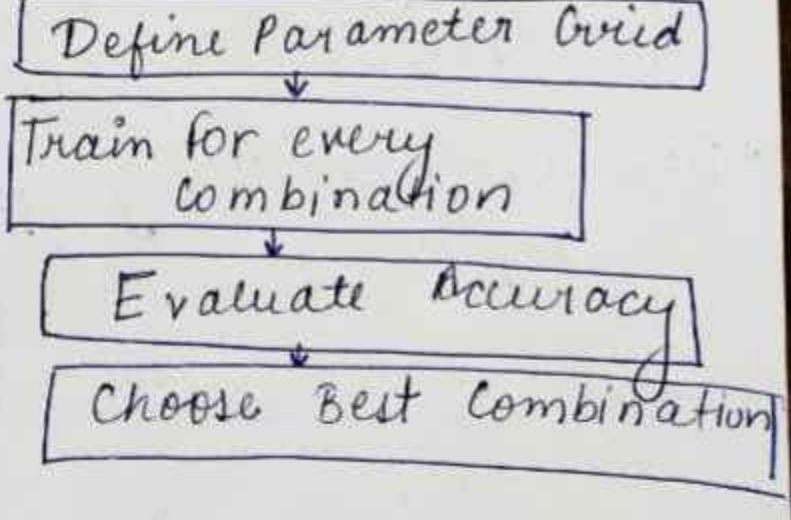
② Grid Search.

Grid search tests all possible combinations of hyperparameters.

Example →

Learning Rate	Batch size
0.1	16
0.1	32
0.01	16
0.01	32

- All combinations are checked
- It is systematic and find good combinations.
- But very slow and expensive for large datasets.



FOR ALL CONTENT

③ Random Search

- Random Search selects random combinations instead of all combinations

Example →
→ Instead of checking 100 combinations. it may test only 10 random combinations.

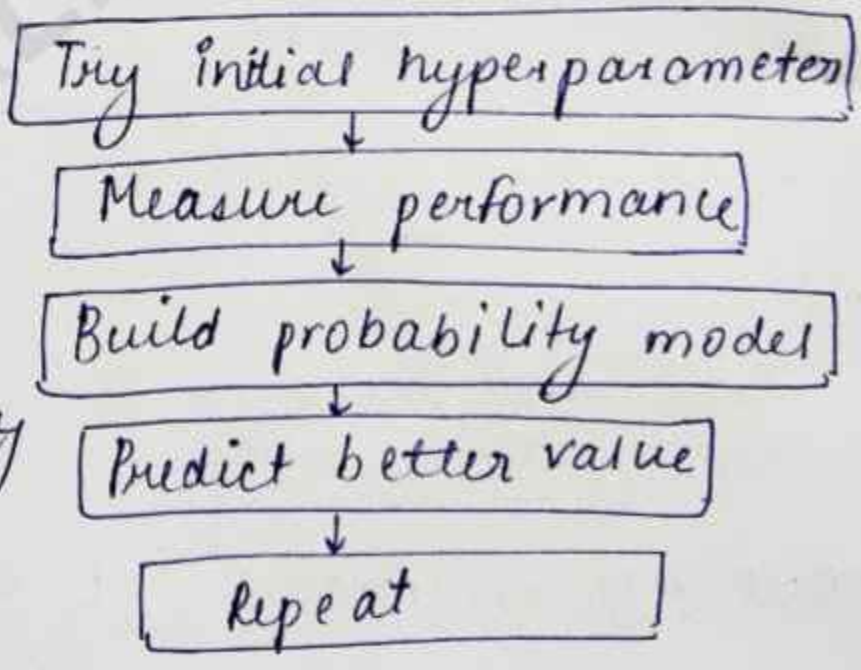
★ It is Faster than grid search and efficient in large space.

④ Bayesian Optimization.

Bayesian optimization uses probability to predict better hyperparameters.

It learns from previous trials.

- It is very efficient and need fewer experiments.
- But mathematically complex



RGM ALL CONTENT

UNIT-3

MACHINE LEARNING

2024

CONVOLUTIONAL NEURAL NETWORKS (CNNs)

A Convolutional Neural Network (CNN) is a special type of artificial neural network. used mainly for processing images, videos and visual data

It is designed to automatically detect important features such as edges, shapes, textures, and objects from input images.

It is one of the most important technique in Deep learning and computer vision.

Main Idea of CNN

A normal neural network treats every pixel separately, but CNN processes images using small filters called kernels.

- These filters move across the image and extract useful information.

FLATTENING in CNN

Flatening is the process of converting a multi-dimensional features map (2D or 3D date) into a single one-dimensional vector.

It is used in a convolutional Neural Network before sending data to the fully connected layer.

Why Flattering is Needed

After convolution and pooling operations, the output is still in matrix form.

- Features map size - 4x4 x3

But the fully connected layer accepts only a 1D vector.

So flattring converts.

- 4x4x3 → 48 values in one row.

Example → Suppose pooling layer output is

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

After Flattening

[1, 2, 3, 4]

Important Points

- Converts 2D / 3D data into 1D data
- Used between pooling layer and fully connected layer
- Does not lose data values
- Makes data ready for classification.

Advantage

- Simple conversion process
- Help connect CNN layer to dense layer
- Easy classification

Disadvantage

- Large feature maps create very large vectors
- Can increase computations.

Role in CNN → CNN layers extract features. while Flattering prepares those features for final classification

Subsampling in CNN

Subsampling is the process of reducing the size of feature map in Deep learning or ML while keeping the important information.

In CNN, subsampling is mainly done using the pooling layer.

It reduces

- Width
- Height
- Number of computations.

without losing important features.

Purpose of Subsampling

- Reduce image dimensions
- Reduce computation time
- Prevent overfiting
- keep important features only

- Subsampling is also known as pooling
- The main goal of subsampling is to make the network more efficiently by reducing the amount of data that must be processed in later layers.

Types of subsampling.

① Max Pooling.

Max pooling selects the largest value from each pooling region

For the example above.

max (1,3,5,6) = 6
max (2,4,1,2) = 4
max (7,8,1,2) = 8
max (3,4,5,6) = 6

Output → 6 4
8 6.

Advantages

- Preserves the strongest features.
- Works well for image recognition tasks

② Average Pooling.

Average Pooling computes the average value within each pooling region.

Example

→ (1+3+5+6)/4 = 3.75

Output → 3.75

Advantages :- • Retains more background information.

• Produces smoother feature maps.

Padding 2025, 2024

- Padding is a technique used in CNN, where extra pixels (usually zeros) are added around the border of an input image or feature map before applying a convolution operation.

- Padding helps control the size of the output feature map and ensures that important information near the image boundaries is not lost.

Why is Padding Needed.

Without padding, the size of the feature map decreases after every convolution.

For example:-

Input image size :- 5X5

kernel size :- 3X3

stride :- 1

Output size = $\frac{(5-3)}{1} + 1 = 3$

Output = 3X3

* The feature map becomes smaller. After several convolution layers, the image may shrink too much.

Padding solves this problem by adding extra pixels around the image.

Example of Padding (original image) 5X5

1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1

After Adding Padding = 1

0 0 0 0 0 0 0 0 0
0 1 1 1 1 1 1 0
0 1 1 1 1 1 1 0
0 1 1 1 1 1 1 0
0 1 1 1 1 1 1 0
0 1 1 1 1 1 1 0
0 1 1 1 1 1 1 0
0 0 0 0 0 0 0 0

The image size becomes → 7X7.

Stride

Stride is the number of pixels by which the convolution filter (kernel) moves across the input image at each step.

Example

- stride=1 → The filter moves 1 pixels at a time.
- stride = 2 :- The filter moves 2 pixels at a time. producing a smaller output feature map.

Effects of Stride.

- Smaller stride(1) :- Captures more details and produces a larger output.

- Larger stride (2 or more) :- Reduce computation and produces a smaller output.

In short
→ "stride is the number of pixels the convolution kernel moves across the input image during the convolution operation."

Convolutional Layer in CNN

A Convolutional Layer is the core building block of a Convolutional Neural Network CNN.

It is responsible for extracting important features such as edge, textures, shapes, and patterns from the input image.

A convolutional layer applies a set of small filters (kernels) to the input image.

Working of a Convolutional Layer.

1. Input Image. - The image is provided as input to the convolutional layer.

2 Kernel (Filter)

→ . A small matrix (e.g 3X3 or 5X5) is used to detect specific features.

3. Convolution Operation.

. The filter moves across the images

. At each position. element - wise multiplica -tion is performed between the filter and the image region.

. The result are summed to produced a single value.

4. Features Map

. The output generated after applying the filter is called a feature map.

Components of a convolutional layer.

- Input :- Image or feature map from the previous layer.
- Kernel (Filter). Detects patterns and features.
- Stride - Determines how many pixels the filters moves each step.
- Padding :- Adds extra pixels around the image to preserve dimensions.
- Activation function :- Introduces non-linearity.

Advantages.

- Reduces the no. of parameters compared to fully connected layers.
- Automatically extracts features
- Improves image recognition performance

Pooling Layer in CNN 2025

Pooling layer is a layer in a CNN that reduces the size of feature maps while preserving the most important information.

It helps decrease computation memory usage and overfitting.

The pooling layer performs a downsampling operation on the feature map generated by the convolutional layer.

Working of Pooling layer.

1. The feature map is divided into small regions (e.g. 2X2)
2. A pooling operation is applied to each region.
3. One value is selected from each ~~each~~ region to create a smaller feature map.

Advantages of Poolin

- Reduces feature map dimensions
- Decreases computational cost
- Reduces overfitting
- Makes the model more robust to small shifts and distortions.

Disadvantages.

- Some information may be lost during downsampling.
- Excessive pooling can reduce import -ant details

Loss Layer

Loss layer is the final layer in CNN. that calculate the error (loss) between the model's predicted output and the actual target output.

The loss layer measures how far the predicted result is from the correct result. The objective of training is to minimize this loss.

Working of a Loss Layer.

1. The CNN produces prediction
2. The prediction is compared with the actual target value.
3. A loss function computes the error
4. The error is propagated backward through the network.
5. Weights are updated to reduce the loss.

Common Loss functions.

1. Cross-Entropy Loss :- Used for classification problems.

For example, Actual class : Cat

Predicted probability :- Cat = 0.9
Dog = 0.1

A higher probability for the ~~higher~~ correct class results in lower loss.

2. Mean Squared Error (MSE)

Used mainly for regression tasks.

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

where.

y_i = Actual value

$\hat{y}_i$ = Predicted value

n = Number of samples.

Importance of the loss layer.

- Evaluates model performance
- Guides the learning process.
- Helps optimize network parameters

Inception Network. 2024

The Inception Network, also known as GoogleNet, is a deep CNN architecture developed by researchers at Google.

It won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) in 2014.

An Inception Network in CNN architecture that performs multiple convolution operations of different sizes (1x1, 3x3 and 5x5) in parallel and then combines their outputs.

This allows the network to capture features at different scales.

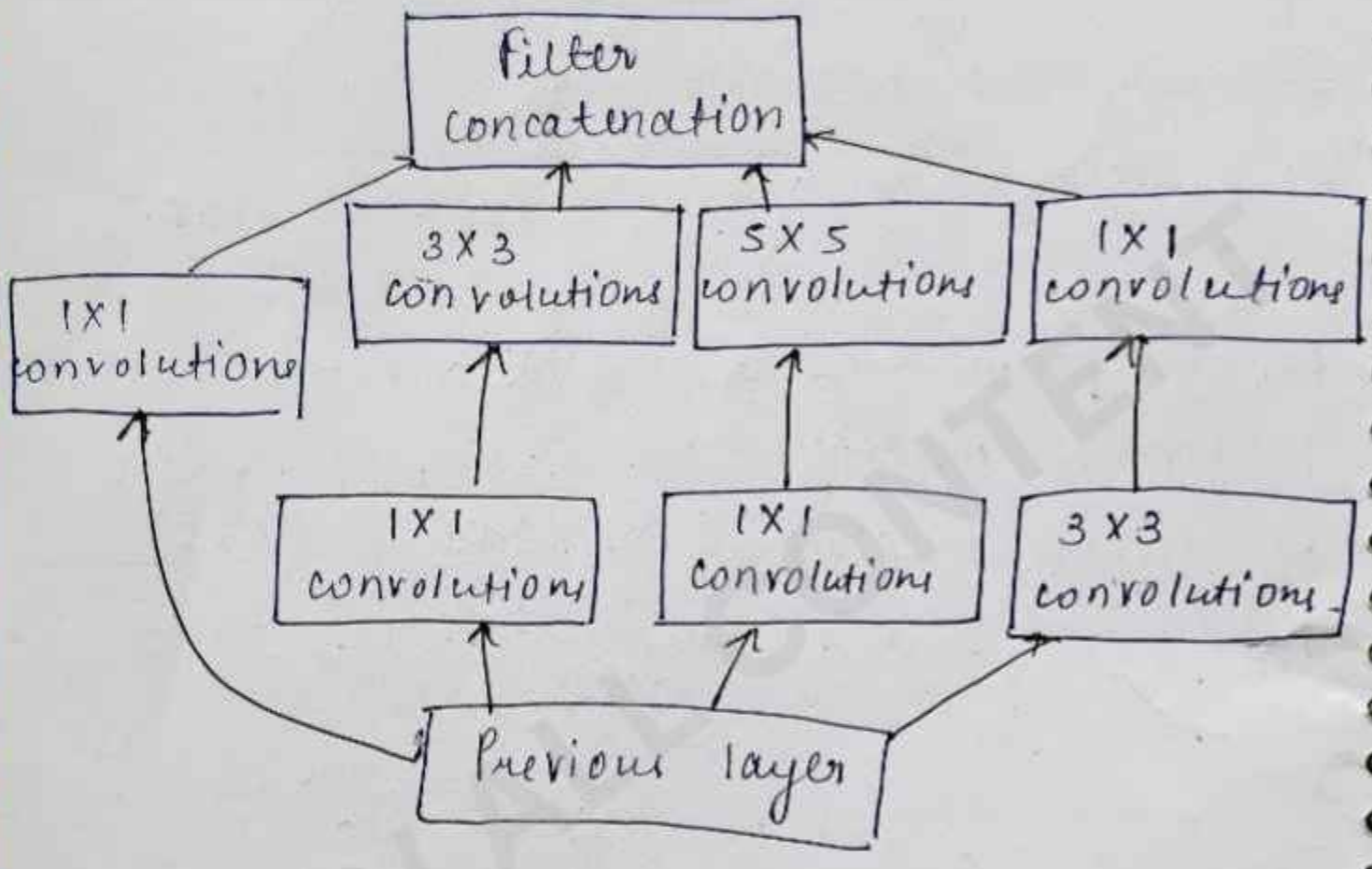
Inception Module.

The main building block of GoogleNet is the Inception Module.

It contains:

- 1x1 Convolution
- 3x3 Convolution
- 5x5 Convolution
- 3x3 Max pooling.

All these operations are performed in parallel, and their outputs are concatena-ted into a single feature map.



Inception Network.

Role of 1X1 convolution

The 1X1 convolution is used before 3X3 and 5X5 convolutions to :-

- Reduces the number of channels
- Lower computational cost
- Reduce memory usage

Transfer Learning.

Tranfer Learning is a ML technique in which a model trained on one task is reused for another related task. Instead of training a new model from scratch. the knowledge learned by a pre-trained model is transferred to the new problem.

Working.

1. Train a model on a large dataset (source task).
2. Save the learned features and weights
3. Use the pre-trained model for a new related task (target task).
4. Retrain only the final layers or fine-tune the model.

Types.

① Feature Extraction - Freeze pre-trained layers and trained only the new output layers.

(2). Fine-Tuning - Retrain some of the pre-trained layers along with the new layers.

Advantages.

1. Requires less training data
2. Reduce training time
3. Improves accuracy
4. Reduces computational cost.

Disadvantages.

1. Not effective if source and target tasks are very different
2. Choosing the wrong pre-trained model may reduce performance

Application

- Image Classification
- Object Detection
- face Recognition
- Medical Image Analysis.

One Shot Learning

One shot learning is a technique in which a model learns to recognize or classify an object from only one training example.

Unlike traditional ML which requires many examples. One-Shot Learning can learn from a very small amount of data.

Working

1. The model is trained to compare similarities between objects.
2. Give one example of a new class. it learns its features.
3. When a new sample arrives. the model compares it with the stored example.
4. If they are similar. it assigns the same class

Example → Suppose a face recognition system sees only one photo of a person. Later. when another photo of that person is shown. the system identifies the person by comparing facial features with the single stored images.

Application
→ Face Recognition
→ Signature Verification
→ Object Recognition.
→ security System.

Dimension Reductions. 2025

Dimension Reduction is the process of reducing the number of input features (variables) in a dataset while keeping as much important information as possible

for example :-

Student	Age	Height	Weight	Marks	Attendance
A	18	176	65	85	90
B	19	168	62	80	88

Here, Age, Height, Weight, Marks and Attendance are 5 dimensions (features) of some features are unnecessary or highly related, we can reduce them to fewer features. This process is called Dimension Reduction.

Need of Dimension Reduction.

Datasets in ML may contain many features

- Too many features can :-
  - Increase Training time
  - Require more memory
  - Cause Overfitting
  - Make Visualization difficult.

- Dimension Reduction helps simplify the data.

Types of Dimension Reduction.

(1) Feature selection.
Select only the most important features and remove the rest.

Example :- From . Age, Height. weight. marks, Attendance.

Keep - Attendance, marks. Age.

Remove less useful features.

Methods - Filter Methods
- Wrapper Methods
- Embedded Methods.

2. Feature Extraction.

Create new features by combining existing features.

Example: Instead of using
- Math Marks
- Physics Marks
- Chemistry Marks.

Create:
- Overall Performance Score.

Advantages:

1. Training is faster
2. Less storage required
3. Reduce overfitting
4. Better model performance
5. Easier data Visualization.

Disadvantages:

1. Some information may be lost
2. Reduce interpretability
3. Choosing the wrong technique can decrease accuracy.

Implementation of CNN like tensor flow

2024

A CNN is a deep learning model widely used for image classification, object detection, and computer vision tasks. TensorFlow provides the keras API to easily build, train and evaluate CNN models.

1. Import Required Libraries.

First import Tensorflow and keras libraries required for building the CNN model.

Python

```
import tensorFlow as tf
from tensorFlow.keras import
datasets, layers, models
```

Purpose - Provides tools for data handling model creation, training and evaluation

2 Load the Dataset

Load the training and testing data.

Purpose - The dataset contains input images and their corresponding labels.

3. Data Preprocessing

Normalize pixel values from 0-255 to 0-1

Purpose :- • Improves training speed
• Helps the model converge faster.
• Reduces numerical instability

4. Build the CNN Architecture.

Create the CNN model using convolution, pooling and dense layer

5. Compile the model

Specify optimizer. loss function, and evaluation metric.

Parameters.

• Optimizer - Updates weights efficiently
• Loss function - Measures prediction
• Accuracy - Evaluate classification performance

6. Train the Model

Train the CNN using the training dataset.

7. Evaluate the Model

~~Train~~ Test the trained model on unseen data.

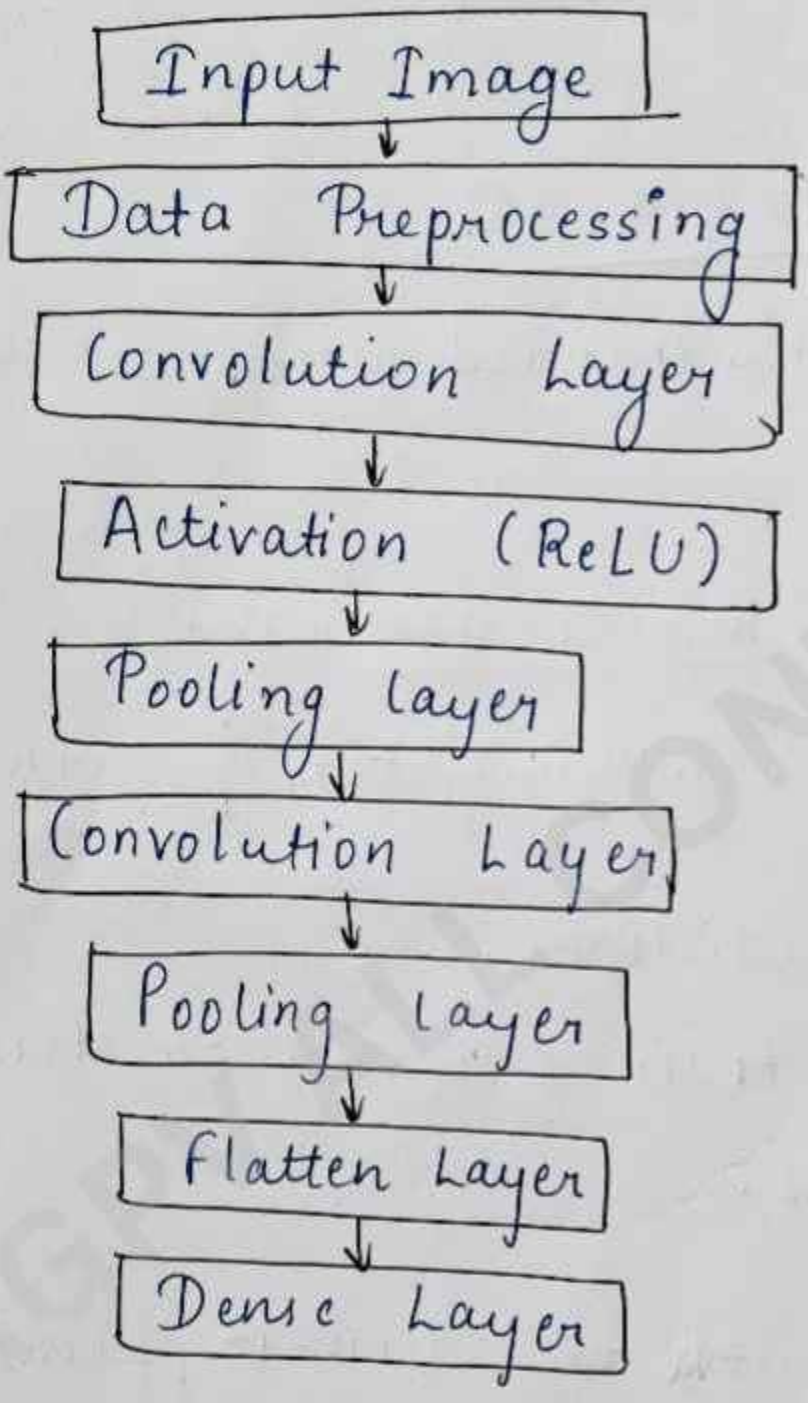
Purpose.

- Measures model accuracy
- Determines generalization performance.

8. Make Predictions.

- Used the trained model to classify new images.
- The class with the highest probability is selected as the final prediction.

Training Process Flow Diagram.



UNIT-4

Machine Learning -

Recurrent Neural Network 2024

A Recurrent Neural Network (RNN) is the type of artificial neural network specially designed to process sequential data.

Unlike traditional neural networks RNN can remember previous information and use it to make decisions about current inputs.

In simple words, RNN has memory which helps it understand the relationship between past and present data.

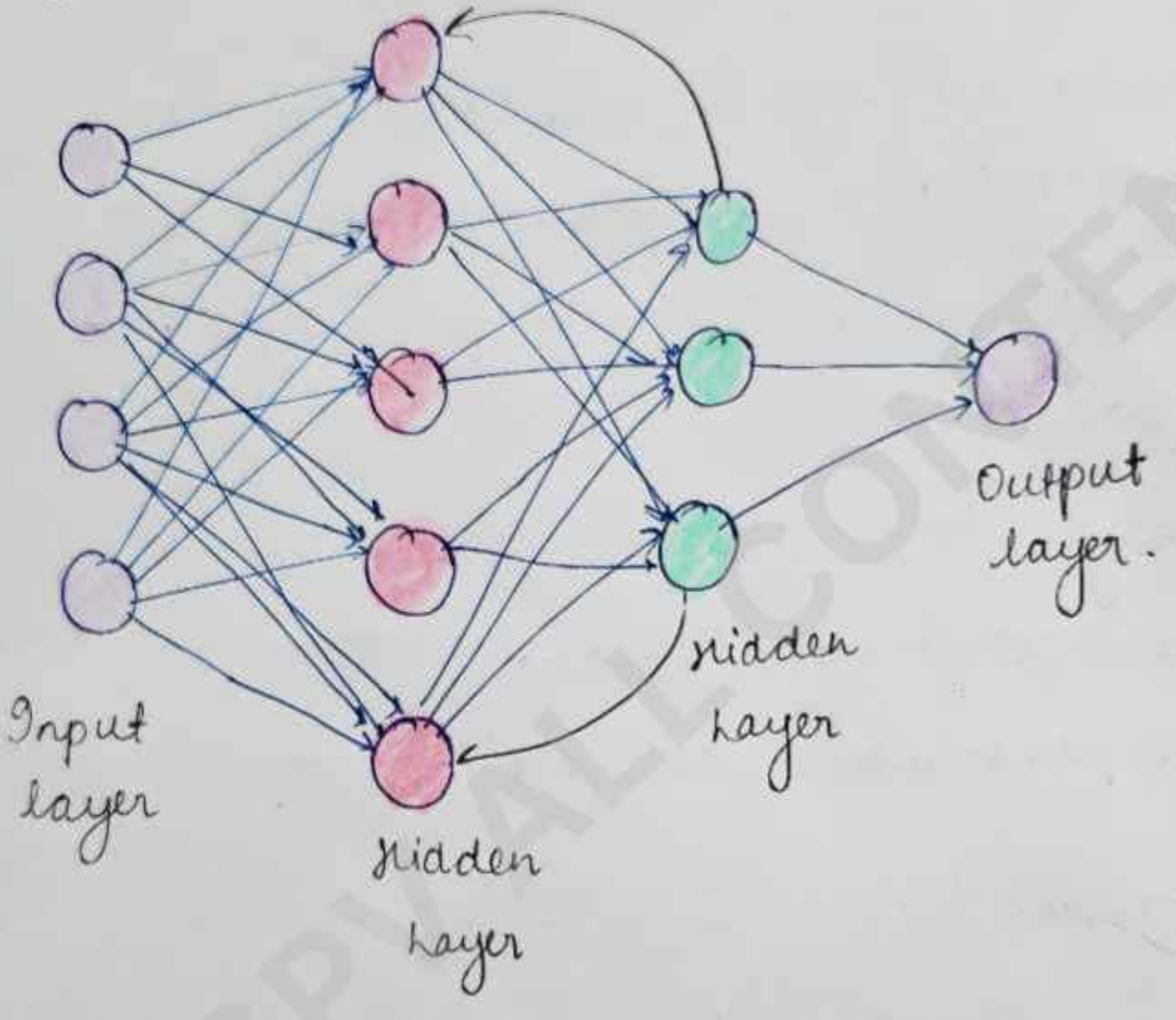
Need of RNN

Normal neural networks process each input independently and cannot remember previous information.

But many tasks require memory, such as.

- Language translation
- Speech Recognition.
- Text Predictions

RNN solves this problem by storing information from previous steps.



Recurrent Neural Network.

Example → Consider the sentence:-

"The movie was not good".

To understand the word "good". the model should remember the earlier word "not"

RNNs use previous content to make better predictions.

Advantages

- Handles sequential data
- Shares parameters across time steps

Limitations.

- Suffer from vanishing and exploding gradients during training.
- Difficult to learn long-term dependencies

To overcome RNN limitations, advanced architectures were developed:-

① LSTM (long-short Term Memory)
- Uses memory cells and gates
- Better at learning long-term dependencies

(2) GRU (Gated Recurrent Unit)

- Simpler than LSTM.
- Faster training with similar peryormance.

Long Short-Term Memory

2024, 2025

LSTM (Long Short-Term Memory) is a special type of Recurrent Neural Network (RNN) designed to remember information for a long period of time.

It was developed to solve the vanishing gradient problem of traditional RNNs.

Why LSTM is Needed?

In a simple RNN, information from earlier inputs may be forgotten after many time steps.

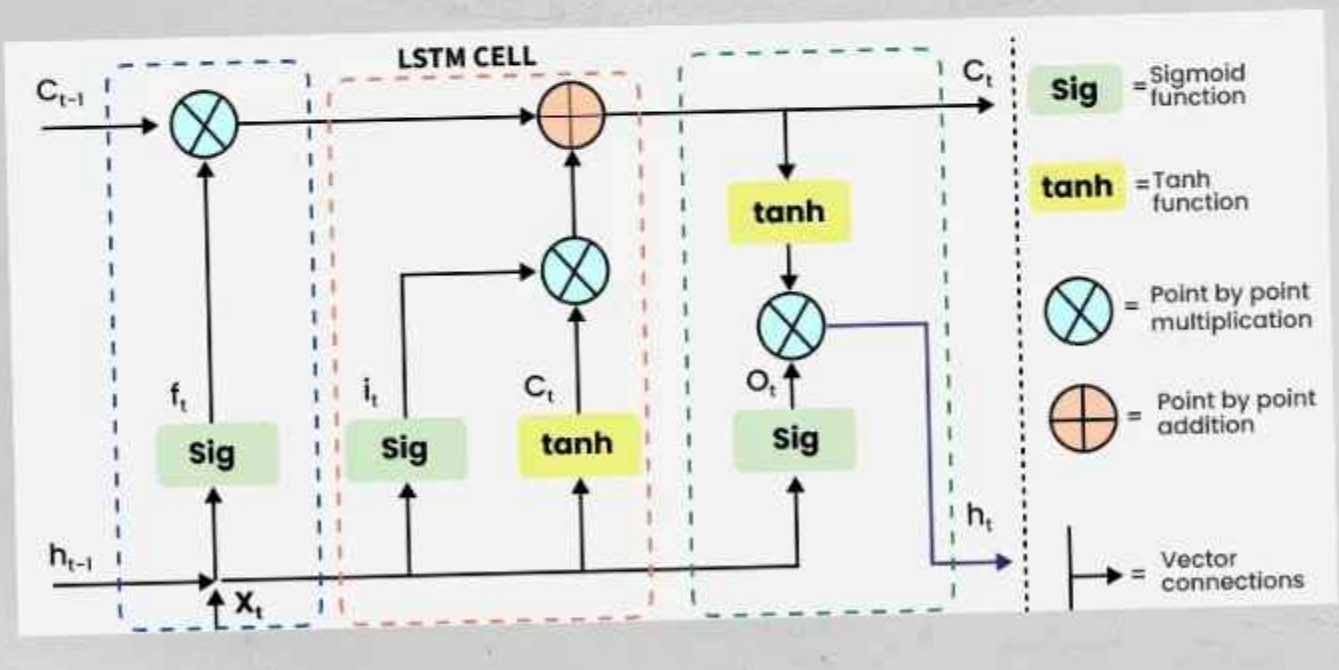
Example

→ "I grew up in France. I speak fluent French."

To understand why "French" is used, the model should remember "France" from earlier in the sentence.

RNNs often struggle with such long-term dependencies, while LSTMs can remember them.

Structure of LSTM.



An LSTM contains a cell state (memory) and three gates.

① Forget Gate - Decides what information should be removed from memory.

② Input Gate - Decides what information should be removed from memory.

③ Output Gate - Decides what information should be ~~stored~~ sent as output.

Working of LSTM.

Step 1 → Removes unnecessary information from the cell state

step 2 - Add useful new information to the cell state.

step 3 - Update the memory by combining old and new information.

step 4. Produce the final output and hidden state.

Advantages

- Remember information for a long time
- Solve vanishing gradient problem.
- Good for sequential data.

- Better performance than simple RNN.

Disadvantages.

- More complex than RNN
- Requires more computation and memory.
- Training can be slower.

Gated Recurrent Unit

A Gated Recurrent Unit (GRU) is a type of Recurrent neural network (RNN) architecture designed to handle sequential data and mitigate the vanishing gradient problem that affects traditional RNNs.

- GRUs use gates to control how information flows through the network, allowing them to remember long-term dependencies more effectively.

Components of a GRU

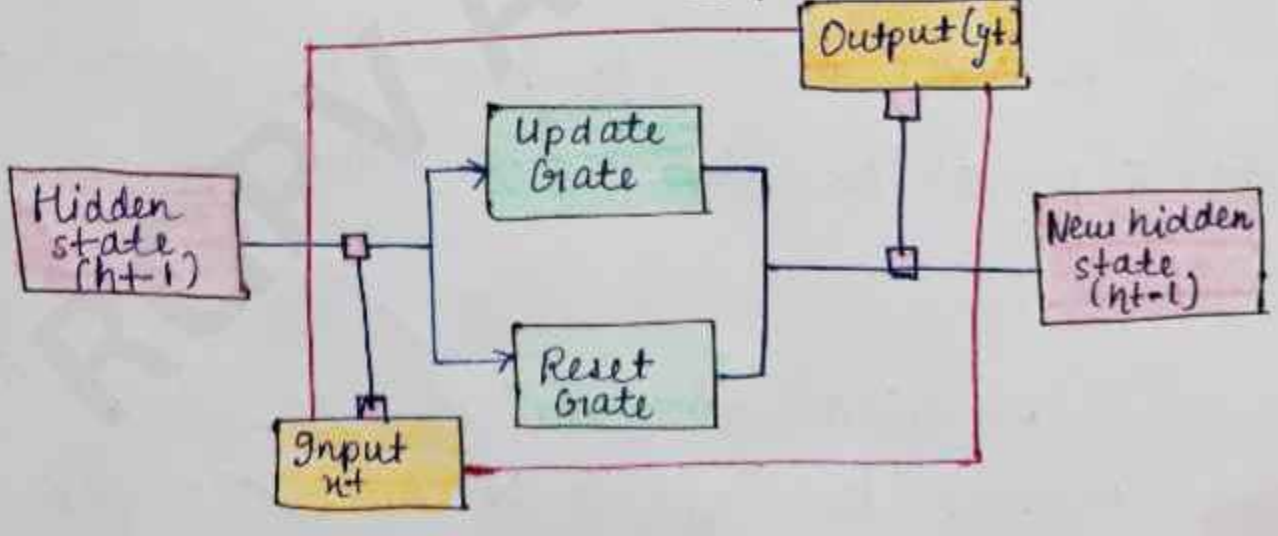
A GRU has two gates :-

1 Update Gate (z_t) • Determines how much of the previous hidden state should be retained.

• Acts similarly to the combination of the forget and input gates in an LSTM

2. Reset Gate (r_t) • Controls how much past information to forget when computing the new candidate state.

Structure



Advantages

- Simpler Architecture than LSTM.
- Fewer parameters, reducing computation and memory requirements.
- Effective for sequence modeling tasks such as:-
  - → Natural Language Processing (NLP)
  - → Speech Recognition
  - → Time-series forecasting
  - → Machine Translation

Attention Model

The Attention model is a mechanism that helps a neural network focus on the most important parts on the input while generating an output.

It was introduced to overcome the limitations of traditional Recurrent Neural Network (RNNs) and Encoder-Decoder models, which struggle with long sequence

Why Attention model is needed.

In a basic Encoder-Decoder model:-

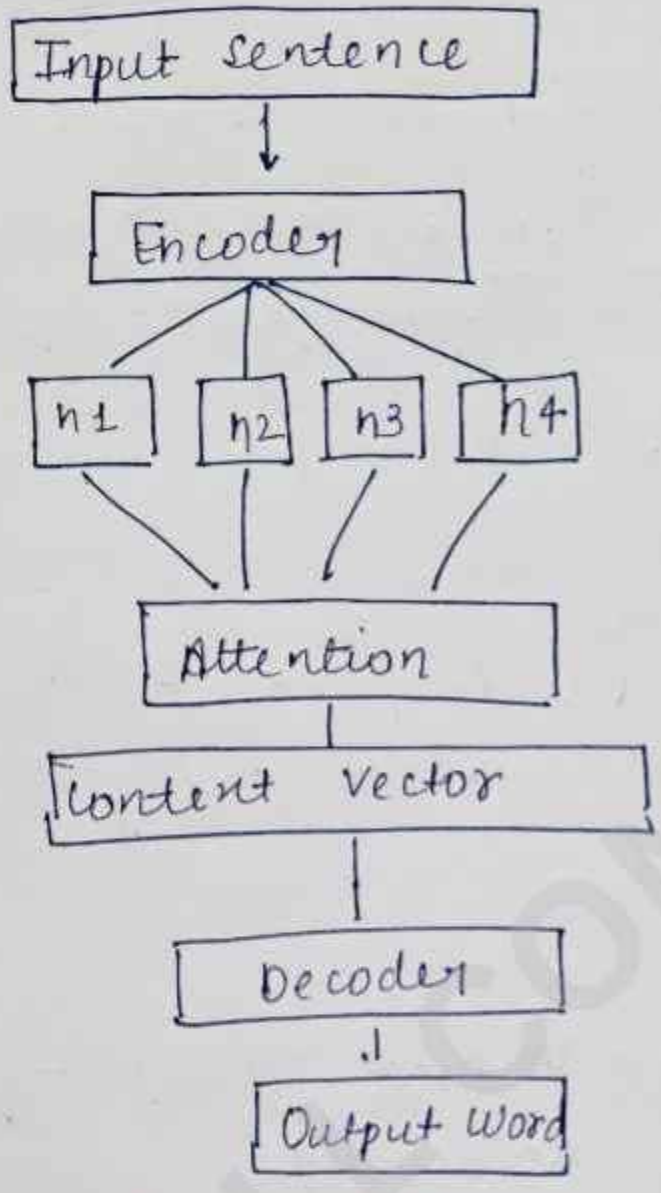
- The encoder converts the entire input sentence into a single fixed-size vector.
- The decoder uses this vector to generate the output.

→ For long sentences, important information may be lost because everything is compressed into one vector.

Example

Input → "The book that I bought yesterday is very interesting."

- When translating this sentence, the model should pay more attention to words like book, bought, and interesting at different decoding steps.
- Attention allows the model to look back at all encoder outputs instead of relying on only one vector.



Simple Diagram of Attention model.

where. h1, h2, h3, h4 are hidden states.

Advantages.

1 Handles long sentence better
2 Improves translation accuracy
3 Focuses on relevant words.
4 Reduces information loss.
5 Forms the foundation of the Transformer architecture.

Short Definition

→ Attention model is a mechanism that enables a neural network to dynamically focus on the most relevant parts of the input sequence while generating each output, improving performance on long-sequence tasks.

Markov Decision Process (MDP)

A Markov Decision Process (MDP) is a 2025 2024 mathematical framework used in Reinforce-ment learning (RL) to model decision-making problems where outcomes depends on the current state and action taken.

Markove Property:-

The future depends only on the current state, not on the sequence of past states.

$$P(S_{t+1} \mid S_t) = P(S_{t+1} \mid S_1, S_2 \ldots S_t)$$

$$P(S_{t+1} \mid S_t) = P(S_{t+1} \mid S_1, S_2 \ldots S_t)$$

This is called markov Property.

Components of MDP

An MDP is represented as :-

(S, A, P, R, γ)

1. State (S) :- Set of all possible state of the environment.

Example :- Position in a maze

2. Action (A) :- Set of action available to the agent.

Example :- Move Up, Down, left, Right

3. Transition Probability (P) :- Probability of moving from one state to another after taking an action.

$P(s', | s, a)$.

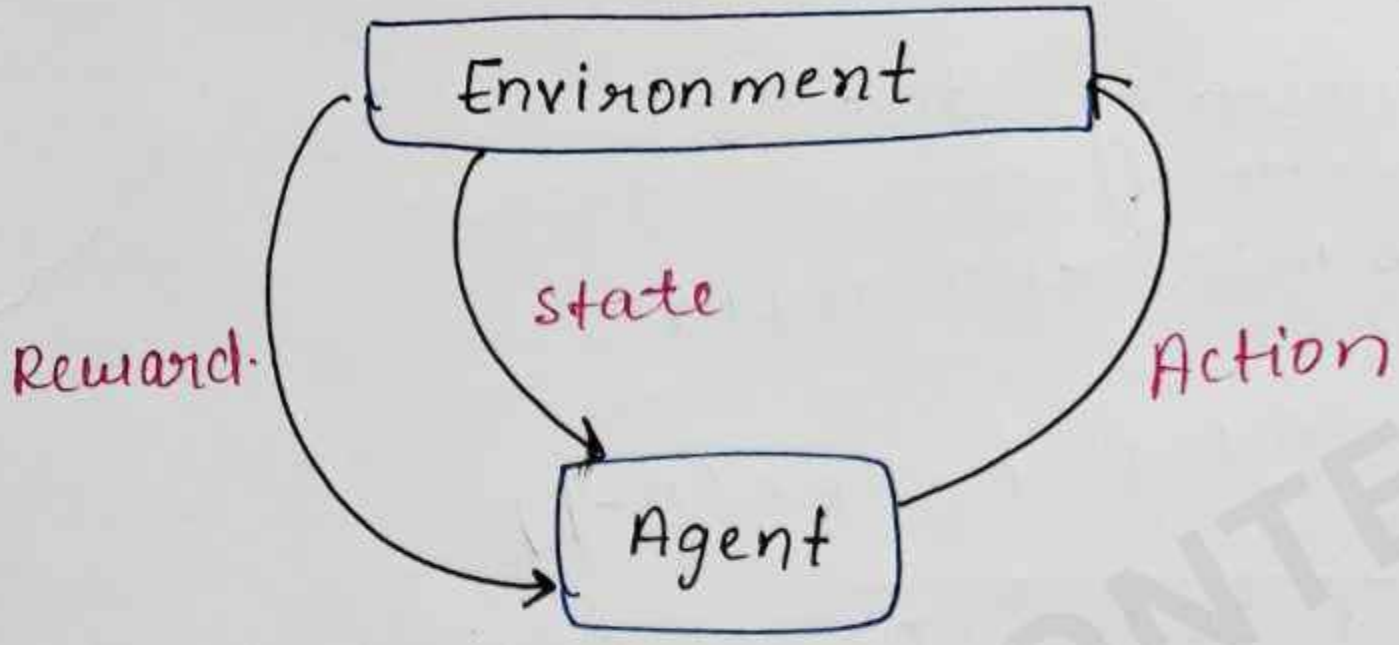
4. Reward Function (R) :- Reward received after performing an action

$R(s, a)$

5. Discount Factor (γ) :- Determines the Importance of future rewards

$0 \leq \gamma \leq 1$

- $\gamma = 0$ → only immediate reward matter
- γ close to 1 → future rewards are important.



Markov Decision Process

Importance of MDP

- Foundation of Reinforcement Learning.
- Models sequential decision-making
- Used in Robotics. game playing, recommendation systems. and self-driving cars!

Reinforcement Learning:-

Reinforcement learning is a type of machine learning in which an agent learns by interacting with an environment. The agent performs actions and receives rewards or punishment. and it learns to choose actions that maximize the total reward over time.

Basic component.

1] Agent → A learner or decision maker.

2] Environment → The place where the agent operates.

3] Action The choices made by the agent.

4] Reward Feedback given to the agent for its actions. (positive or negative)

5] State The current situation of the environment.

Working of Reinforcement Learning:

1. The agent observes the current state of the environment.
2. It choose an action
3. The environment responds with a reward or penalty.
4. The agents updates its strategy to maximize future rewards.

Example:

A robot learning to walk.

- If it moves correctly, it gets a positive reward
- If it falls, it get negative reward.

→ Over time, the robot learns the best way to walk.

Application

- Robotics
- Game playing (like chess or video games)
- Self - driving cars.
- Recommendation systems.

Advantages

- Learns from interaction with the environment.
- Can solve complex decision - making problems.
- Improves performance over time.

Disadvantages

- Requires a large amount of training time
- Designing the reward system can be difficult
- Sometimes it may learn inefficient behaviors

Bellman Equation.

The Bellman Equation is a fundamental equation in Reinforcement learning (RL) that expresses the value of a state as the immediate reward plus the discounted value of future ~~reward~~ states.

State Value function.

$$V(s) = R(s) + \gamma \sum_{s'} P(s'|s) V(s')$$

Where:

- $V(s)$:- Value of state s
- $R(s)$:- Immediate reward
- γ = Discount Factor (0-1)
- $P(s'|s)$ = Probability of moving to next state s'.

Meaning ⇒ Value of Current state = Immedi-ate Reward + Discount-ed future Reward.

The Bellman equation breaks a complex problem into smaller subproblems, making it easier to find the optimal policy.

Importance.

- Foundation of Reinforcement learning
- Used in Value Iteration and Policy Iteration.
- Helps an agent find the optimal strategy

Value Iteration & Policy Iteration

2024

Both are methods used to solve a Markov Decision Process (MDP) and find the optimal policy

1. Value Iteration

Value Iteration repeatedly updates the value of each state until the values converge

Steps

1. Initialize all state values.
2. Update values using the Bellman equation.
3. Repeat until values stop changing significantly
4. Derive the optimal policy from the final values

Advantages.

- Simple to implement
- Find the optimal policy

Disadvantages.

- May requires may iterations.

2. Policy Iteration

Policy Iteration directly improves a policy until it becomes optimal.

Two main Points

a) Policy Evaluation :- Calculate the value of states under the current policy.

b) Policy Improvement :- Choose better actions based on the computed values.

- Repeat these two steps until the policy no longer changes.

steps

1. Start with a random policy
2. Evaluate the policy
3. Improve the policy
4. Repeat until convergence.

Advantages

Usually converges in fewer iterations

Often faster than value iteration

Disadvantages.

- Each iteration can be computationally expensive

Difference

Value Iteration	Policy Iteration
1 Updates values directly	1 updates policy directly
2 One step (value update) repeated	2 Two steps :- evaluation + improvement
3 More Iteration	3 Fewer Iteration
4 Simpler	4 Slightly more complex

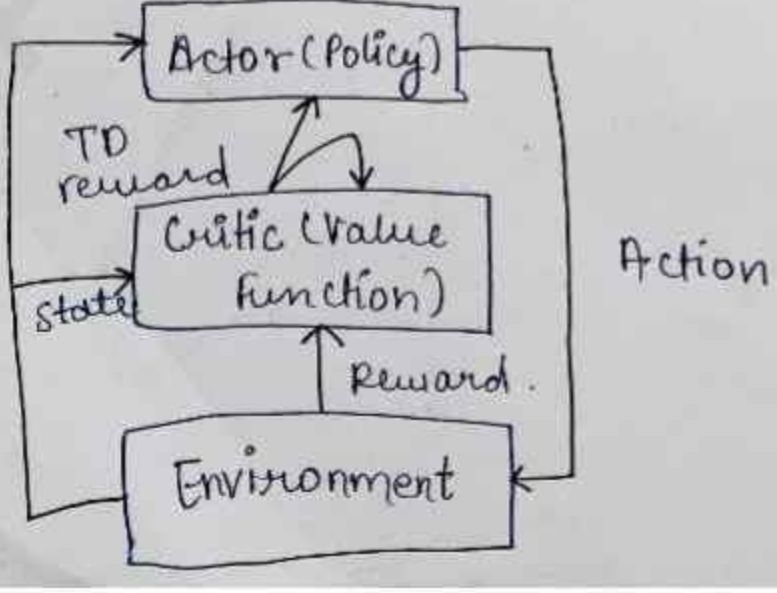
NOT FOR COMMERCIAL CONTENT

Actor - Critic Model 2024

A Actor - Critic Model is a Reinforcement Learning (RL) method that combines Policy - Based Learning and Value - Based learning

It has two component.

1. Actor :-
   - Decides which action to take
   - Learns the policy $\pi(a|s)$.
   - Interacts with the environment.
2. Critic :-
   - Evaluates the action chosen by the actor.
   - Estimates the value function $V(s)$.
   - Provides feedback to improve the actor.



Steps

1. Actor observes the current state
2. Actor selects an action
3. Environment returns reward and next state.
4. Critic evaluates how good the action was.
5. Critic sends feedback.
6. Actor updates its policy and improves future actions.

Example.

Imagine a robot learning to walk.

- Actor - Decides how to move its legs
- Critic - Checks whether the movement improved balance and speed.
- if good → positive feedback
- if bad → negative feedback

Over time, the robot learns to walk efficiently

Advantages

learns faster than many basic RL methods.

Handles large state spaces

combines strength of policy and value methods

suitable for continuous actions.

Application

- Robotics
- self driving cars
- Game Playing
- Resource allocation

Q - Learning

2025

Q- Learning is a model - free Reinforcement learning algorithm that learns the best action to take in each state by estimating Q- values (Quality Values).

A Q- value represents the expected future reward of taking an action in a parti - cular state.

Q - Value

Q (s, a).

where

- s = state
- a = Action
- Q(s,a) = Expected reward for taking action a in state s.

Q- Learning update Equation.

$$Q(s,a) = Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s,a)]$$

Where

- α (alpha) = Learning rate
- γ (gamma) = Discount factor
- r = Immediate reward
- s' = Next state
- $\max Q(s', a')$ = Maximum future Reward.

Working steps.

1. Observe current state s.
2. Choose an action a.
3. Receive reward(r) and next state (s').
4. Update the Q-value using the Q-learning equation.
5. Repeat until the optimal policy is learned.

Q-Table Example.

State	Left	Right
S_1	2	5
S_2	3	8

The agent chooses the action with the highest Q-Value.

Example.

In S1, choose Right because 5 > 2.

Advantages.

- Simple and easy to implement
- Does not require a model of the environment.
- Learns the optimal policy through experience.

Disadvantages.

- Large state spaces require large Q-tables.
- Learning can be slow for complex problems.

Application.

- Robotics
- Game Playing
- Path Finding
- Resource Management.

SARSA

(State - Action - Reward - State - Action)

SARSA is a model-free Reinforcement Learning algorithm that learns the action value function Q(s,a) based on the action actually taken by the agent

The name SARSA comes from:.

$$S \rightarrow A \rightarrow R \rightarrow S \rightarrow A$$

- State
- Action
- Reward
- Next state
- Next Action

working steps

1. Observe current state s.
2. Choose action a
3. Take action and receive reward r.
4. Observe next state s'
5. Choose next action a'
6. Update Q-value
7. Repeat

PERSONAL CONTENT

2024

Q Describe the significance of n-gram precision and brevity penalty in BLEU score calculation.

Ans → BLEU - (Bilingual Evaluation Understudy) is a metric used to evaluate the quality of machine-translated text by comparing it with one or more reference translations.

N-gram Precisions → N-gram precisions measures how many words or sequences of words (n-gram) in machine-generated translation match those in the reference translation.

- 1-gram (unigram) - Checks individual words
- 2-gram (Bigram) - Checks pair of consecutive words.

- 3-gram (Trigram) - Check sequences of three words
- 4-gram. Check sequences of four words.

Importance →

- Evaluates the accuracy of translated words and phrases.
- Higher-order n-grams capture fluency and correct word order.
- Prevents a system from getting a high score by simply using correct words in the wrong sequence.

Example: Reference :- The cat is on the mat

Candidate :- The cat is on mat.

- Many words match, so unigram precision is high, but some higher-order n-grams are missing.

Q **Brevity Penalty(BP)** :- A translation system could achieve high precision by producing very short translations containing only common matching words. To avoid this. BLEU uses a Brevity Penalty.

Formula

$$BP = \begin{cases} 1 & c > r \\ e^{(1-r/c)} & c \leq r \end{cases}$$

where c = length of candidate translation
r = length of reference translation

Importance

Penalize translations that are too short

Encourages complete translations

Balance precision with adequate translation length.

Significance in BLEU score

- N-gram precision measures translation accuracy and fluency.
- Brevity Penalty ensures the translation is not excessively short
- Together, they provide a balanced evalution of machine translation quality

NOT FOR COMMERCIAL CONTENT

MACHINE LEARNING

UNIT - 01

Introduction to Machine learning.

Machine learning is a branch of Artificial Intelligence (AI) that allows computers to learn from data and make decisions or predictions. without being explicitly programmed.

Instead of writing rules. we give data. and the computers learn pattern from it.

Machine learning is a subset of Artificial Intelligence.

It is useful when

- Data is large
- Rules are complex
- Patterns are difficult to write manually.

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Artificial Intelligence

Any technique that enables computers to mimic human intelligence, using logic, if-then rules, decisions trees and machine learning (including deep learning)

Machine learning

↑ Subset of AI that includes that include statistical techniques that enable machines to improve at tasks with experience. This include deep learning.

Deep learning

The subset of ML that allow software to train itself to perform tasks, like speech and image recognition, by exposing multilayered neural networks to large data.

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Important terms of Machine learning.

① Algorithm → A step-by-step method used by a computer to solve a problem or learn from data.

Example → Decision Tree, Linear Regression.

② Dataset → A collection of data used to train and test a machine learning model.

Example → Images, numbers

③ Training Data → The data used to ~~check~~ teach the model so it can learn patterns.

④ Test Data → Data used to check how well the model works after traning.

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Scope of Machine Learning.

The scope means the area where machine learning can be used.

① Healthcare → It helps doctors detect diseases, analyze medical images, and predict patient risks.

② Finance and Banking → Used for fraud detection, credit scoring, and stock market prediction.

③ E-commerce → Recommends products to users based on their interests (like online shopping recommendations).

④ Self-Driving Cars → It helps vechicles to recognize roads, objects and traffic signals.

⑤ Natural Language Processing → Used in chatbots, language translation, and voice assistants.

⑥ Education → Personalized learning systems and intelligent tutoring systems.

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Limitations of Machine Learning

Despite its advantages, Machine Learning also has some limitations.

① Needs Large Amount of Data → Machine learning models require a lot of data to learn properly.

② High Cost → Training machine learning models requires powerful computers and infrastructure.

③ → Time-consuming → Training and testing models can take a long time.

④. Data Quality Problem → If the training data is incorrect or biased, the model will give wrong results.

⑤ Security Risks → Machine learning systems can be attacked or manipulated with fake data.

⑥ Predictor variables → It is a feature(s) of the data that can be used to predict the output.

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Advantages / scope of ML.

① Automation of Tasks.

Machine learning can automatically perform tasks without human intervention, it reduces manual work and increases efficiency.

② Handles Large Data.

ML can analyze and process very large amounts of data quickly and accurately.

③ Improves with experience.

ML systems learn from data and improves their performance over time.

④ Accurate Predictions.

ML models can make accurate predictions and decisions based on patterns in data.

⑤ Wide Application.

Machine learning is used in many fields such as healthcare, finance, education, transportation, and e-commerce.

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Types of Machine Learning.

→ A machine can learn to solve a problem by any one of the following three approaches :-

These are the ways in which a machine can learn :-

- Supervised learning
- Unsupervised learning
- Reinforcement learning.

1 Supervised Learning :-

→ It is a type of machine learning in which the model is trained using labelled data.

→ This means the input data already has the correct output (target), and the algorithm learns the relationship between input and output.

→ The machine learns from examples with correct answers.

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Example → Suppose we want to predict wheather an email is spam or not spam.

Email features	Output
Contains many ads	spam
From known person	Not spam

The machine studies many such examples and learn to classify new emails.

How supervised Learning works.

1. Training data is given with input and correct output.
2. The algorithm learns the pattern between input and output.
3. The model is trained.
4. The trained model predicts the output for new data.

Types of Supervised Learning.

1 Classification → Output is a category or class.

- Example → Spam detection, disease detection.

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2→ Regression → Output is a continuous value.

Example → Predicting house price or temperature.

Application of supervised learning.

- Email spam detection
- Face Recognition
- Medical Diagnosis
- Stock price predictions.
- Image classification.

2. Unsupervised Learning.

Unsupervised learning is a type of machine learning in which the model is trained using data without labeled outputs.

The algorithm analyzes the data and finds hidden patterns or structures in the dataset automatically

Example → Suppose we have data of many customers but we do not know their categories. The algorithm studies the data and groups of customers with similar behavior together. This process is called clustering.

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Types of Unsupervised learning.

1 Clustering. - It groups similar data points together.

Example :- Customer segmentation

2. Association :. It defines and finds relationships between variables in the data.

Example → Market basket analysis (people who buy bread often buy butter).

Common Algorithm.

- K-means Clustering
- Hierarchical clustering
- DBSCAN
- Apriori Algorithm.

Application.

① Customer segmentation
② Recommendation systems.
③ Fraud detection.
④ Market analysis.

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Advantages.

1. No labeled data is required
2. It helps discover hidden patterns in data.
3. Useful for analyzing large datasets.

Disadvantages.

1. Results can be difficult to interrupt
2. Accuracy is harder to evaluate.
3. Sometimes it may finds irrelevant patterns.

③ Reinforcement Learning.

Reinforcement learning is a type of machine learning in which an agent learns by interacting with an environment. The agent performs actions and receives rewards or punishment. and it learns to choose actions that maximize the total reward over time.

Basic component.

1] Agent → A learner or decision maker.
2] Environment → The place where the agent operates.

COPY ALL CONTENT

3] Action The choices made by the agent.

4] Reward Feedback given to the agent for its actions. (positive or negative)

5] State The current situation of the environment..

Working of Reinforcement Learning.

1. The agent observes the current state of the environment.
2. It choose an action
3. The environment responds with a reward or penalty.
4. The agents updates its strategy to maximize future rewards.

Example.

A robot learning to walk.

- If it moves correctly, it gets a positive reward
- If it falls, it get negative reward.

→ Over time, the robot learns the best way to walk.

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Application

- Robotics
- Game playing (like chess or video games)
- Self - driving cars.
- Recommendation systems.

Advantages

- Learns from interaction with the environment.
- Can solve complex decision - making problems.
- Improves performance over time.

Disadvantages

- Requires a large amount of training time
- Designing the reward system can be difficult.
- Sometimes it may learn inefficient behaviors

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Regression

REGRESSION

Regression is a method in machine learning that finds the relationship between independent variables (inputs) and a dependent variable (output) to predict a continuous value.

Regression is used when the output is a number not a category.

Example.

Some common examples :-

- Predicting house price based on size and location.
- Predicting temperature for tomorrow.
- Predicting salary based on experience.
- Predicting sales of a product.

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How Regression works.

1. The model takes training data (input + output)
2. It finds the mathematical relationship
3. bitween them
4. It creates a best-fit line or curve
5. This line is used to predict new values

Example formula (Linear Regression).

$$y = mx + c$$

where

y = predict output
m = slope
x = input variables
c = intercept

Types of Regression.

1. Linear Regression
2. Polynomial Regression
3. Support vector Regression
4. Decision tree regression
5. Random forest regression.

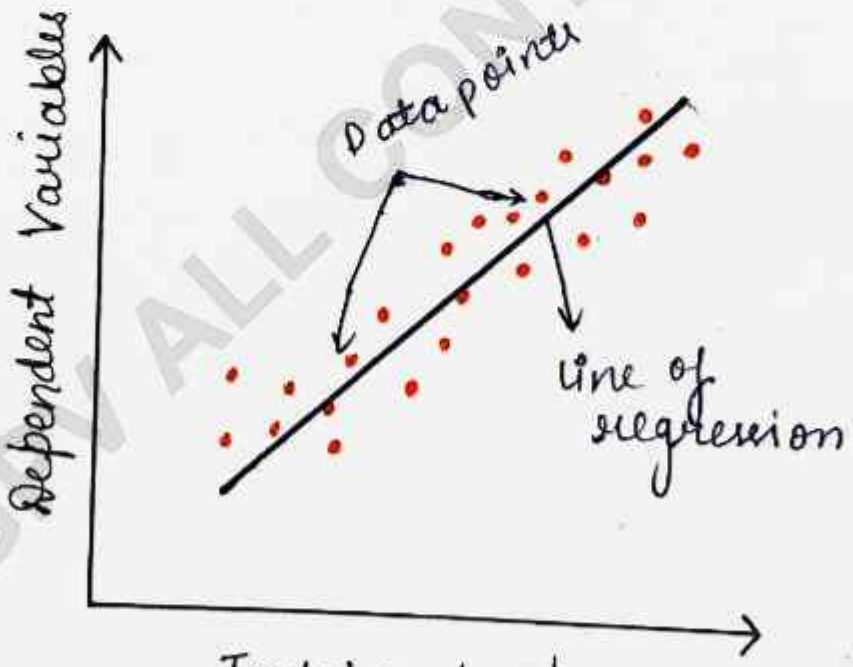
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1. LINEAR REGRESSION

Linear Regression is a regression algorithm used to model the linear relationship between independent variables (X) and a dependent variable Y

It predicts values using a straight line formula.

$$Y = mX + c$$



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Working

1. The algorithm takes training data.
2. It finds the best-fit straight line that minimize prediction error.
3. This line is used to predict new values.

Example.

Predicting house price based on area.

Area = 1000 sq ft

Price = ₹ 30 Lakh

Linear regression finds the relationship between area and price

POLYNOMIAL REGRESSION.

Polynomial Regression is a type of regression used when the relationship between (X) and output (Y) is not linear (not a straight line) but curved.

It is actually a extension of linear regression, but it models data using polynomial equation.

Formula → $Y = a + bx + cx^2 + dx^3 \ldots kx^n$

Y = predicted output

x = input variable

a, b, c, d = coefficients.

n = degree of polynomial.

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Example - (degree 2)

$$Y = a + bx + cx^2.$$

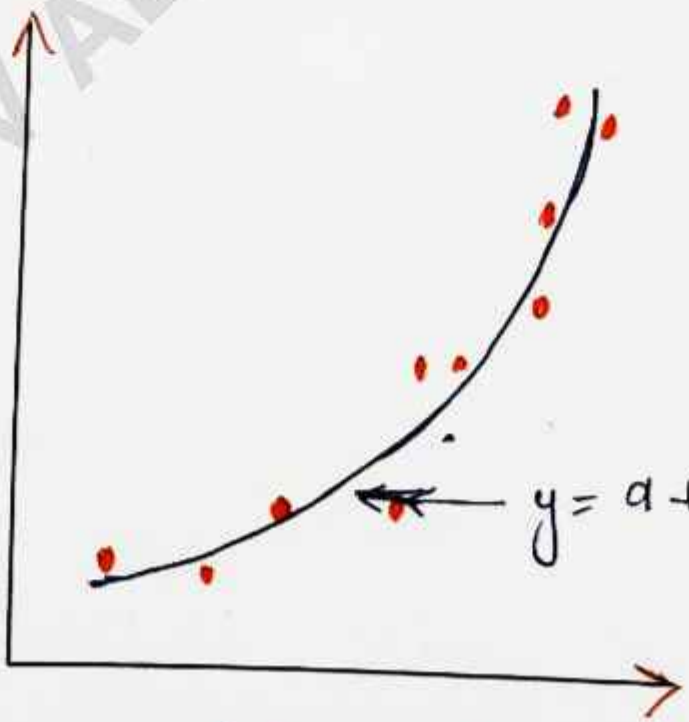
Use of Polynomial Regression.

Linear regression fails when data is curved.

Example :- Growth of plant over time
Population growth.
Temperature variation.

These follow a curve, not a straight line.

Graph →



Polynomial model.

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Working -

1. Take input data (X, Y)
2. Convert X into polynomial features
   $X \rightarrow X, X^2, X^3 \ldots$
3. Apply linear regression on these new features.
4. The model fits a curve instead of a straight line
5. Use the curve to predict new values.

Degree of Polynomial.

The degree (n) controls the shape of the curve

Degree 1 → Straight line (linear)

Degree 2 → U-shape curve (parabola)

Degree 3 → More complex curve.

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SUPPORT VECTOR REGRESSION.

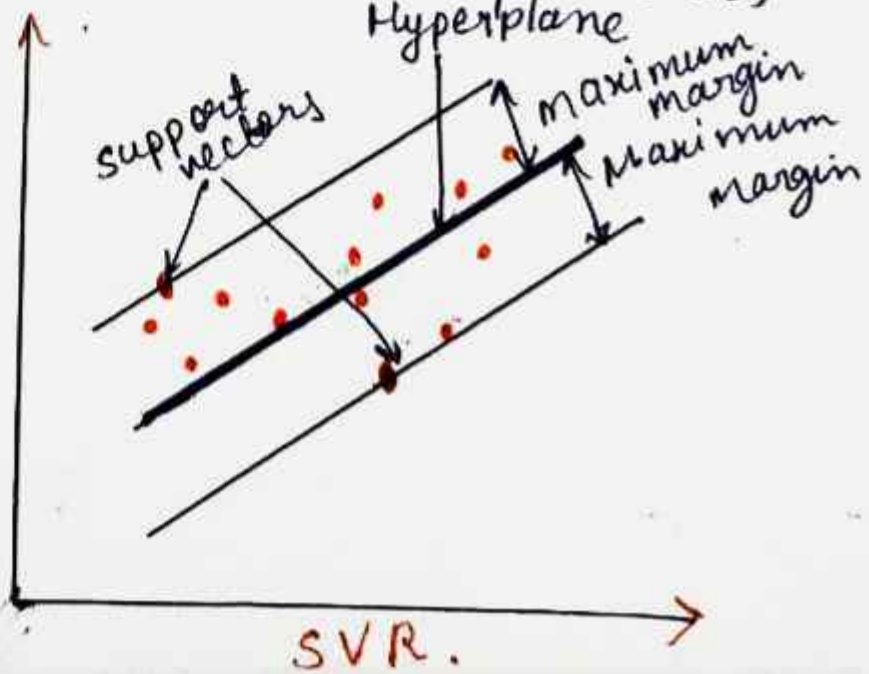
SVR is a regression algorithm based on support vector machines (SVM) that predicts continuous values by finding a function that fits the data within a fixed error margin (ε)

- Instead of minimizing total error, SVR tries to keep error within a limit.

SVR does not fit all data points exactly. It tries to:

- Fit a line (or curve)
- Keep most data points within a margin (ε)

→ This margin is called the epsilon (ε) tube.



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DECISION TREE REGRESSION.

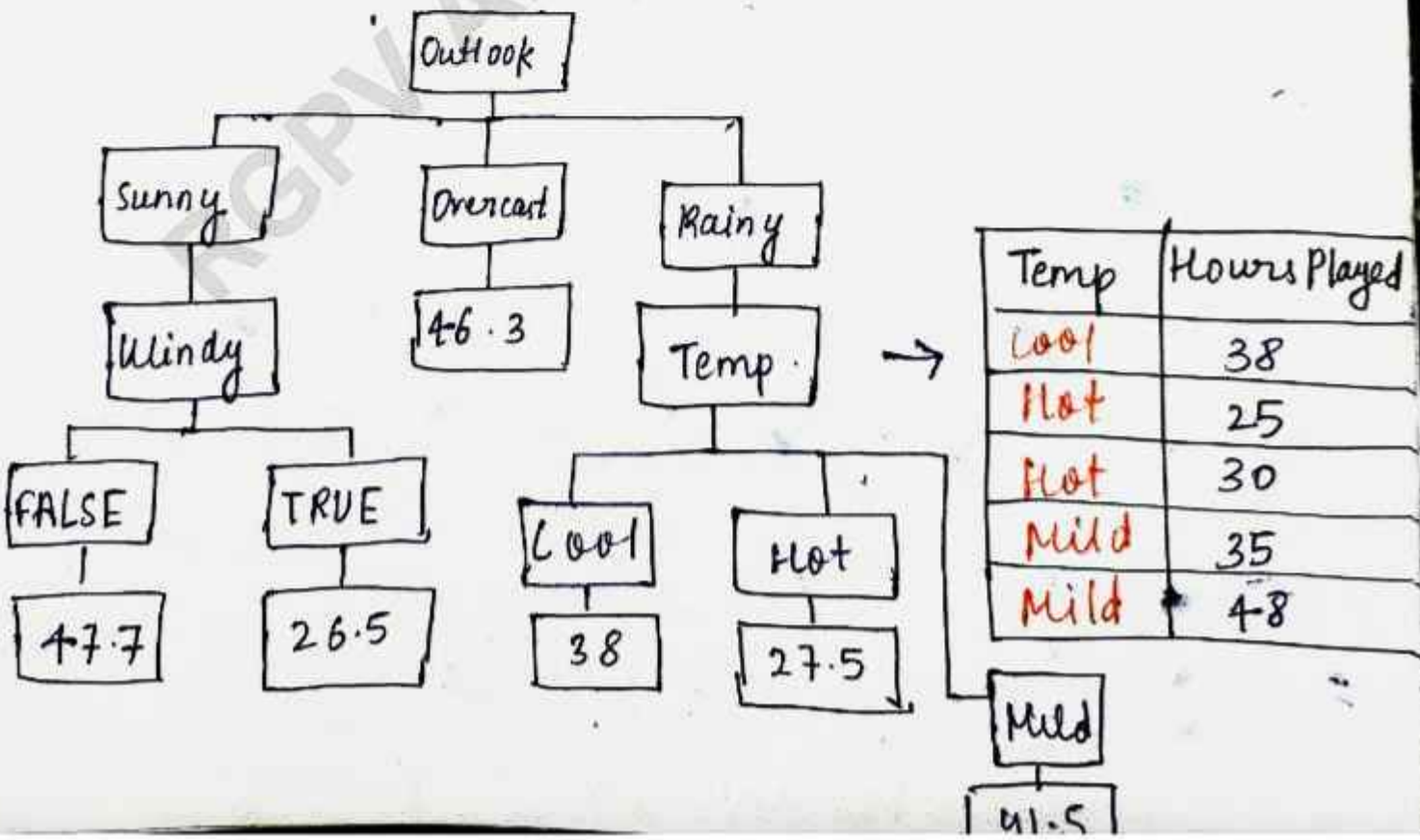
It is a machine learning algorithm that predicts continuous values by splitting the dataset into smaller parts using a tree like structure.

It creates decisions step-by-step and give a numerical output.

Instead of fitting a line (like linear regression)

- It Divides data into regions.
- Assign a constant value to each region.

So prediction is based on which region the input falls into.



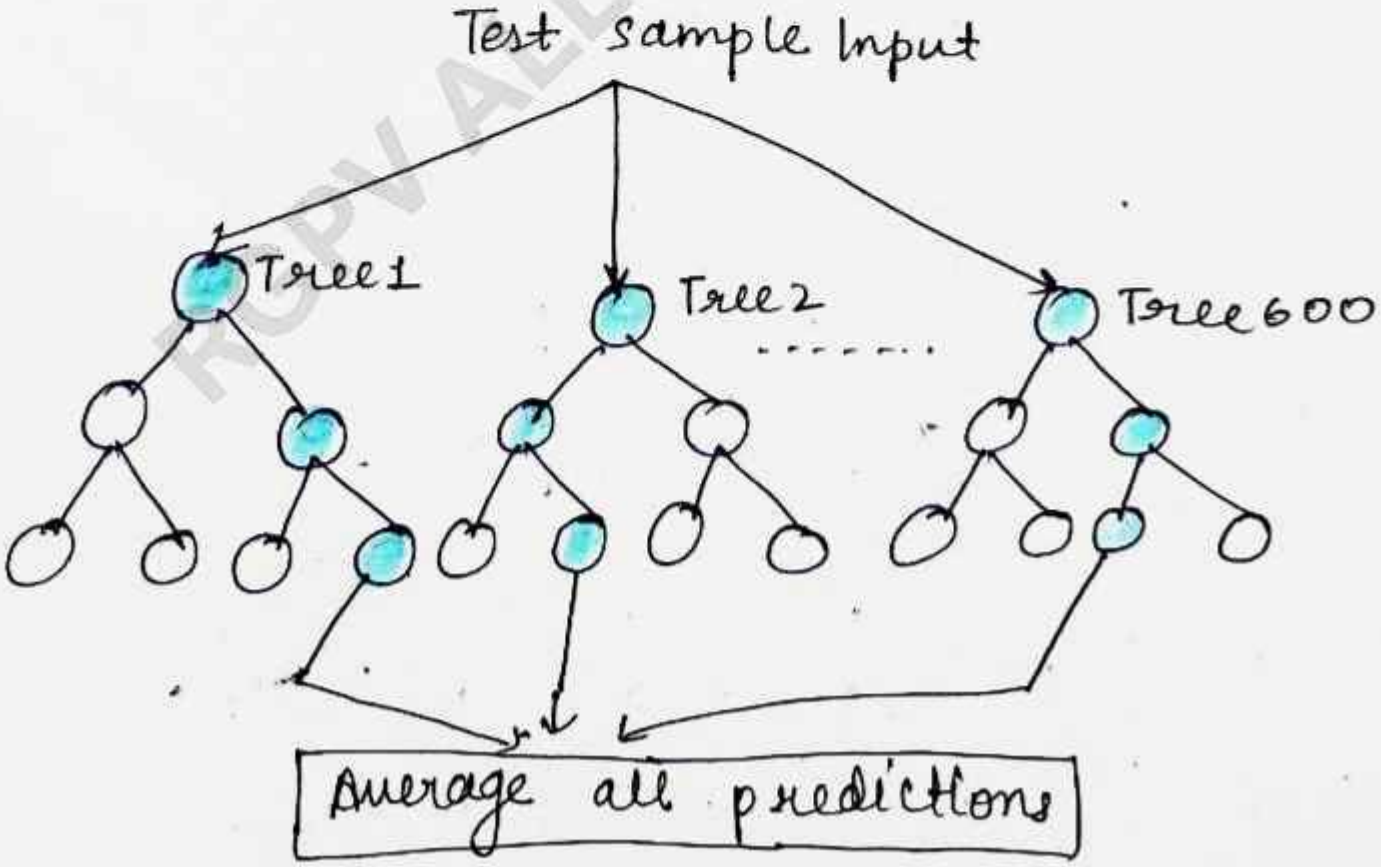
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RANDOM FOREST REGRESSION

Random forest Regression is an algorithm that predicts continuous values by combining the result of multiple decision trees.

- Instead of using one tree, it uses a forest (many trees).

- one decision tree ⇒ may overfit.
- Many trees together ⇒ better accuracy.
- final prediction ⇒ average of all tree predictions



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PROBABILITY

Probability is the mathematical foundation of Machine learning used to model uncertainty, randomness, and prediction confidence.

In real-world data, Data is noisy. Outcomes are uncertain and exact answer is not possible.

So, ML uses probability to say:
This prediction is 80% likely.

$$0 \leq P(E) \leq 1$$

0 → event will never happen
1 → event will definitely happen

Basic Probability Terminology.

① Random Experiment: An experiment whose outcome cannot be predicted exactly.

Example ① Tossing a coin
② Rolling a dice
③ Selecting a customer randomly.

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② Sample Space (S) :- The set of possible outcomes of an experiment

Example → Rolling a dice.

$S = \{1, 2, 3, 4, 5, 6\}$

③ Event (E) - A subset of a sample space.

Example → Even (A) → getting an even number.

$A = \{2, 4, 6\}$.

Common type of Probability.

① Classical Probability → It is used when all outcomes are equally likely.

$$P(E) = \frac{\text{favorable outcomes}}{\text{Total outcomes}}$$

Example → Probability of getting Head in a coin toss

$$P(H) = \frac{1}{2}$$

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② Empirical (experimental) Probability.
Based on past data and observation.

$$P(E) = \frac{\text{No. of time event occurs}}{\text{Total Trials}}$$

Example→ Outcome of 100 customers, 60 purchased.

$$P(\text{Purchase}) = \frac{60}{100} = 0.6$$

③ Subjective Probability.
Based on personal experience or judgement.

Example→ Probability of stock market rising tomorrow.

PROBABILITY DISTRIBUTION.

It describes how the values of a random variable are distributed along with their probabilities.

A distribution tells us which values occur, how often, and with what probability

Importance→
It help us to → ① Understand data behavior
② Identify patterns & trends
③ Make prediction and soon

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STATISTICS

Statistics is the foundation of Machine Learning because it helps us understand data, make predictions, and evaluate models.

Statistics is the science of collecting, analyzing, Interpreting, and presenting data.

In Machine Learning, statistics is used to

- Understand patterns in data
- Make Predictions
- Handle Uncertainty
- Evaluate model performance.

→ In simple words :-

ML = Data + Statistics + Algorithms.

Types of Statistics.

① Descriptive ~~data~~ Statistics.

② Inferential Statistics.

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(1) Descriptive Statistics.

Descriptive Statistics is the branch of statistics used to Organize data, Summarize data and describe the main features of data

- It does not make predictions.
- It only tells us what the data look likes.

Importance.

Before building any ML model. we must.

- Understand the dataset
- Detect patterns
- Find errors or outliners.

(2) Inferential Statistics:-

It is the branch of statistics that uses sample data to draw conclusions, make predictions, or take decisions about a population.

sample → subset of population, small & manageable, denoted by n.

main component → Estimation, Hypothesis Testing, Regression analysis, Analysis of Variance

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LINEAR ALGEBRA.

Linear Algebra is a branch of mathematics that deal with vectors, matrices and linear equations.

It is very important in fields like machine learning, data science, computer graphics and engineering.

☆ Linear algebra studies :-

Vectors → quantities with direction and magnitude.

Matrices → rectangular arrays of numbers

Linear equations → equation forming straight lines.

Example of linear equation:

$$2x + 3y = 5.$$

Example of linear Algebra in Machine Learning.

1. Dataset and Data Files.
2. Images and Photographs.
3. Linear Regression.
4. Principle component Analysis.
5. Deep learning.
6. Recommender system.

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CONVEX OPTIMIZATION.

Convex Optimization is a very important topic in machine learning and mathema-tics, especially for solving problem where we want the best (optimal) solution.

Convex optimization is a type of optimization problem where :-

① The objective function is convex

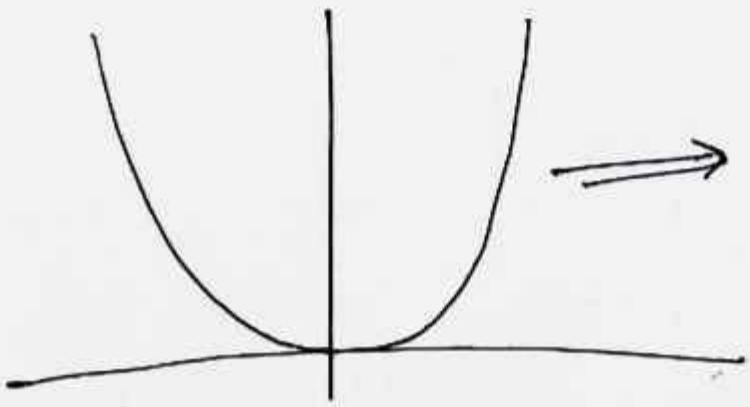
② The feasible region is also convex.

convex function → A function is convex if the line between any two points line above or on the curve.

Mathematical condition

$$f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta) f(x_2)$$

Example = $f(x) = x^2$



$\Longrightarrow y = x^2$

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DATA VISUALIZATION.

Data Visualization is the process of transfor-ming raw data into graphical representati-ons like charts, graphs, and dashboards to make information.

Easy to understand, Insightful and Actionable.

It is a key part of EDA (Exploratory Data Analysis) in Machine Learning.

Need of Data Visualization.

Data visualization is used when:

① Easy Understanding → Complex data becomes simple.

② Identify patterns → Trends and patterns are easy to see.

③ Better decision making → Helps in analyzing information quickly.

④ Saves times Visual data is faster to read than tables.

⑤ Detect errors - Outliners or unusual values become visible.

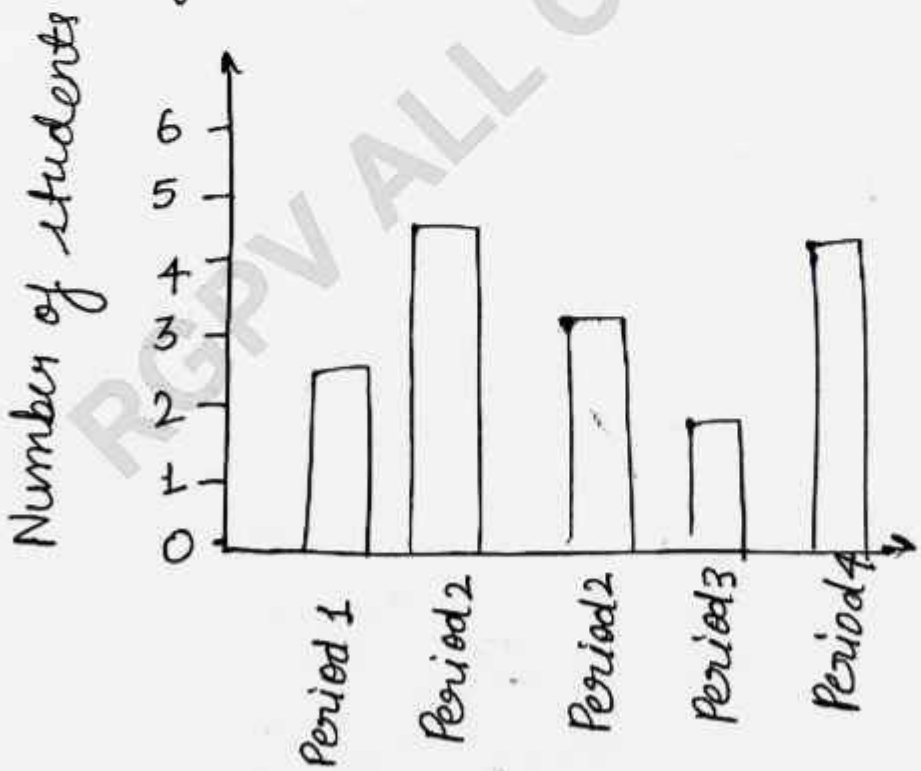
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Common types of Data Visualization

1. Bar Chart
2. Line Chart
3. Pie Chart
4. Histogram
5. Scatter plot.

BAR CHART

The Bar Chart is a graphical representation of data using rectangular bars where the length or height of each bar represents the value of the data.



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A bar chart has two axes :-

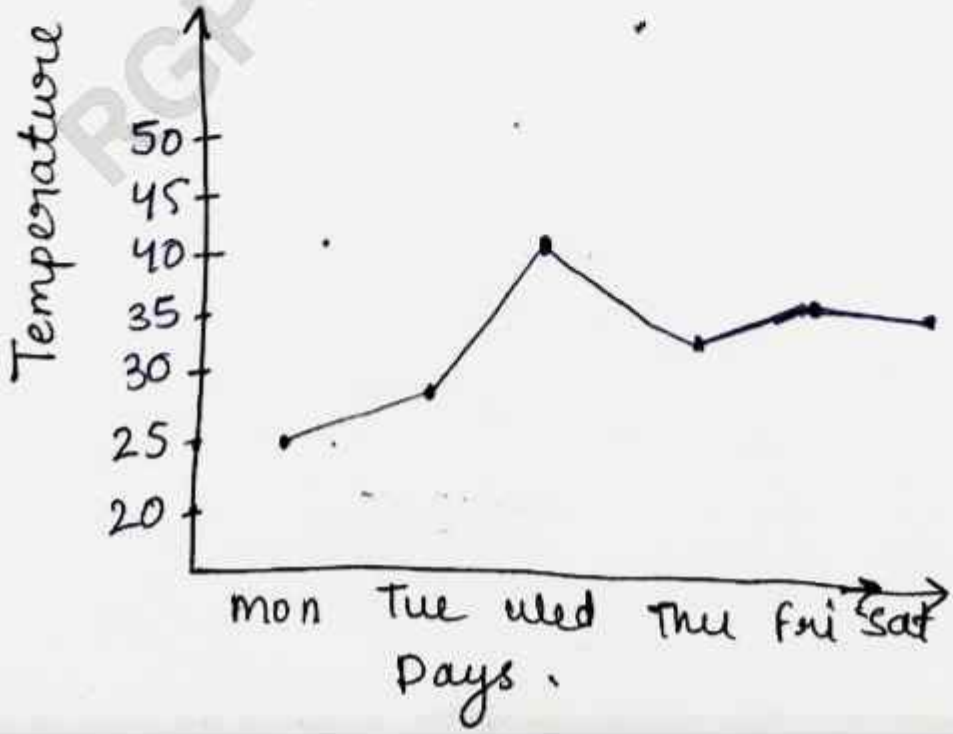
- X-axis (Horizontal axis) → shows categories (subject, products, years, etc).
- Y-axis (Vertical axis) → shows values or numbers

LINE CHART

A line chart is a type of graph that shows changes in data over time by connecting data points with a continuous line.

A line chart represents data using points connected by straight lines to show trends or changes.

Example → Temperature change during a week.



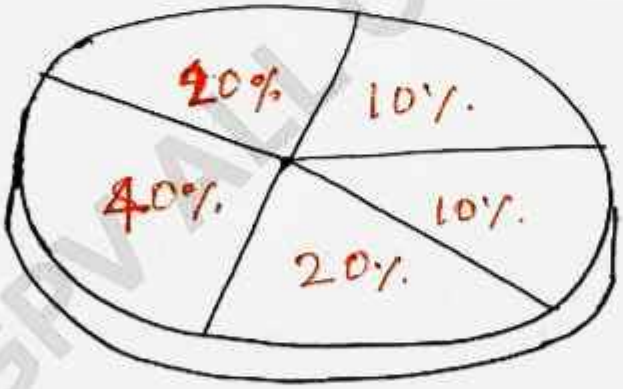
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PIE CHART

A Pie chart is a circular graph used to represent data as parts of a whole. The circle is divided into different slices. and each slice shows the percentage or proportion of a category.

A pie chart shows how a whole quantity is divided into different parts.

Example. Distribution of students ~~on~~ in different streams.



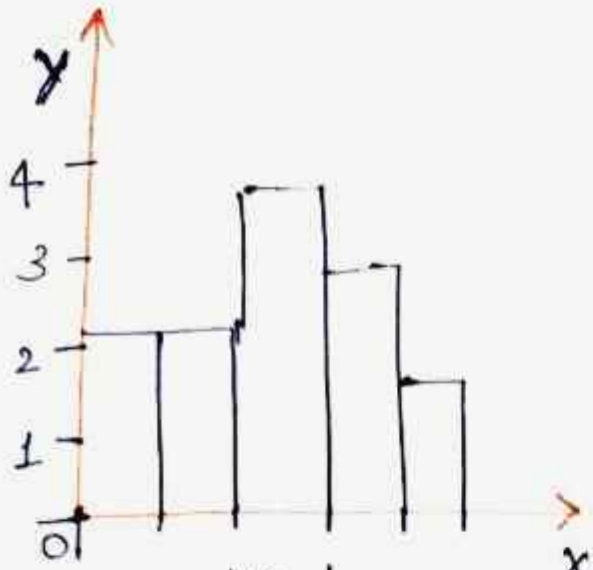
Percentage category.
40% CSE
10% IT
10% Civil
20% AI
20% ME.

HISTOGRAM

A histogram is a type of graph used to represent the frequency distribution of continuous numerical data using rectangular bars.

→ A histogram shows how often data values occur within certain ranges (intervals)

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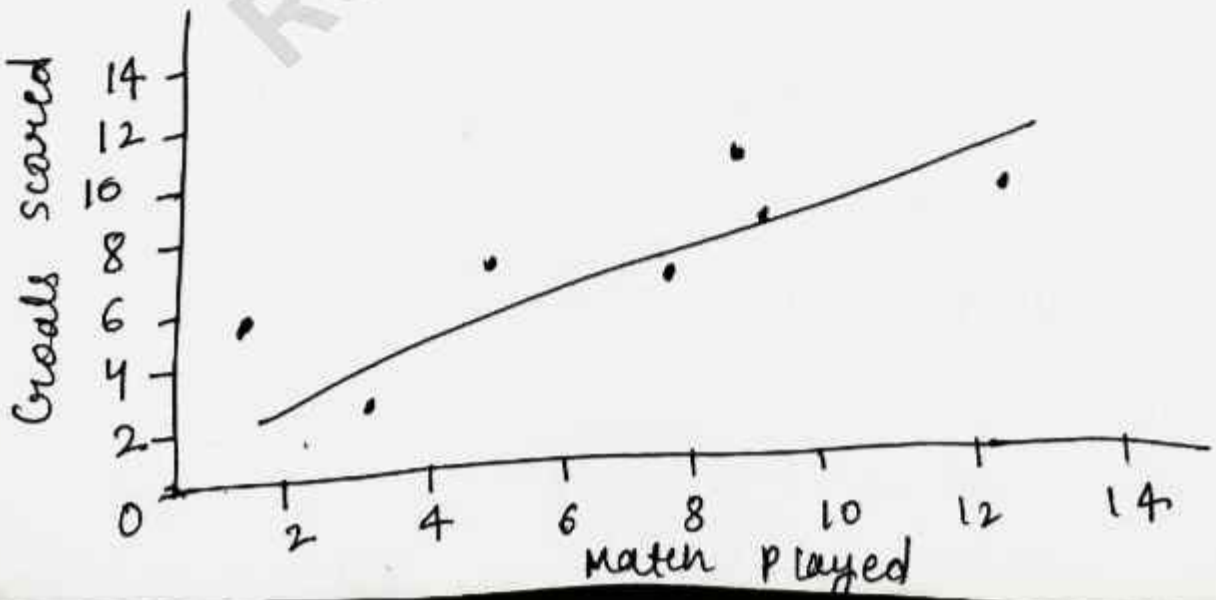


Histogram

SCATTER PLOT.

A scatter plot is a type of graph used to show the relationship between two variables by plotting data points on a coordinate plane.

A scatter plot displays data as points to show how two variables are related.



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HYPOTHESIS FUNCTION.

A Hypothesis function is a mathematical function used by a machine learning model to predict the output value from input data

Hypothesis function = a formula used by the model to make predictions.

It represents the relationship between input (features) and output (target values)

Example→ Predicting house price based on size:

Input → Size of house

Output → Price of house.

The model uses a hypothesis function to estimate the price.

Example of hypothesis function→

Linear Regression.

HYPOTHESIS TESTING

Hypothesis Testing is a statistical method used to decide whether a statement about data is true or false based on sample data

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Hypothesis testing checks if a claim about data is correct.

Types of Hypothesis

① **Null Hypothesis** → The default assumption that there is no effect or no difference

Example → H_0:- The new ML model does not improve accuracy.

② **Alternative Hypothesis** (H_1 or H_a)

The opposite statement that we want to prove.

Example: The new ML model does not improve accuracy.

Example in Machine Learning.

Suppose we build a new ML algorithm and want to test if it performs better.

- H_0 → New model accuracy = Old model accuracy.
- H_1 → New model accuracy > Old model accuracy.

After testing → If result is statistically significant → New model is better.

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DATA DISTRIBUTION

Data Distribution describes how data values are spread or arranged across different ranges or categories.

Data distribution shows how frequently different values appear in a dataset.

Example → Marks of students in a class may be distributed like :-

Marks Range	Number of students
0 - 20	2
21 - 40	5
41 - 60	10
61 - 80	8
81 - 100	3

This table shows how the marks are distributed among students.

Importance of Data Distribution.

Data distribution is important because it helps to :-

RGPV ALL CONTENT

- Understand the pattern ~~the~~ of data.
- Identify trends and variations.
- Detect outliners (unusual values)
- Choose the correct machine learning model.
- Perform statistical analysis

DATA PREPROCESSING

Data Preprocessing is the process of cleaning, transforming and organizing raw data before using it in a machine learning algorithms.

It means preparing raw data so that it can be used effectively for machine learning.

Example → If a dataset has missing values, duplicate records, or incorrect formats, preprocessing is used to fix these issues.

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Need for Data Preprocessing.

Data Preprocessing is important because real-world data may have.

1. Missing values
2. Noisy data (errors)
3. Duplicate data.
4. Duplicate Data formats.
5. Irrelevant information.

Without preprocessing, machine learning models may produce incorrect predictions.

DATA AUGMENTATION

Data augmentation is a technique used in machine learning and deep learning to increase the size of a dataset by creating new training data from existing data.

It is mainly used when available dataset is small.

It means generating additional data from existing data to improve model training.

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Example → If we have 100 images of cats, we can create more images by rotating, flipping and cropping.

NORMALIZATION OF DATA SETS.

Normalization is a data preprocessing technique used to rescale numerical features so they fall within a common range.

It converts different-scale data into a uniform scale without changing relationships between values.

The scale of normalization usually between 0 and 1.

Importance of Normalization –

In datasets, features can have very different ranges.

- Age → 18 to 60
- Salary → 10,000 to 1,00,000.

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- Machine learning models may give more importance to larger values (like salary).
- Normalization fixes this by bringing everything to the same scale.

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UNIT-2

MACHINE LEARNING

1. Linearity Vs non Linearity
2. Activation functions like sigmoid, ReLU.
3. Gradient Descent
4. Multilayer Network
5. Back Propagation.
6. Auto encoders
7. Batch Normalization.
8. L1 and L2 Regularization.
9. Tuning Hyper parameters.

Important Topics

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UNIT-2

Machine Learning.

Linearity Vs non linearity

Linearity A model is linear when the relationship between input and output is a straight line (or a linear combination of inputs)

Mathematical form -

$$y = w_1 x_1 + w_2 x_2 + \ldots + b$$

Main Points.

Output changes proportionally with input

It is easy to compute and interpret.

Decision boundary = straight line (2D) or plane (3D).

Example →

- Predicting salary based on experience.
- linear regression.

RGPV ALL CONTENT

Non-Linearity

A model is non-linear when the relationship is curved or complex, not a straight line.

Example → $y = x^2$
or
$y = \sin(x)$

Main Points.

- It captures complex patterns
- Decision boundary - curve or irregular shape.
- Needed for real-world problems like images, speech.

Importance of Non-Linearity.

If we only use linear models:-

Even multiples layers behaves like a single linear model

suppose → $y = w_2(w_1 x)$

This is still linear!

So, without non-linearity, deep learning is useless.

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Activation function.

Neuron → A neuron is the smallest unit of a neural network that.

Take input.
Processes it
Produces an output

It is inspired by human brain neuron

Activation function - An activation function is a mathematical function applied to neuron's output :-

$$y = f(Wx + b)$$

Y = Activation (Σ (weight * input) + bias).

Activation function decides, whether neuron should activated or not by calculating weighted sum and adding bias with it to introduce non-linearity in the output of neuron.

It used to determine the output of neural network like yes or no, It maps the resulting values in between 0 to 1 or -1 to 1 etc. (depending upon the function).

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Activation function are an extremely important feature of the artificial neural networks.

Without Activation Function

- Neural network becomes linear.
- Cannot solve complex problems.

With Activation function.

Introduces non-linearity.

And Enables:

- Image recognition
- Speech processing
- NLP tasks.

Important Types of Activation function

1. Sigmoid Activation function
2. ReLU (Rectified Linear Unit)

RGPV NOTES CONTENT

① Sigmoid.

The sigmoid activation function is a mathematical function that converts any real value into a value between 0 and 1.

It is also called the logistic function

It is easy to work with and has all the nice properties of activation functions, it's non-linear, continuously differentiable, monotonic, and has a fixed output range.

Mathematical formula.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

where

- x = input to the neuron
- e = Euler's constant

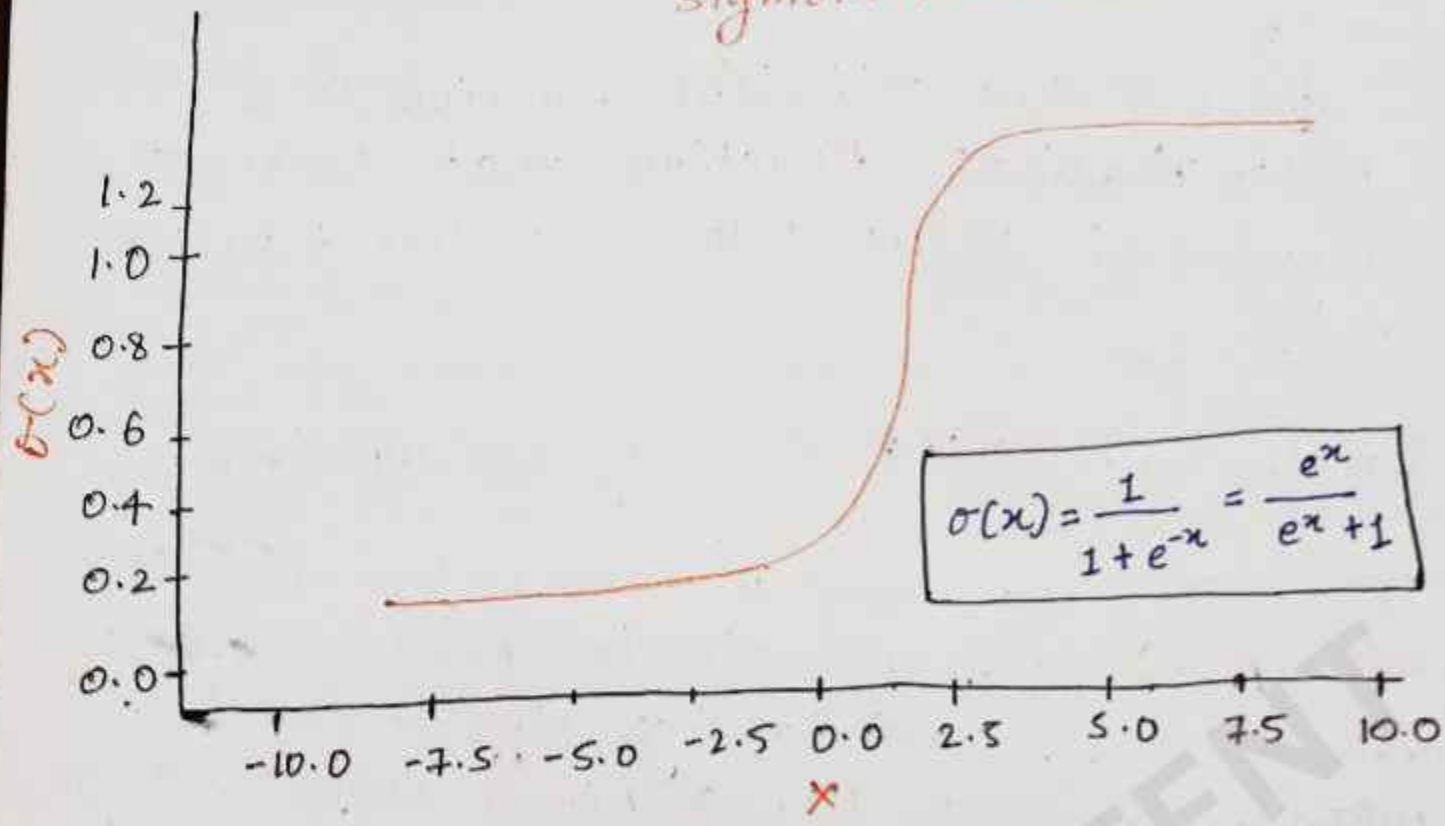
Output range →

$$0 < \sigma(x) < 1$$

- Output is always between 0 and 1
- Never exactly 0 or 1

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Sigmoid function



$$\sigma(x) = \frac{1}{1+e^{-x}} = \frac{e^x}{e^x+1}$$

Derivative

$$\sigma'(x) = \sigma(x)(1-\sigma(x))$$

Easy to compute
Used in backpropagation.

Advantages / Pros

1→ Produces values between 0 & 1, so it can be interpreted as probability. and useful in binary classification.

2- The function is continuous and smooth.

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3. Good in output layer especially in Yes / no problems and True / False predictions.

4. It is Monotonic Function means always increasing.
   - Higher Input → higher output.
   - no sudden jumps

Disadvantages / Cons.

(1) Vanishing Gradient Problem occurs means for large positive or negative values.
Gradient becomes very small (≃0).

(2) A extreme values:
   - Output becomes almost constant (0 or 1)
   - because of this model stops learning.

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(2) ReLU

- ReLU (Rectified Linear Unit) is a non-linear activation function
- It provides the same benefits as sigmoid but with better performance.
- It transforms input as:

$$f(x) = \max(0, x)$$

- If input is negative → output is 0.
- If input is positive output is same value

Importance.

→ Without activation

- Neural network becomes linear.

→ With ReLU.

- It introduces non-linearity.
- Helps model learn complex patterns.

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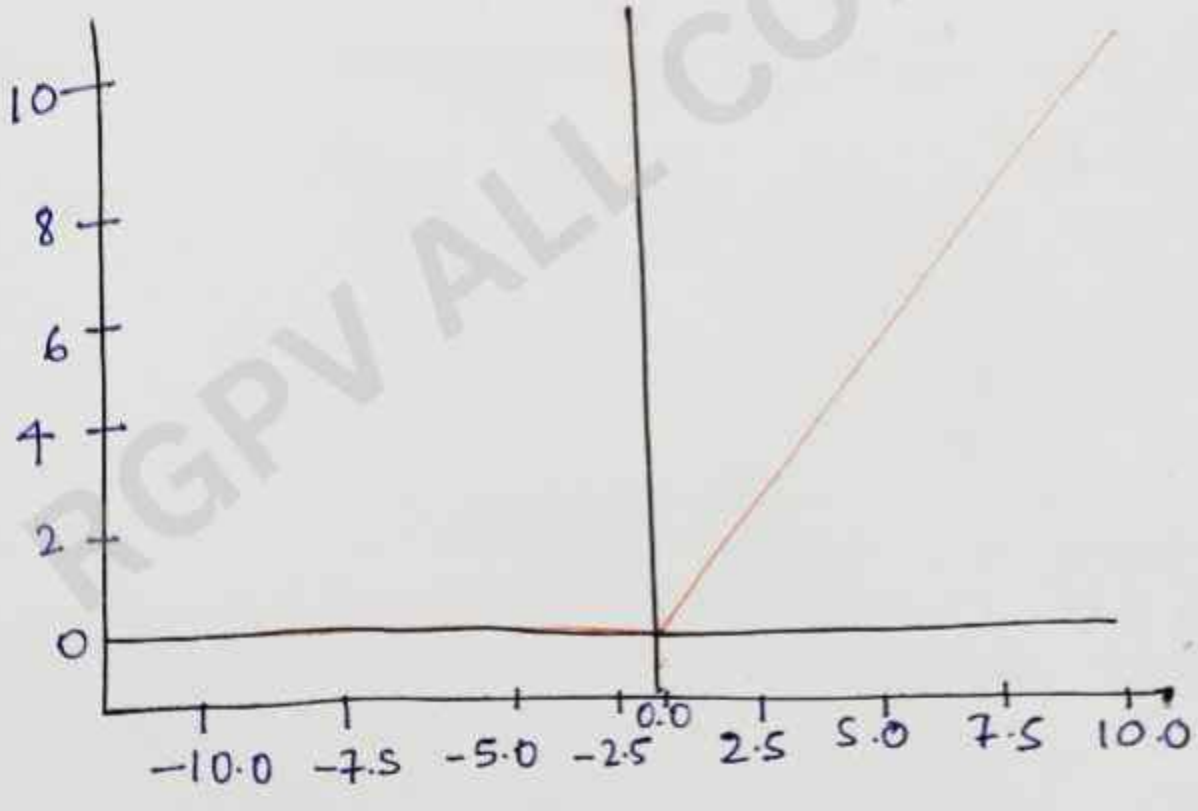
$$f(x) = \max(0, x).$$

where

- x is the input to the neuron.
- The function returns x if x is greater than 0.
- If x is less than or equal to 0, the function return 0.

The formula can also written as:

$$f(x) = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$



ReLU Activation function

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Parametric ReLU (PReLU)

→ Learns slope automatically for negative values.

ELU (Exponential Linear Unit.

→ Smooth version of ReLU.

where is ReLU used

→ Mostly in

- Hidden layers of neural networks
- Deep learning models.

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What are Weights and Bias in Neural Networks.

A neuron (like a small calculator) works like this :-

$$\text{Output} = (w \cdot x) + b$$

where.

x : input
w = weight
b = bias.

Weight

Weight is a numerical parameter in a neural network that determines the strength and importance of an input feature in producing the output.

weight tells - How much influence does the input have on the final result?

Representation

for single input neuron :-

$$y = w \cdot x$$

+ for multiple inputs.

$$y = w_1x_1 + w_2x_2 + w_3x_3 \ldots\ldots w_nx_n$$

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Where.

x_1, x_2, x_3 = inputs

w_1, w_2, w_3 = corresponding weights.

Each input has its own ~~input~~ weight.

Role of Weight

Weights play a very important role in neural network.

Feature Importance

High weight - input is very important

Low weight - input is less important.

Decision Making

- Output of neuron depends mainly on weights
- They help the model decide correct predictions.

Pattern learning

- Neural network learns pattern by adjusting weights.

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Real life example

→ Suppose we are predicting student performance.

Inputs:

Study hours = x_1

Sleep = x_2

Practice = x_3

Weights

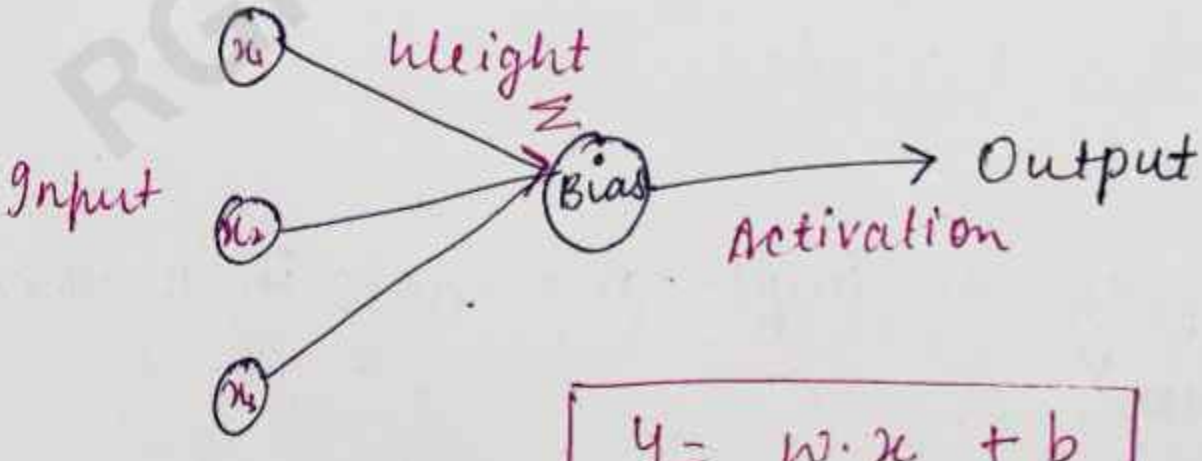
$w_1 = 3$

$w_2 = 1$

$w_3 = 2$

$y = (3 \times x_1) + (1 \times x_2) + (2 \times x_3)$

Study has heighest importance (weight = 3)



$$y = w \cdot x + b$$

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Bias

A bias is an additional parameter in a neural network that is added to the weighted sum of inputs to shift the output.

In simple words → Bias is a value that helps the model adjust the output independently of input.

Mathematical Representation -

For a neuron.

$$y = w \cdot x + b$$

for multiple inputs.

$$y = w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n + b$$

where b = bias.

Role of Bias

Bias play an important role in neural networks.

① shift the output

Bias move the output up and down.

REPRODUCTION ALL CONTENT

② Increases Flexibility. Helps model to fit data better.

③ Work without Input
Even if all inputs are zero, output can still be non-zero.

Real life example.

Suppose predicting salary:.

Inputs:.

Experience = x

weight = 5

Bias = 10 (base salary

$$y = 5x + 10$$

→ Even if experience is 0. salary = 10.

→ Bias = fixed starting value.

Bias in neural network.

- Every neuron has its own bias
- Bias is added after multiplying weights.

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Flow

① Multiply inputs with weights
② Add Bias
③ Apply activation function.

* Activation function → Applied to the output of a neuron ~~should be~~ ~~activated~~ to determined the final output.

It decides whether a neuron should be activated (ON) or not.

Loss FUNCTION

A function that measures the difference between actual output and predicted output.

Loss function tells how wrong the model's prediction is.

Why loss function is important.

Loss function is very important beause

① Measure Error - Tells how far prediction

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is from correct answer.

② Guides learning → Helps model improve by reducing error.

③ Used in optimization

- Works with Gradient Descent to update weights.

It tells the model.

- If prediction is close to actual value → Low Loss.

If prediction is far from actual value → High Loss

- The main goal of training is to minimize the loss.

Simple Example → Suppose actual house prince = ₹ 50 lakh

Model predicted = ₹ 45 lakh.

Then error → difference between actual and predicted.

Loss function calculate this error mathematically.

Common Loss functions.

1. Mean Squared Error (MSE).

This is used in Regression.

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$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

- squares the error
- Large errors get more penality

2. Mean Absolute Error (MAE)

used in Regression

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \tilde{y}_i|$$

- Uses absolute difference
- Less affected by outliners.

3. Binary Cross Entropy.

Used in Binary classification (Yes/No, Spam/Not spam).

$$L = -(y\log(p) + (1-y)\log(1-p))$$

Difference Between Lose and cost

- Loss = error fore one training example
- Cost = average lose for all examples

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GRADIENT DESCENT

Gradient Descent is an optimization algorithm used in Machine learning (ML) to find the best values of parameters (weights and biases) that minimize the error of a model.

- It helps the model learn from data step by step.

Simple Idea.

→ Imagine you are standing on a mountain and want to reach the lowest point.

you cannot see the whole mountain. so you:

1. Check the slop around you.
2. Move a little downward.
3. Repeat until you reach the bottom.

This is how Gradient Descent works in ML.

Main Purpose / Role in ML

Gradient Descent is used to minimize the cost function (or Loss function).

- High cost → model predictions are bad.
- Low cost → model predictions are good.

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formula

$$\theta = \theta - \alpha \frac{\partial J(\theta)}{\partial \theta}$$

where:

- θ = parameter (weight)
- α = learning rate
- $J(\theta)$ = cost function.

★ we subtract because we want to move toward the minimum error.

Steps of Gradient Descent.

1. Initialize Random weights
2. Calculate prediction
3. Find error using cost function.
4. Compute Gradient (slop)
5. Update weights.
6. Repeat until error become minimum.

PGV ALL CONTENT

Types of Gradient Descent.

1. Batch Gradient Descent.
2. Stochastic Gradient Descent.
3. Mini-Batch Gradient Descent.

(1) Batch Gradient Descent

In this complete training dataset is used to calculate the gradient of the cost function before updating the model parameters.

In this method. the algorithm processes all training examples together as a single batch and then performs one update of weights in each integration.

Working process.

- Take the entire training dataset
- Calculate predicted output for all examples.
- Compute total error (cost)
- Calculate gradient using all data.
- Update weight once
- Repeat until convergence.

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Characteristics

- Use full data set in every integration
- Produces smooth and stable convergence.
- Computationally expensive for large datasets.
- Require high memory.

Advantages

1. Accurate, gradient calculation.
2. Stable movement toward minimum
3. Easy mathematical analysis.

Disadvantages.

1. Very slow for large datasets.
2. High computations time
3. Not-suitable for real-time learning

PGRV ALL CONTENT

Example :- If a dataset contains 50,000 records. Batch Gradient Descent uses all 50,000 records before updating weights once.

2. Stochastic Gradient Descent (SGD)

:- It is a variation of gradient Descent where only one randomly selected training example is used to calculate the gradient and update the model parameters at a time.

- The word "stochastic" means random.
- Instead of waiting to process the entire dataset. SGD updates weights immediately after processing each training example.
- This makes learning faster but less stable.

Working Process.

1. Select one training examples.
2. Calculate predictions and error
3. Compute gradient
4. Update weights immediately.
5. Repeat for all examples.

RDPV ALL CONTENT

Characteristics

- Uses one data point at a time
- Fast updates
- Random fluctuations during learning.
- Less memory usage

Advantages

1. Faster training
2. Work well for huge datasets.
3. Can escape local minima
4. Suitable for online learning

Disadvantages.

1. Noisy updates
2. Convergence is less stable
3. Error graph fluctuate

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Example → If dataset has 50,000 records :-

- Weights update 50,000 times in one epoch.
- Each update uses only one record.

3. Mini - Batch Gradient Descent.

- Mini Batch Gradient Descent is a Hybrid approach that combines Batch Gradient Descent and Stochastic Gradient Descent.

- In this method, the training dataset is divided into small groups called mini-batches. and each mini - batch is used to update the weights.

- It balance the speed of SGD and the stability of Batch GD.

- This is the most commonly used optimizat-ion method in modern deep learning.

working process.

1. Diride data set into small batches
2. Select one mini - batch
3. Calculate gradient for that batch
4. Update weights
5. Repeat for remaining batch

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Characteristics.

- Uses small subset of data
  - faster computation
  - more stable than SGD
  - Efficient memory usage

Advantages

- fast and efficient.
- Stable convergence
- Suitable for GPU Processing
- Most practical method.

Disadvantages.

- Batch size selection is important.
- Improper batch size may reduce performance.

Example

→ If dataset has 50,000 records and batch size is 500:

- Data Divided into 100 mini-batches.
- Weights update 100 times in one epoch.

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MULTILAYER NETWORK

A Multilayer Network means a neural network that has more than one layer between input and output

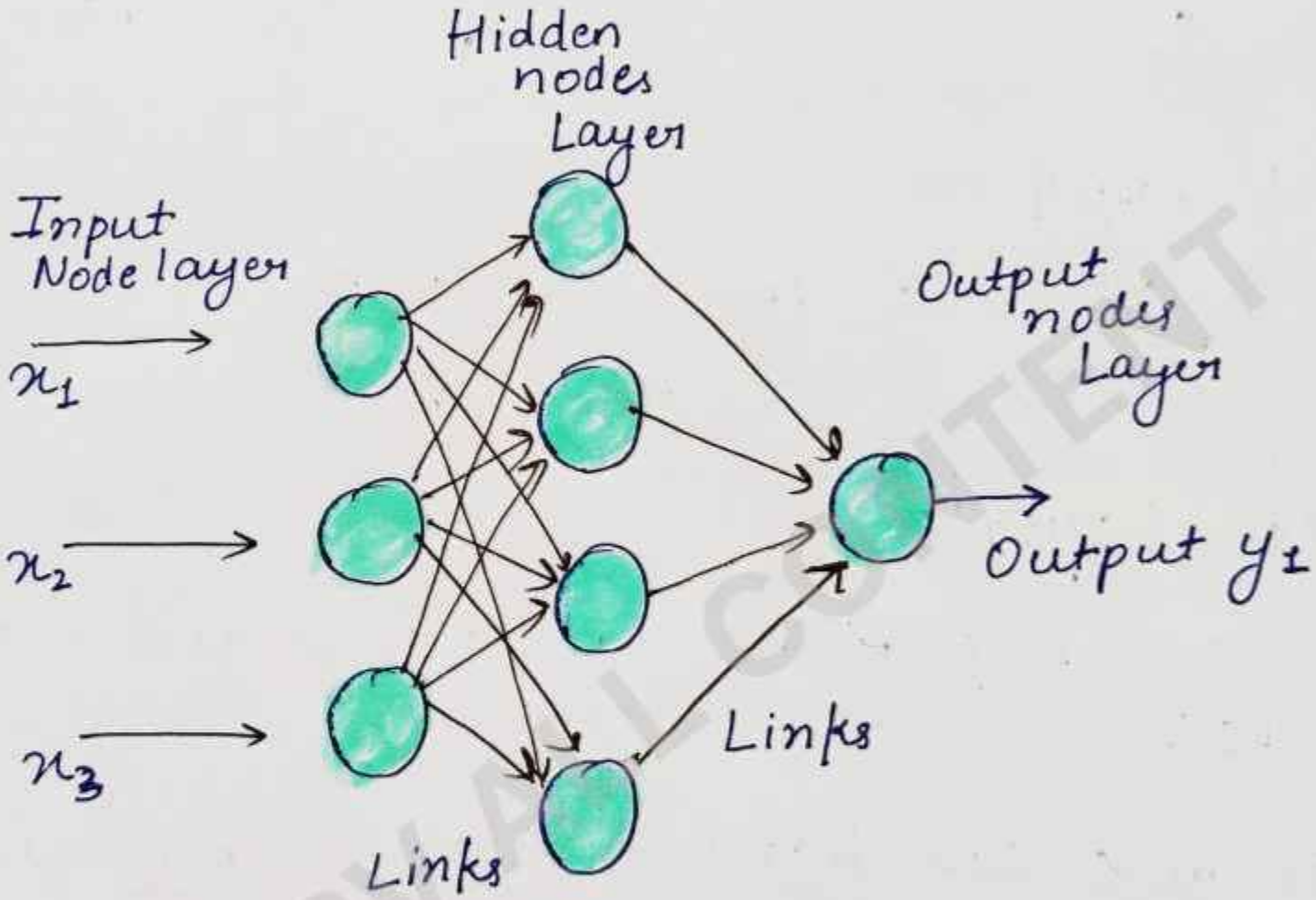
It is also called a Multilayer Neural Network or Multilayer Perceptron (MLP)

Structure of Multilayer Network

A multilayer network mainly has 3 types of layers:-

1. Input layer
   - Take input data from the user
   - Example:- image pixels, numbers, text etc.
2. Hidden Layer(s)
   - These layers perform calculations and learn patterns.
   - A network can have one or many hidden layers.
   - More hidden layers = deeper learning.
3. Output Layer:
   Gives the final result or prediction.
   Example:- . Cat or Dog classification..
   . Spam or Not Spam.

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Working of Multilayer Network.

Suppose we want to identify whenever an image is a cat or dog.

Step 1. Input Layer - The image data is given as input.

Step 2:- The Hidden layers:- detect edges, shapes, ears, eyes, patterns etc.

- Each neuron performs calculations using: weights, bias, activation function.

Step 3 Output Layer:- Finally, the network predicts: Cats or Dog.

Mathematical Idea

→ Each neuron calculates:-

$$y = f(\Sigma wx + b).$$

where:-

- x = input
- w = weight
- b = bias
- f = activation function
- y = output.

Why Multilayer Network are used.

Single-layer networks can solve only simple problems.

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- Multilayer networks can solve :-
- image recognition.
- speech recognition.
- language translation.
- medical diagnosis.
- Recommendation system etc.

Advantages

1. Learns complex patterns
2. High accuracy.
3. Works well for large datasets.
4. Used in Deep learning.

Disadvantages.

1. Requires more training time.
2. Need large data
3. Computationally expensive
4. Can overfit sometimes.

Real-Life Application.

- Face Recognition
- self-driving cars
- Voice assistants
- Handwriting recognition.
- Fraud detection.

FOR ALL CONTENT

BACK PROPAGATION

Back Propagation is a learning Algorithm used in neural networks to reduce error and improve accuracy.

It helps the network learn by calculating error, sending the error backward and updating weights.

It is mainly used in Multilayer Neural Networks.

- **Back** :- error moves backward.
- **Propagation** :- spreading information.

So Back Propagation means sending the output error backward through the network to improve learning.

Steps of Back Propagation.

① **Forward Propagation** → Input is passed through the network to get output

Example:-

Input → Hidden layer → Output.

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2. Calculate Error

~~to~~ compare : Actual Output and Predicted output.

Error formula :-

Error = Actual Output - Predicted Output.

3. Backward Propagation.

The error is sent backward from :-

- Output layer to
- Hidden layer.

The network checks :-

- which weights caused more error.

4. Update weight

Weight are adjusted to reduce error.

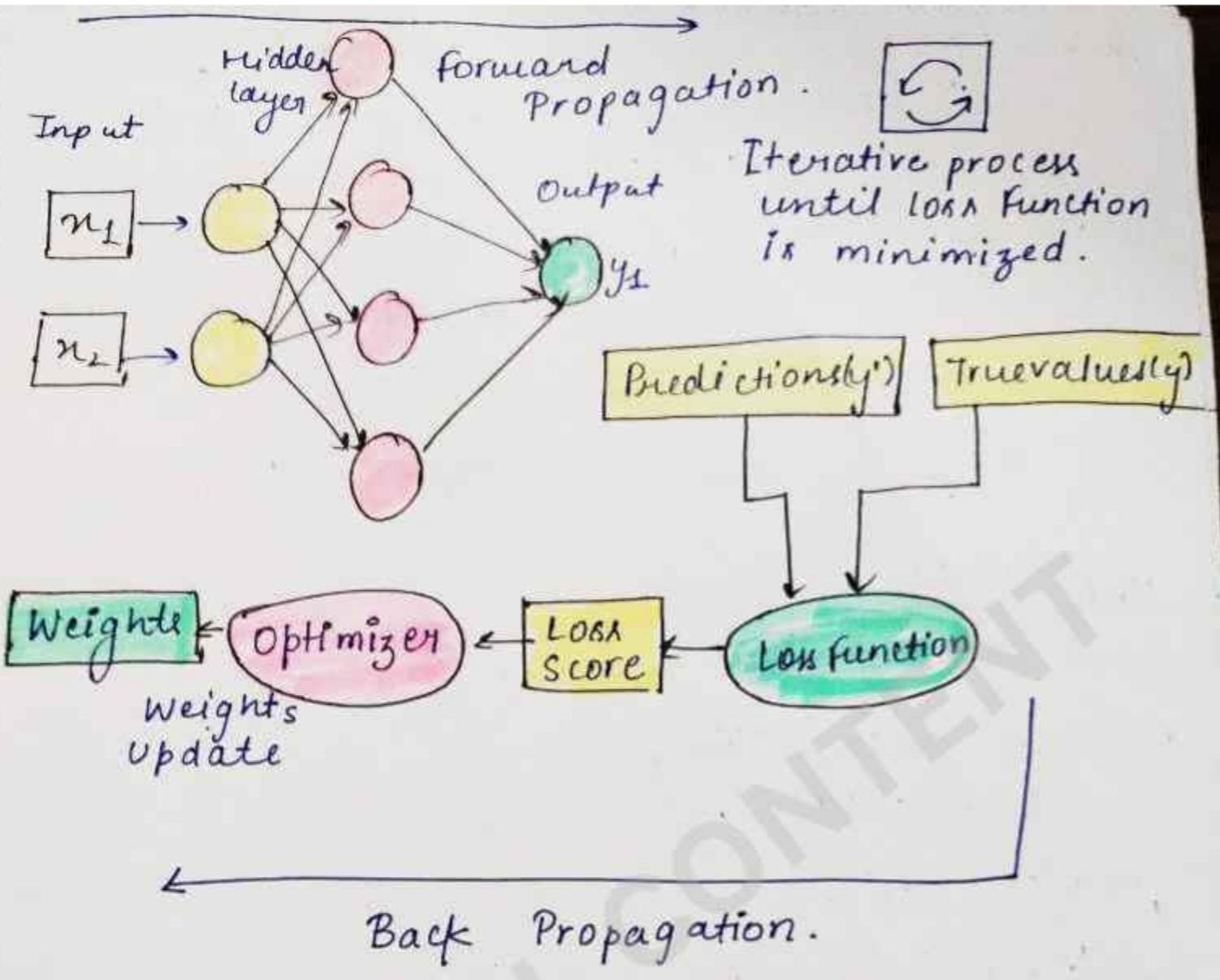
Weight update formula :-

$$W_{new} = W_{old} - \eta \frac{\partial E}{\partial W}$$

where :-

- W → weight.
- η → learning Rate.
- E → error.

SIGNAL CONTENT



Importance:-

- It helps the neural network:-
- learn from mistakes.
- Improve predictions.
- Increase accuracy.

★ without back propagation, deep learning would not work properly.

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Advantages

- Improve Accuracy
- Learn complex patterns
- Work well in deep learning.
- Reduce error automa-tically.

Disadvantages.

- Training can be slow ~~Learn complex patterns~~
- Needs large data.
- Require High computation power.

Applications.

Back Propagation is used in:-

- Image Recognition.
- Speech Recognition
- Medical dignosis
- Self-driving cars
- Chatbots

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AUTO ENCODER

- An Autoencoder is a type of neural network used to learn important features of data, Compress Data and Reconstruct the original data.
- It works automatically, so it is called Auto Encoders.

Main Idea

An autoencoder tries to:
Convert input data into a smaller form and then rebuild the original data from it.

Example
- Compressing an image.
- Removing noise from Photos.
- Features extraction.

Structure of Autoencoder.

An autoencoder has 3 main parts:-

1. Encoder
   - Compresses input data into smaller representation.
   - Extract important features.

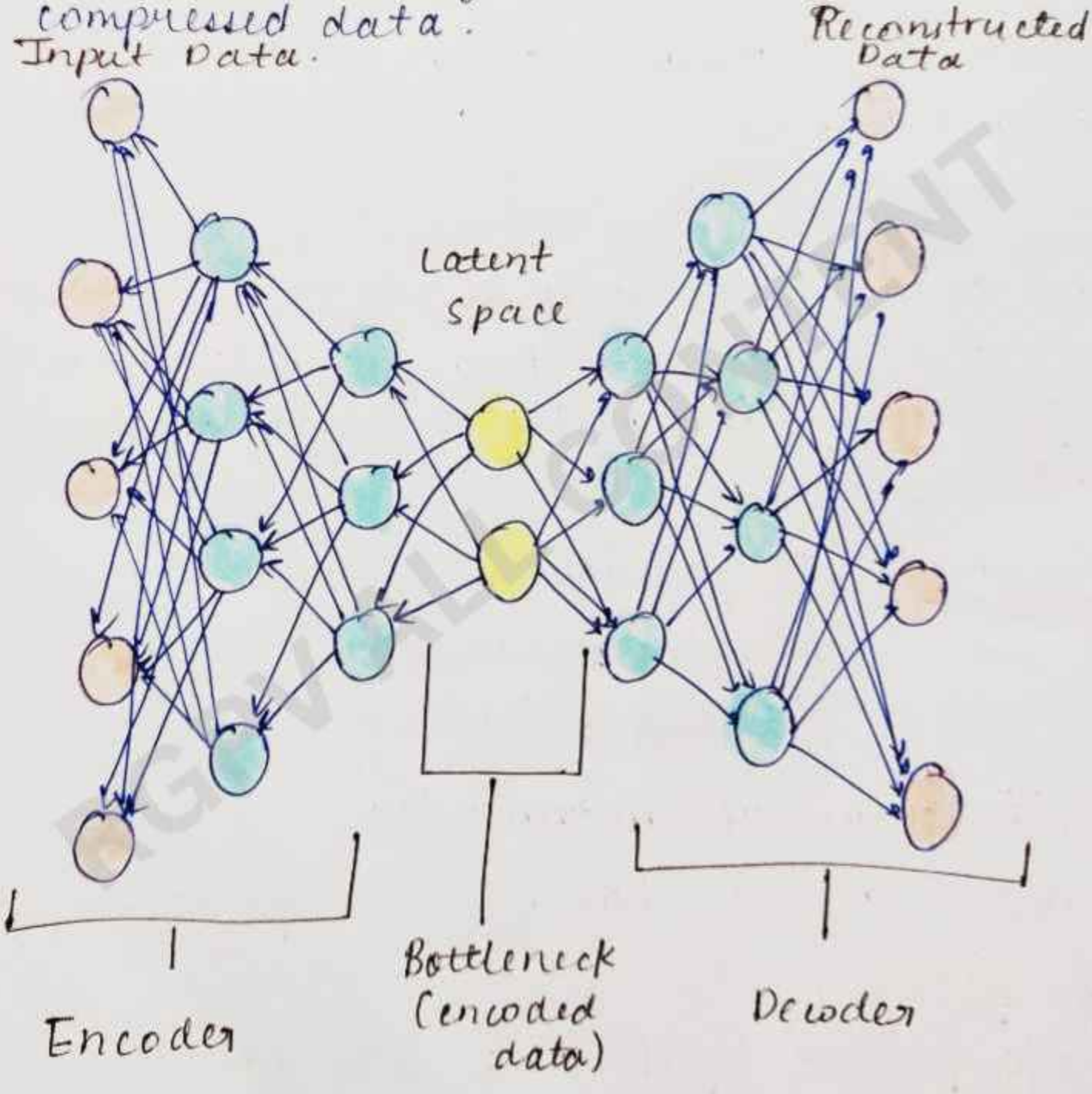
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2. Bottleneck (Latent space)

- Small compressed form of data.
- Stores only important information.

3. Decoder

- Reconstructs original data from compressed data.



Working of Autoencoder.

Suppose an image is given as input.

step 1 Encoder - The encoder compress the image.

- Large image → smaller feature representation.

Step 2 Bottleneck - Only important information is stored.

step 3 Decoder - The Decoder rebuilds the image from compressed data.

step 4 Compare Output - The reconstructed image is compared with original image.

The network learns by reducing reconstruction error.

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Types of Auto encoders

(1) Convolution Auto encoders
(2) Sparse Auto encoders
(3) Deep Auto encoders

(1) Convolution Auto Encoders

It uses Convolutional Neural Network (CNN) layers for learning image features.

A convolutional autoencoder:

★ Compresses the image using convolutional layers.
★ Learns important image features.
★ Reconstructs the original image.

→ It is mainly used for:-

- Image compression.
- Noise removal
- Feature extraction
- Image reconstruction

It has two main parts

(1) Encoder - The encoder:-

- uses convolution and pooling layers
- reduces image size.
- extracts important features.

Example → Image → feature Maps → Compressed Representation.

② Decoder :- The decoder :

- Reconstructs the image.
- uses deconvolution / upsampling layers.
- produces output similar to original image.

② Sparse Autoencoder :-

- In this only a few neurons are active at one time.
- It learns only important features of data
- Even if many neurons exist, the network forces most neurons to remain inactive. This creates efficient learning.

Features

- learn useful patterns.
- Reduces unnecessary learning.
- Improves feature extraction.

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(3). Deep Autoencoder

A Deep Autoencoder is an encoder with many hidden layers.

It is used for learning complex patterns in large datasets

features.

- It learns deep features
- Provide Better compression of data.
- It also Handles complex data.

Advantages of Autoencoder.

- Reduces data size
- Learns features automatically.
- Remove noise
- Useful for dimensionality reduction.

Disadvantages of Autoencoder.

- Training can be slow
- Needs large data.
- Sometimes reconstructed output is blurry.

REPLY ALL CONTENT

BATCH NORMALIZATION

Batch Normalization is a technique in ML used to make training neural networks faster, more stable, and more reliable.

Batch normalization (often called BatchNorm) normalizes the inputs of each layer so that they have.

- mean ≈ 0
- variance ≈ 1

This is done for each mini-batch during training

Need of Batch Normalization.

During data traing, the distribution of activations changes as weights updated - this is called internal covariate shift.

BatchNorm helps by :-

- stabilizing these distributions
- allowing higher learning rates
- reducing sensitivity to initialization
- acting as a mild regularizer.

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How it Works:

for each feature in a mini-batch:-

1. Compute mean

$$\mu_B = \frac{1}{m} \sum_{i=1}^{m} x_i$$

2. Compute Variance.

$$\sigma_B^2 = \frac{1}{m} \sum_{i=1}^{m} (x_i - \mu_B)^2$$

3. Normalize

$$\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$$

4. Scale and shift (learnable parameters):

$$y_i = \gamma \hat{x}_i + \beta$$

- γ (gamma) = scale
- β (beta) = shift
- ϵ (epsilon) = small constant for numerical stability.

Where it is used.

BatchNorm is commonly applied:

- after linear/dense layers.
- after convolution layers (in CNNs)
- before activation functions (most common practice)

PRIVATE CONTENT

Training Vs Inference.

- Training - uses batch statistics (mean & variance of current mini-batch)
- Inference - uses running averages collected during training

Benefits / Advantages.

- Faster convergence
- Most stable gradients
- Less overfitting (slight regularization effect)
- Enable deeper networks.

Limitations / Disadvantages.

- Depends on batch size (small batches - noisy estimates)
- Less effective in some architecture. (e.g. RNNs).
- Alternative exist like
  - Layer Normalization
  - Group Normalization

ROYAL CONTENT

DROPOUT

Dropout is a regularization technique. used to reduce overfitting in neural networks.

It works by randomly turning off some neurons during training.

During training, dropout randomly removes some neurons from the network temporarily.

These neurons

- do not participate in Forward propagation
- do not participate in Back propagation

→ This forces the network to learn more about features

Need of Dropout

Without dropout

- neural networks may memorize training data.
- model become overfitted
- poor performance on new idea.

Dropout helps the model generalize better.

PGP ALL CONTENT

L1 and L2 Regularization.

Regularization:- It is a technique in ML and Deep learning used to reduce overfitting

Overfitting happens when a model learns the training data too well and perform poorly on new data.

Regularization controls the complexity of the model by adding a penality to the loss function.

Need of Regularization.

→ Suppose a model has very large weights

→ Large weights make the model
- too sensitive to small changes.
- memorize training data.
- perform poorly on test data.

→ Regularization keeps weights smaller and simpler

BCA ALL CONTENT

General formula.

The new loss function becomes.

Total loss = Original Loss + Regularization Term

The regularization term penalizes large weights.

L1 Regularization.

L1 Regularization adds the sum of absolute value of weights to the loss function.

$$\text{Loss} = \text{Original Loss} + \lambda \sum_{i=1}^{n} |w_i|$$

where

w_i = weights.

λ = regularization strength

Large λ → Stronger penality

How L1 Works

L1 pushes some weights exactly to 0.

That means.

- unimportant features are removed
- only important features remain

so L1 performs automatic feature selection.

NOT FOR ALL CONTENT

Example of L1

Suppose weights are.

- [5, 0.2, 0.01, 7]

→ After L1 regularization:

- [4.5, 0, 0.6]

→ some weight become zero.

Properties of L1

1. Sparse model → many weight become zero.
2. Feature selection → Unnecessary features are removed automatically.
3. Simpler model → Model becomes easier to understand.

Advantages

- Reduces overfitting
- Removes useless features
- Good for high-dimensional data
- Creates compact models

AGRVAL CONTENT

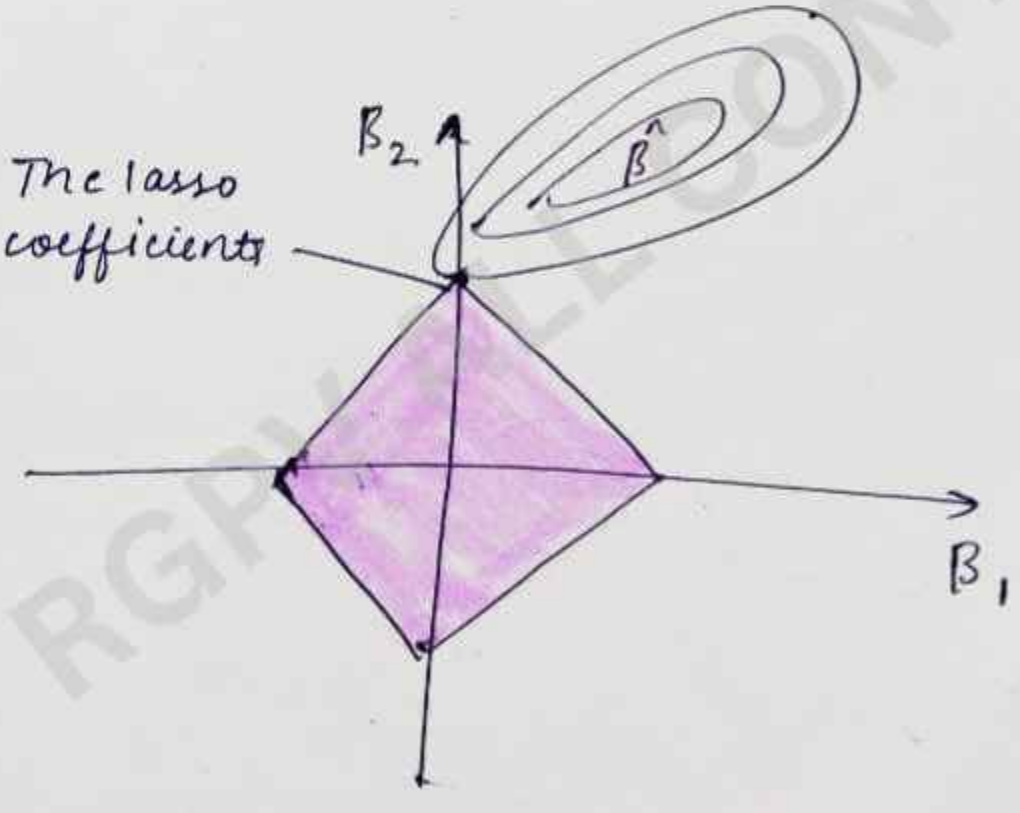
Disadvantages.

- Training may become unstable
- Problem with highly correlated features
- Can remove useful features accidentally.

Lasso Regression.

L1 regularization in linear regression is called lasso Regression.

Lasso means → Least Absolute Shrinkage and Selection operator.



LASSO

RGHT CONTENT

L2 Regularization.

L2 regularization adds the ~~square of~~ sum of squared weights to the loss function.

formula.

$$\text{Loss} = \text{Original loss} + \lambda \sum_{i=1}^{n} w_i^2$$

How L2 Works

L2 reduces weights gradually.

Weights become:-

- small
- close to zero.

But usually not exactly zero.

Example :- Original weights

[5, 0.2, 0.01, 7]

After L2:

[4.2, 0.15, 0.008, 5.8]

All weights shrink smoothly

Properties of L2

1. Weight Shrinkage → Large weights are reduced.

RGPV ALL CONTENT

2. Smooth learning - Training becomes more stable.

3. No feature Removal - Features stay in the model.

Advantages of L2:-

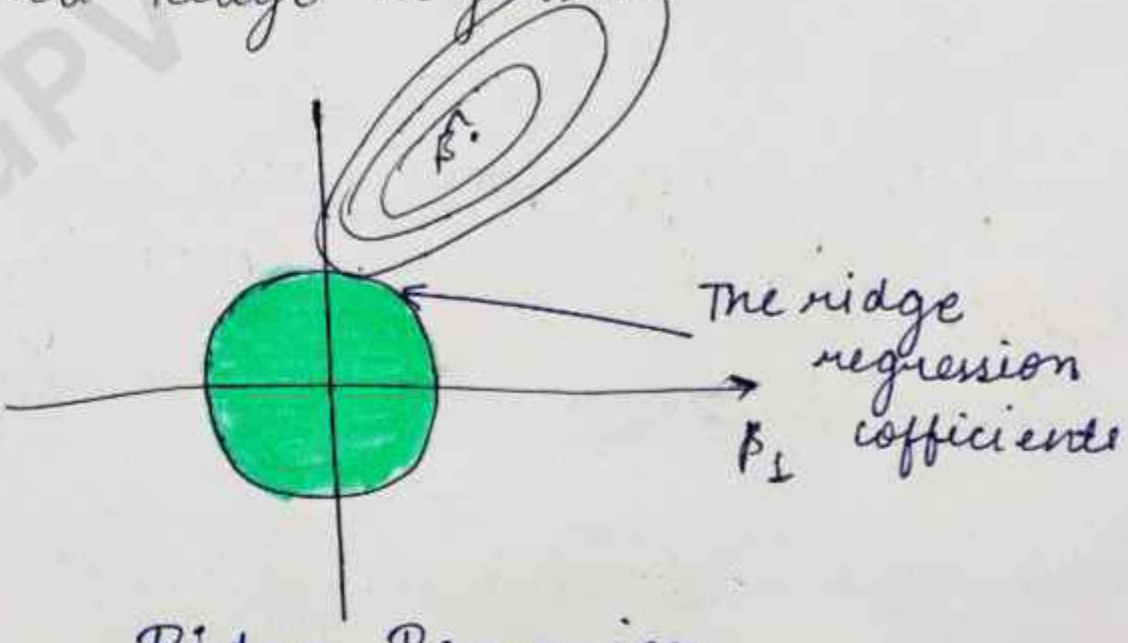
- Prevent Overfitting effectively
- Stable training
- Works well in deep learning
- Handles correlated features better

Disadvantages of L2:-

- Does not perform features selection.
- Keeps all features in model.

Ridge Regression

L2 regularization in linear regression is called Ridge Regression.



Ridge Regression

RGPV FULL CONTENT

Hyperparameter Tuning

Hyperparameter tuning is a process in ML used to find the best values of hyperparam -eters so that a model gives maximum accuracy and performance.

Hyperparameters are settings that are chosen before training the model.

They control:

- learning process
- Model complexity
- training behavior

They are not learned automatically from data.

Examples of Hyperparameters.

Hyperparameters		Meaning
1 Learning Rate	→	Controls step size during training
2 Batch Size	→	Number of samples processed together.

PGPV ALL CONTENT

3 Number of Epochs → How many time data passes through model.

4 Number of Hidden Layers → Depth of neural network

5 Number of Neurons → Units in each layer

6 Regularization Parameter (λ) → control overfitting

Importance

→ Correct Hyperparameters help to :-

- Improve accuracy
- Reduce overfitting
- Speed up training
- Increase model stability.

→ Bad Hyperparameters may cause :

- poor accuracy
- slow learning
- overfitting.

Example

Suppose learning rate is :-

- Too high :- model may overshoot minimum.
- Too low :- training becomes very slow.

So we find the best value.

PGDALECONTENT

Types of Parameters

① Model Parameters → Learned automatically during training.

Examples :- weights & biases.

② Hyperparameters

Set manually before training

Example-learning rate, epochs, batch size.

Techniques of Hyperparameter Tuning.

Some Best techniques of Hyperparameter tuning are given below :-

① Manual search

② Grid search

③ Random search.

④ Bayesian Optimization.

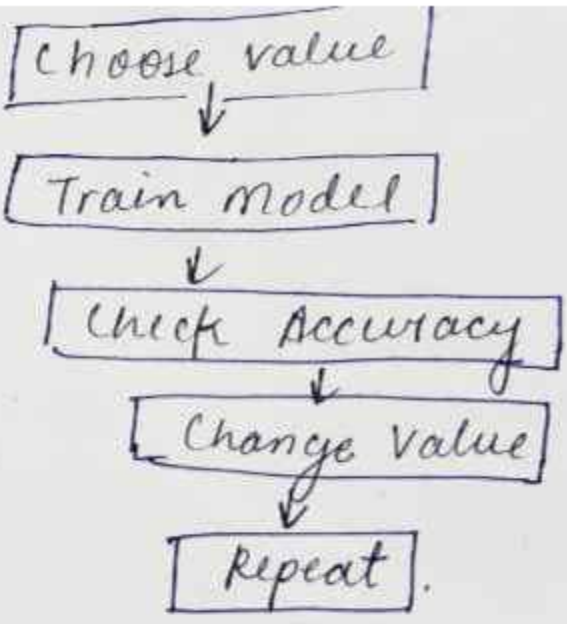
1 Manual search

In this technique, the user manually changes hyperparameters and checks performance.

Example :- learning rate 0.1
- then 0.01
- then 0.001

RCP ALL CONTENT

- It is simple and easy for beginners
- But time consuming and depend on user experience.



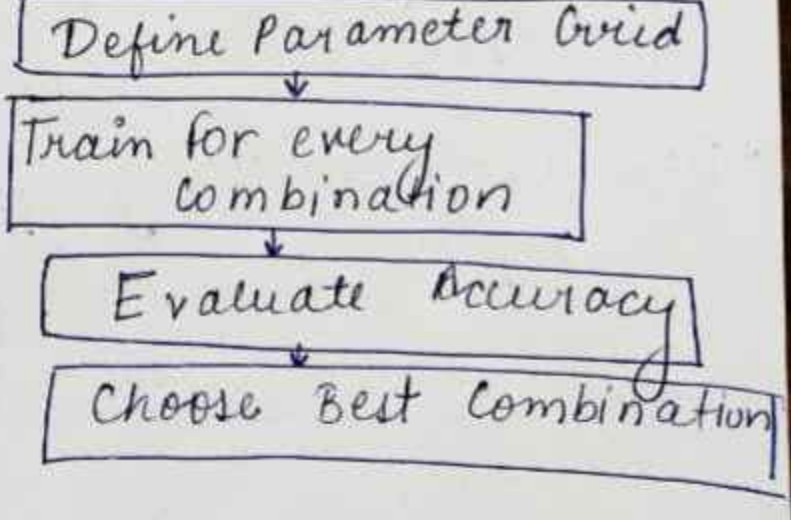
② Grid Search.

Grid search tests all possible combinations of hyperparameters.

Example →

Learning Rate	Batch size
0.1	16
0.1	32
0.01	16
0.01	32

- All combinations are checked
- It is systematic and find good combinations.
- But very slow and expensive for large datasets.



FOR ALL CONTENT

③ Random Search

- Random Search selects random combinations instead of all combinations

Example →
→ Instead of checking 100 combinations. it may test only 10 random combinations.

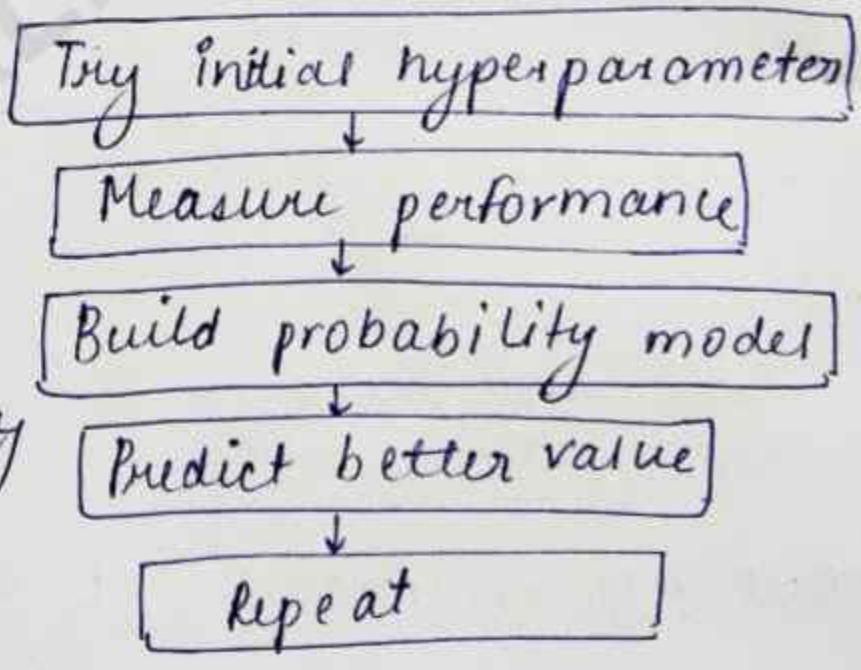
★ It is Faster than grid search and efficient in large space.

④ Bayesian Optimization.

Bayesian optimization uses probability to predict better hyperparameters.

It learns from previous trials.

- It is very efficient and need fewer experiments.
- But mathematically complex



RGM ALL CONTENT